

# RamboQuant

Complete Design Guide

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# RamboQuant — Complete Design Guide

| Generated     | Revision                 | Website                                     |
|---------------|--------------------------|---------------------------------------------|
| See PDF cover | See <code>git log</code> | <a href="https://ramboq.com">ramboq.com</a> |

**Tech Stack** — SvelteKit · Litestar · PostgreSQL · async SQLAlchemy 2.x · KiteTicker WebSocket · Kite / Dhan / Groww adapters · MCP · Gemini.

## About the author

**Ramana R Ambore, FRM** — Platform Architect & Quantitative Developer for **RamboQuant LLP** · *AI-augmented · built with Claude Code*.

**30+ years** across mainframe modernization and cloud-native financial platforms, including **19 years at Fidelity Investments**. Currently **Principal System Analyst** at Fidelity — leading billing-platform modernization on AWS + Snowflake and distributed fee-calculation engines — and concurrently the **platform architect + quantitative developer** building **RamboQuant** end-to-end (live at [ramboq.com](https://ramboq.com)), applying a **forward-deployment-engineer ethos** (end-to-end ownership, tight feedback loops, ship-what-works) and leveraging **Claude Code** as a force multiplier for platform-scale solo output.

**FRM** (GARP, 2022) and **CFA Level 3** candidate: the derivatives-risk, options-pricing (Black-Scholes / Greeks), and portfolio-analytics theory from those programs materialize directly as RamboQuant's derivatives layer, hedge-proxy  $\beta$  regression, and units-based NAV accounting. The Master's in Computer Science + Fidelity engineering discipline (distributed systems, event-driven architecture, legacy-modernization patterns) keeps a multi-broker, real-time, single-operator platform correct and fast. **NTT Innovation Award** recipient (top-40 global innovator). Based in Merrimack, NH.

**Profile:** [ramanaambore.me](https://ramanaambore.me)

## Experience & recognition

|                       |                                                                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Current role</b>   | Principal System Analyst · Fidelity Investments — billing-platform modernization on AWS + Snowflake; distributed fee-calculation engines. |
| <b>Industry depth</b> | 30+ years FinTech, mainframe to cloud-native — derivatives risk, options pricing (Black-Scholes, Greeks), multi-leg strategy analytics.   |
| <b>Recognition</b>    | NTT Innovation Award — top-40 global innovator, selected for financial-services engineering.                                              |

## Credentials

FRM (GARP, 2022) · CFA Level 3 · Master's, Computer Science · Six Sigma Green Belt · IBM Certified DB2 DBA · Sun Certified Java Programmer.

## Links

- **Website:** [ramboq.com](https://ramboq.com)
- **Profile:** [ramanaambore.me](https://ramanaambore.me)
- **Contact:** through the [ramboq.com](https://ramboq.com) contact page
- **Repo:** [github.com/RamanaAmbore/algo-trader](https://github.com/RamanaAmbore/algo-trader)

## RamboQuant platform scope

Builds and maintains the RamboQuant platform end-to-end: a production application covering multi-broker order routing, real-time market data pipelines, options analytics, portfolio tracking, and operator + investor-facing tooling.

**Full tech stack:**

| Layer         | Technologies                                                                                                                                                                                            |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend      | SvelteKit · Svelte 5 (runes) · Vite · ag-grid-community · @tanstack/svelte-query · Tailwind · Playwright (e2e)                                                                                          |
| API           | Litestar 2.x · <b>msgspec.Struct</b> schemas (10× faster than pydantic) · uvicorn (single-worker) · asyncio                                                                                             |
| Persistence   | PostgreSQL 17 · async SQLAlchemy 2.x · asyncpg · <b>polars</b> (routes) · pandas (broker SDK boundary only)                                                                                             |
| IPC           | <b>mmap</b> at /dev/shm/ramboq_ticks (KiteTicker → API tick pipeline) · UDS at /tmp/ramboq_conn.sock (main API ↔ conn_service) · SSE (live LTP push) · WebSocket · in-process BroadcastBus              |
| Queues        | In-process asyncio + EventQueue (algo/agent/order/mcp_audit) + write_queue (disk + db bulk-batched workers) · <b>ARQ + Redis</b> background worker as a separate systemd unit ( ramboq_worker.service ) |
| Brokers       | kiteconnect · dhanhq · growwapi · pyotp (Kite 2FA TOTP) · KiteTicker WebSocket · Fernet-encrypted credentials at rest                                                                                   |
| Intelligence  | Gemini ( google-genai ) market summaries · MCP server (17 read + 2 persist + 6 write-gated tools) · fpdf2 (monthly investor statements) · babel (i18n / number formatting)                              |
| Security      | JWT HS256 (24h TTL) · PBKDF2-SHA256 password hashing · cryptography.Fernet (broker creds encryption) · maxminddb (visitor geolocation)                                                                  |
| Deploy        | systemd ( ramboq_api , ramboq_conn , ramboq_dev_api , ramboq_worker , ramboq_hook ) · nginx reverse proxy · Cloudflare DNS · webhook.ramboq.com HMAC-authenticated auto-deploy hook                     |
| Observability | Perf-snapshot cron + admin dashboard · web-vitals runtime capture (Playwright) · radon (cyclomatic complexity) · pytest-json-report (test-duration tracking) · vulture (dead-code detection)            |

## About this document

The full developer onboarding document. Read top-to-bottom to understand the codebase end-to-end; reference specific sections for ongoing work. Flow diagrams use Mermaid. Tech-stack rationale (why / what / how / where) is interleaved with each subsystem rather than collected separately — easier to learn in context.

**Goal:** anybody who reads and understands this document should be able to modify and enhance features by making the actual code changes. Each subsystem section names the files; **Part IX** at the end is a cookbook of common change recipes with exact-diff-level guidance.

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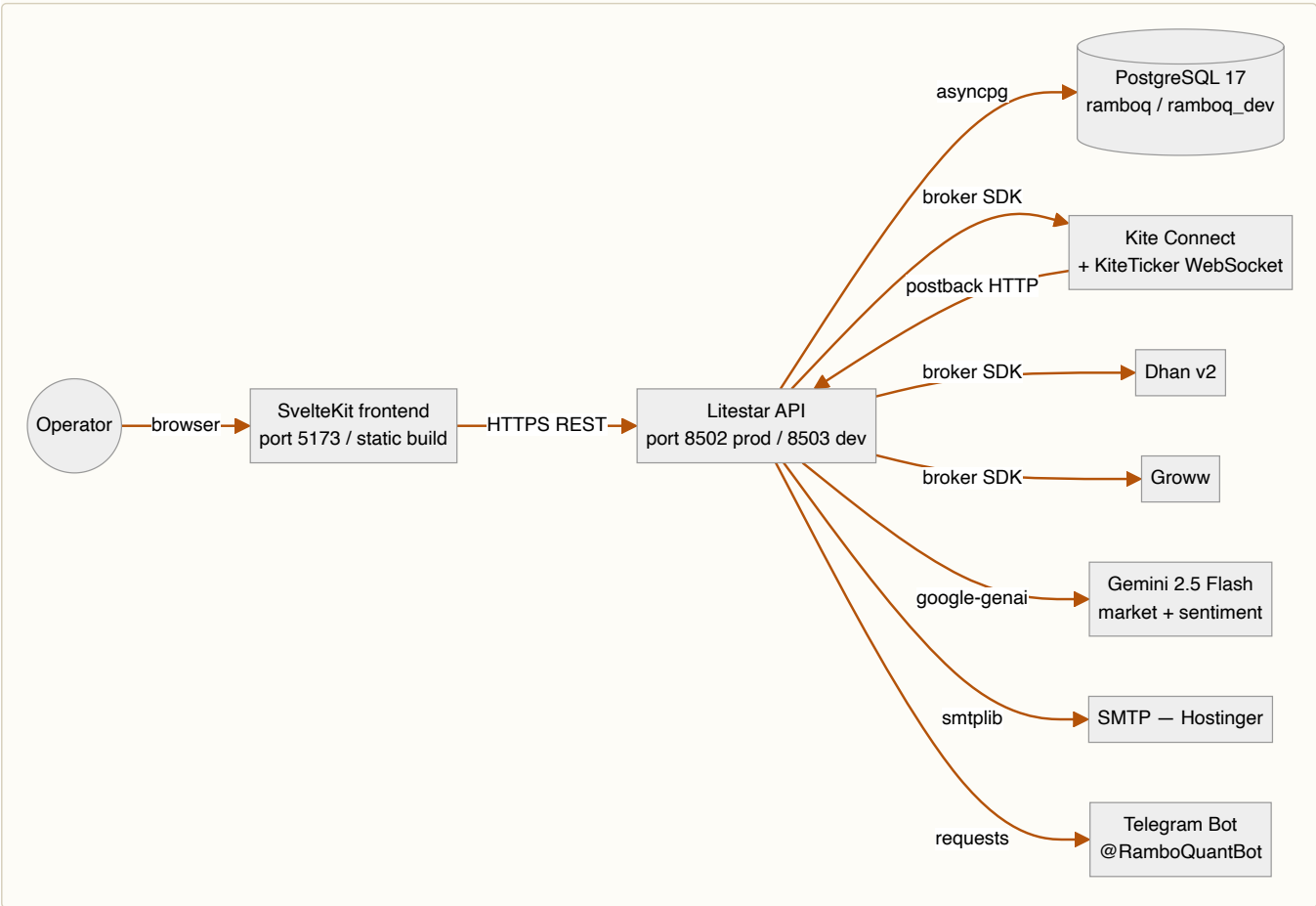
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Tech-stack rationale boxes appear inline as ⚙️ **TECH: WHY · WHAT · HOW · WHERE** callouts throughout. Look for the gear glyph.

# Part I — Foundation

## 1. Architecture overview



| Layer      | Tech                                                          | Key files                           |
|------------|---------------------------------------------------------------|-------------------------------------|
| Frontend   | SvelteKit + Svelte 5 runes + ag-Grid + hand-rolled SVG charts | frontend/src/                       |
| API        | Litestar 2.x + msgspec.Struct schemas                         | backend/api/                        |
| DB         | PostgreSQL 17 + SQLAlchemy 2.x async + asyncpg                | backend/api/database.py , models.py |
| Brokers    | Vendor SDKs behind a unified Broker ABC                       | backend/brokers/                    |
| Background | asyncio tasks spawned at app startup                          | backend/api/background.py           |

## 2. Tech stack — at a glance

Each choice has a **why/what/how/where** callout inline where relevant (marked by ⚙️). Key stacks:

| Layer         | Tech                                           | Why                                                                       |
|---------------|------------------------------------------------|---------------------------------------------------------------------------|
| API           | Litestar 2.x + msgspec                         | ~10× faster JSON encode/decode than pydantic on big payloads              |
| DB            | PostgreSQL 17 + SQLAlchemy 2.x async + asyncpg | Fast, reliable, static typing, JSONB for attached_gtts_json blob          |
| Frontend      | SvelteKit + Svelte 5 runes + ag-Grid           | Smaller bundle, native reactivity, row virtualization for 1000+ ticks/sec |
| Charts        | Hand-rolled SVG (no Chart.js)                  | 150KB saved, tighter control, palette integration                         |
| Concurrency   | asyncio + single uvicorn worker                | Kite token affinity, in-process locks, background task state              |
| Notifications | Telegram (Bot API), SMTP (Hostinger)           | Free, reliable, works everywhere                                          |
| WebSocket     | KiteTicker (Twisted reactor → SSE bridge)      | Sub-second LTP updates without burning rate limit                         |

Full rationale for each technology appears as **TECH** callouts throughout this doc (e.g. §3.1 broker abstraction, §4.2 KiteTicker threading, §4.3 background task lifecycle).

### 3. Core architectural principles

#### 3.1 Single source of truth at the broker boundary

The `Broker` abstract base class (`backend/brokers/base.py`) is the **only** place vendor differences should leak. Every route, agent, and background task talks to a `Broker` instance via `get_broker(account)` from the registry.

**TECH** — **WHY** Vendor SDKs disagree on EVERYTHING (qty units, status strings, GTT shape). Letting that disagreement propagate past the adapter boundary creates bug surface area in every consumer. **WHAT** The ABC declares ~20 methods (`place_order`, `modify_order`, `cancel_order`, `orders`, `holdings`, `positions`, `funds`, `ltp`, `quote`, `historical_data`, `place_gtt`, `modify_gtt`, `cancel_gtt`, `get_gtts`, `instruments`, `profile`, `holidays`, `order_status`, `trades`, `basket_order_margins`). **HOW** New broker? Implement every method, translate to Kite shape in `_normalise_*` helpers, register in `_ADAPTERS`. Capability gaps go in `BrokerCapabilities` — never inline `if broker_id == "groww"`. **WHERE** `backend/brokers/base.py` (ABC); `backend/brokers/adapters/kite.py` / `dhan.py` / `groww.py` (implementations); `backend/brokers/capabilities.py` (matrix).

#### 3.2 Idempotency is the default

Every path that places a broker order or GTT can fire twice — postbacks arrive twice, chase terminals race postbacks, reconcile sweeps re-fire attaches. Four patterns make this safe:

| Pattern                                             | Where                                      | What it guards                                               |
|-----------------------------------------------------|--------------------------------------------|--------------------------------------------------------------|
| <code>attached_gtts_json IS NULL</code> check       | <code>_fire_template_attach_on_fill</code> | Double-place TP/SL/Wing at broker                            |
| <code>_TEMPLATE_ATTACH_LOCKS[parent_row_id]</code>  | Same function                              | Concurrent races within the same row                         |
| <code>_KILLED_ORDER_IDS</code> dict with 60-min TTL | <code>chase.py</code>                      | Operator kills landing on stale <code>broker_order_id</code> |
| <code>MAX(prior, cumulative)</code> clamp           | <code>_record_partial_fill</code>          | Restart causing cumulative to be added again                 |

**When adding a new fill-time side-effect, ask:** can my handler fire twice for the same parent? If yes, what's the idempotency check?

#### 3.3 Database is authoritative; in-memory is fast-path

The single uvicorn worker (`--workers 1` in prod — see §4.1) means in-process locks are sufficient, but the DB is still the source of truth. After a restart, every chase loop recovers via `recover_from_db` and re-derives state. Don't store anything operationally meaningful in in-process state without a DB write to back it up.

The `attached_gtts_json` column is a deliberate small-state JSON blob rather than a foreign-key normalized table:

- ✓ Atomic write per parent — no half-attached state visible to readers
- ✓ Easy to refactor the GTT spec shape (just version the JSON inside)
- ✗ Harder to JOIN against; we accept this because GTT inspection is rare

#### 3.4 Async by default, sync when forced

Everything API-facing is `async def` over `asyncpg`. Broker SDK calls are sync — Kite/Dhan/Groww use `requests` under the hood — so we wrap them in `asyncio.to_thread(...)` to keep the event loop unblocked. The threadpool sizing is the default (32 workers); we've never seen it saturate because broker API calls are sub-second.

**Anti-pattern to avoid:** `broker.method()` directly in an `async def` route handler. Even if it returns "fast," a single 2-second hang stalls every other request on that worker.

### 3.5 Demo mode = signed-out + prod branch

Demo isn't a separate code path — it's a runtime guard at the API boundary ( `backend/api/auth_guard.py` ) plus a frontend flag pulled from context. The same routes serve authenticated + demo traffic; the guard masks accounts and blocks writes. This means **a feature works in demo the moment it works for read-only sessions** — there's no separate "demo enablement" step to forget.

### 3.6 Singleton pattern — one instance, everywhere

Several long-lived, expensive-to-construct components are implemented as **process-wide singletons**. The pattern is used deliberately where all of the following are true: (a) construction is expensive (network handshake, mmap open, DB warm), (b) there is exactly one canonical instance per process, (c) many callers need the *same* instance so state stays consistent. Fits nicely with the single-uvicorn-worker guarantee from §4.1 — no cross-worker cache-coherence problem.

**Canonical singletons:**

| Singleton             | Module                                                 | Purpose                                                                                                                                                                                                                                                                                        |
|-----------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Connections           | <code>backend/brokers/connections.py</code>            | Broker session manager (Kite / Dhan / Groww). Owns credentials (decrypted from <code>broker_accounts</code> table via Fernet), 2FA, token refresh, IPv6 binding. <code>connections.Connections()</code> returns the same instance every time; <code>rebuild_from_db()</code> mutates in place. |
| TickerManager         | <code>backend/brokers/kite_ticker.py</code>            | KiteTicker WebSocket wrapper. <b>One WebSocket per process</b> — Kite rejects duplicates. Lives inside <code>conn_service</code> when <code>RAMBOQ_USE_CONN_SERVICE=1</code> ; owns the mmap tick writer at <code>/dev/shm/ramboq_ticks</code> .                                               |
| MmapTickReader        | <code>backend/brokers/mmap_ticker.py</code>            | Main-API tick reader. <code>get_mmap_reader()</code> returns the module-level <code>_singleton</code> . Polls the shared-memory version-word every 50ms; publishes deltas to <code>BroadcastBus</code> .                                                                                       |
| BroadcastBus          | <code>backend/api/broadcast.py</code>                  | In-process pub/sub. Every SSE client + WebSocket connection subscribes to the same bus for tick + <code>book_changed</code> + <code>position_filled</code> events.                                                                                                                             |
| GrammarRegistry       | <code>backend/api/algo/grammar_registry.py</code>      | Agent condition-tree token catalog. Reloaded on <code>/api/admin/grammar/reload</code> (mutates the singleton in place; existing agents pick up the new tokens on their next <code>run_cycle</code> ).                                                                                         |
| MarketLifecycle       | <code>backend/api/algo/market_lifecycle.py</code>      | Per-exchange open / close / close_settled event bus. Polled every 30s by <code>_task_market_lifecycle</code> . Handlers registered via <code>market_lifecycle.register(event, callback)</code> at import time.                                                                                 |
| MetricRegistry (perf) | <code>backend/api/persistence/perf_snapshots.py</code> | Perf-snapshot writer coordinator — buffers metrics until nightly <code>_task_perf_snapshot</code> flushes them.                                                                                                                                                                                |

**Convention:** singleton modules expose a lowercase accessor ( `connections.Connections()` , `get_mmap_reader()` , `broadcast_bus()` ) rather than a bare module-level global — the accessor lets us swap the implementation in tests (e.g., replace `_singleton` with a fake) and defers construction until first use.

**Anti-patterns to avoid:**

- **Don't** create local instances of these classes anywhere in route handlers or background tasks. Always go through the accessor. Multiple `Connections()` instances would each mint a fresh Kite session and the second one would evict the first from Kite's session registry.
- **Don't** cache the singleton reference across `rebuild_from_db()` boundaries — the accessor is cheap; long-lived local references miss config changes.
- **Don't** rely on singleton state during startup — `on_startup` order matters. See `backend/api/app.py: on_startup` for the canonical wire-up sequence.

**Testing note:** pytest fixtures reset `_singleton = None` in `conftest.py::reset_singletons` between tests so isolation holds. Tests that need a real singleton use the `real_connections` fixture.

## 4. Concurrency model

### 4.1 Why one uvicorn worker?

`--workers 1` in prod is **intentional** for three reasons:

- **Kite session affinity:** multiple workers would invalidate each other's Kite tokens because Kite enforces one active session per IP.
- **In-process locking is enough:** all locks are `asyncio.Lock` instances; we never need multiprocessing coordination.
- **Background tasks need shared state:** the trail-stop poller's in-memory state ( `_TEMPLATE_ATTACH_LOCKS` , the ticker manager's `_tick_map` ) is process-scoped. Multi-worker would require Redis or similar.

If we ever scale horizontally we'd need to externalize: tokens → DB, locks → DB advisory locks or Redis, ticker state → a separate fanout service.

## 4.2 KiteTicker threading

KiteTicker runs Twisted internally — **all WebSocket callbacks fire on a Twisted reactor thread**, not the asyncio event loop. The TickerManager bridges this:

- Twisted thread writes `_tick_map[token] = ltp` under a `threading.Lock`
- Async handlers read via `get_ltp(token)` — same lock, briefly held

The lock is non-reentrant and the critical section is O(1) so no deadlock risk.

🔗 **TECH** — WHY Twisted reactors can't see asyncio's event loop, so we can't `await` from a tick callback. Lock-protected dict is the simplest viable bridge. WHAT `TickerManager._tick_map: dict[int, float] + _tick_lock: threading.Lock`. HOW Tick handler does with `self._tick_lock: self._tick_map[token] = ltp`. Async reader does the same under the lock, briefly. WHERE `backend/brokers/kite_ticker.py`.

If you add anything that runs on the Twisted side, never call `asyncio.run_coroutine_threadsafe` without testing both directions of the round-trip. The reactor doesn't know about asyncio's event loop.

## 4.3 Background task lifecycle

All background tasks are spawned in `app.on_startup` via `asyncio.create_task(...)`. They run forever; cancellation only happens at app shutdown. Each task is responsible for its own error handling — an uncaught exception kills the task silently. Every task body should be:

```
async def _task_X():
    while True:
        try:
            ...real work...
        except Exception as e:
            logger.exception(f"_task_X iteration failed: {e}")
            await asyncio.sleep(interval)
```

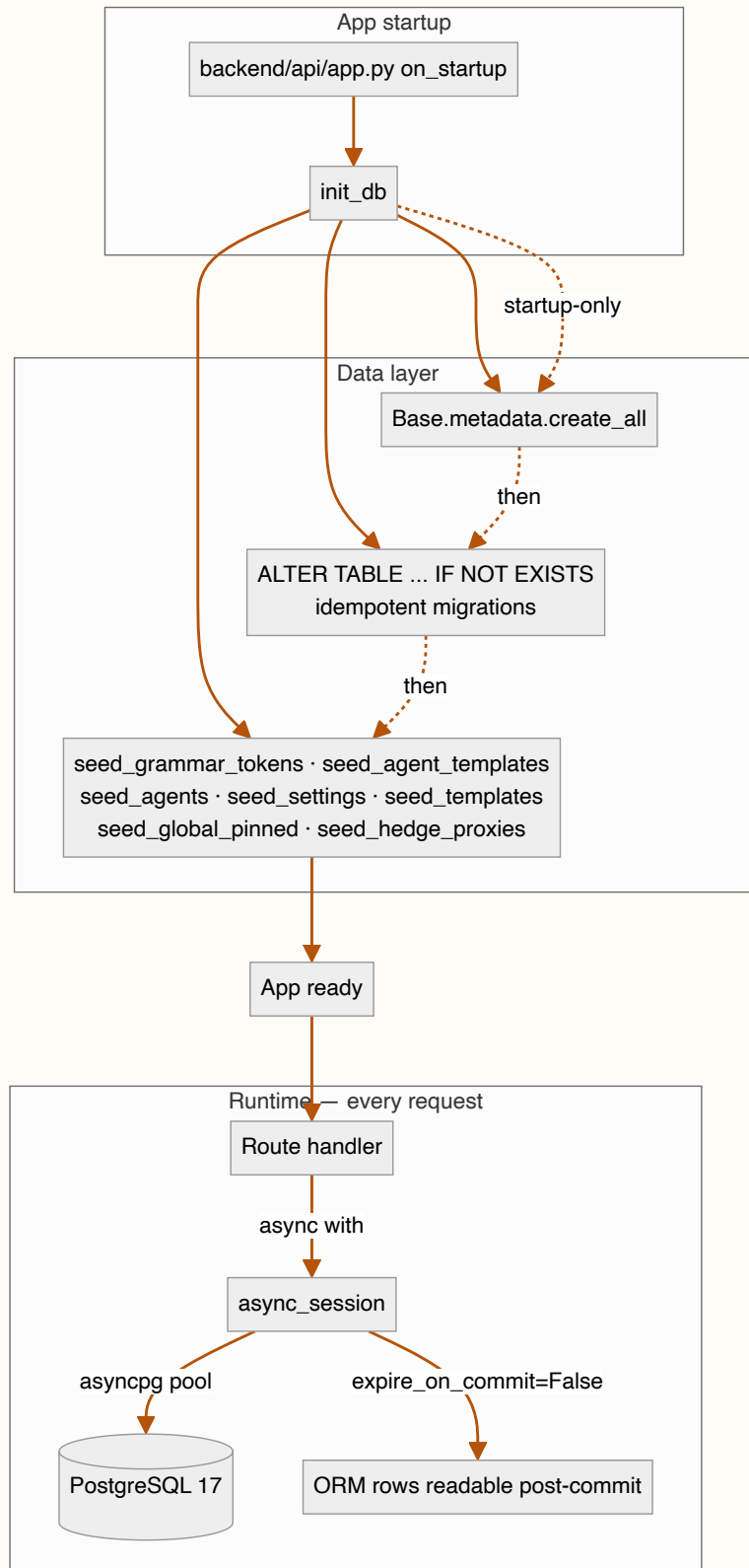
The `try/except` around the loop body is non-negotiable. We've burned hours debugging "why did the trail stop go silent" only to discover an unhandled `KeyError` ate the task three days earlier.

## 4.5. Data layer — implementation detail

This section is the "if you're modifying the data layer, here's the actual shape" reference. The data layer = SQLAlchemy 2.x async ORM + PostgreSQL + asyncpg driver. Read this end-to-end before changing any table.



#### 4.5.1 Topology



## 4.5.2 File map

| File                               | Purpose                                                                                                                                 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| backend/api/database.py            | Engine + session factory + <code>init_db</code> (the only place we touch DDL)                                                           |
| backend/api/models.py              | Every SQLAlchemy declarative model. One file by convention so the data shape is one grep away.                                          |
| backend/api/schemas.py             | <code>msgspec.Struct</code> wire types. Mirror of models for HTTP responses.                                                            |
| backend/shared/helpers/settings.py | SEEDS list + cached settings reader ( <code>get_int</code> / <code>get_float</code> / <code>get_bool</code> / <code>get_string</code> ) |
| backend/api/algo/templates_seed.py | <code>SYSTEM_TEMPLATES</code> + the seeder                                                                                              |
| backend/api/algo/grammar.py        | <code>_SYSTEM_TOKENS</code> + grammar registry seeder                                                                                   |
| backend/api/cache.py               | In-memory TTL cache with per-key locking (NOT a substitute for the DB; cache invalidates on PATCH)                                      |

## 4.5.3 Engine + session factory

`database.py` is small and worth reading in full. Key shape:

```
# backend/api/database.py
engine = create_async_engine(
    DATABASE_URL,          # postgresql+asyncpg://...
    echo=False,
    pool_size=5,           # max connections kept warm in the pool
    max_overflow=10,       # extra one-off connections when pool exhausted
)

async_session = async_sessionmaker(
    engine,
    expire_on_commit=False, # load-bearing – see 4.5.5
    class_=AsyncSession,
)

async def init_db() -> None:
    """Idempotent: safe to run on every startup."""
    async with engine.begin() as conn:
        await conn.run_sync(Base.metadata.create_all)
        # ALTER TABLE ... IF NOT EXISTS for every column added after
        # the table's initial creation. We don't use Alembic.
        await conn.execute(text(
            "ALTER TABLE algo_orders ADD COLUMN IF NOT EXISTS filled_quantity INTEGER"
        ))
        # ... many more such ALTERS ...
    # Seeders run after the DDL block – see §4.5.7
```

⚠ **TECH — Why `expire_on_commit=False`** — **WHY** After `commit()`, SQLAlchemy's default is to expire all ORM attributes on the committed rows, so the next attribute access triggers a fresh SELECT. That's catastrophic in our codebase because several handler paths commit then immediately read attributes (chase reconcile attach queue, `retry_template`, `postback fallback match`). With `expire-on-commit` we'd issue redundant SELECTs per commit. **WHAT** Setting this to `False` keeps the in-Python row state intact after commit. **HOW** Set globally on the `async_sessionmaker`. Never override per-session — consistency matters. **WHERE** `backend/api/database.py:async_session`.

## 4.5.4 Models — how to add / modify

`backend/api/models.py` is the canonical schema. Every table is a `Mapped[]`-typed class:

```
# backend/api/models.py
class AlgoOrder(Base):
    __tablename__ = "algo_orders"

    id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
    broker_order_id: Mapped[str | None] = mapped_column(String(64), nullable=True, index=True)
    account: Mapped[str] = mapped_column(String(32), index=True)
    symbol: Mapped[str] = mapped_column(String(64), index=True)
    quantity: Mapped[int] = mapped_column(Integer)
    filled_quantity: Mapped[int | None] = mapped_column(Integer, nullable=True)
    status: Mapped[str] = mapped_column(String(32), default="OPEN", index=True)
    mode: Mapped[str] = mapped_column(String(8), default="live", index=True)
    template_id: Mapped[int | None] = mapped_column(
        ForeignKey("order_templates.id", ondelete="SET NULL"), nullable=True
    )
    attached_gtts_json: Mapped[str | None] = mapped_column(Text, nullable=True)
    created_at: Mapped[datetime] = mapped_column(
        DateTime(timezone=True), server_default=func.now(), index=True
    )
    # ... ~30 more fields, see actual file ...

    __table_args__ = (
        Index("ix_algo_orders_mode_status", "mode", "status"), # composite, Sprint E
    )
```

#### Rules when adding a column:

1. Add the `Mapped[]` declaration to the model class.
2. Add an idempotent `ALTER TABLE algo_orders ADD COLUMN IF NOT EXISTS ...` to `init_db`. **Never write an Alembic migration** — our pattern is idempotent ALTERs at startup.
3. New column should be `nullable=True` with sensible default unless you're guaranteed to backfill all rows.
4. Composite indexes go in `__table_args__` and need an `ALTER` to ensure existence on rebuild. The convention: `Index("ix_<table>_<cols>", "col1", "col2")`.

#### Rules when adding a table:

1. Subclass `Base`, set `__tablename__`.
2. Declarative create happens automatically via `Base.metadata.create_all`.
3. Add seeded rows via a new seeder function called from `init_db`.
4. Add the corresponding `msgspec.Struct` in `schemas.py` if the table is operator-visible.

### 4.5.5 Session lifecycle in handlers

The canonical handler pattern:

```
@get("/example", guards=[auth_or_demo_guard])
async def example(self, request: Request) -> ExampleResponse:
    async with async_session() as s:
        # Reads + writes happen here
        rows = (await s.execute(
            select(AlgoOrder).where(AlgoOrder.status == "OPEN")
        )).scalars().all()
        # Mutate
        for r in rows:
            r.last_seen = datetime.now(timezone.utc)
        # Single commit at the end of a logical group
        await s.commit()
    return ExampleResponse(rows=[_to_info(r) for r in rows])
```

#### Anti-patterns to avoid:

- **✗** Holding a session across `await` to a broker SDK call. The broker call could take 5 seconds; the session holds a connection from the pool the whole time. Wrap the broker call in `asyncio.to_thread` OR exit the `async with` block first and re-open after.
- **✗** Committing inside a loop without batching. Each commit is a round-trip — if you have 100 rows to update, batch into one commit at the end.
- **✗** `select(AlgoOrder)` without a WHERE clause and no LIMIT. Full-table scans land in production logs eventually; always paginate operator-visible queries.
- **✗** Mutating an ORM row from one session and reading from another within the same request. Use one session per logical unit of work.

### 4.5.6 Idempotent migrations pattern

We don't use Alembic. Every schema change is an `ALTER TABLE ... IF NOT EXISTS` in `init_db`:

```

async def init_db():
    async with engine.begin() as conn:
        await conn.run_sync(Base.metadata.create_all)
        await conn.execute(text(
            "ALTER TABLE algo_orders ADD COLUMN IF NOT EXISTS filled_quantity INTEGER"
        ))
        await conn.execute(text(
            "ALTER TABLE algo_orders ADD COLUMN IF NOT EXISTS sl_trail_pct NUMERIC(8, 4)"
        ))
    # ... dozens more ...

```

This pattern was chosen because:

- **✓** Zero ops overhead — no migrations folder, no version cursor, no rollback worry.
- **✓** Deploy is just `git pull + restart`. The new column appears the moment the restart finishes.
- **✓** Branch-switching works — if dev is ahead of main, switching back to main doesn't break (the column exists; the code that uses it is gone).
- **✗** No DOWN migrations. We don't drop columns; if we want to remove a field, we stop reading from it but leave the column in place.
- **✗** Cannot rename columns easily. The workaround: add a new column, dual-write for one deploy, switch reads, then stop writing the old.

The pattern is suitable for small-team prod with infrequent destructive changes. If we ever need ten-engineer concurrent migrations, this stops scaling and we'd move to Alembic.

#### 4.5.7 The seeders — bootstrapping defaults

Seven seeders run at startup. Six fire from inside `init_db()` (in `backend/api/database.py`); `seed_hedge_proxies` is registered as its own `on_startup` coroutine in `app.py` so it runs after the DDL block:

| Seeder                            | Where it lives                                                                                                          | What it seeds                                                                                                                                                | Operator-<br>overridable?                                                                                                                            |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>seed_grammar_tokens</code>  | <code>backend/api/algo/grammar.py</code>                                                                                | Grammar tokens<br>(metric/scope/op/channel/format/template/action_type)                                                                                      | Toggleable<br><code>is_active</code> ;<br>cannot delete<br>system tokens                                                                             |
| <code>seed_agent_templates</code> | <code>backend/api/algo/template_registry.py</code>                                                                      | Reusable agent templates referenced by built-in agents                                                                                                       | Toggleable; refresh<br>on boot                                                                                                                       |
| <code>seed_agents</code>          | <code>backend/api/algo/agent_engine.py</code>                                                                           | 10 built-in agents (6 loss-, 3 <i>expiry</i> -, 1 manual)                                                                                                    | Editable; seeder<br>force-resets<br><code>schedule</code> field +<br>force-inactives<br>built-in summary<br>agents on every<br>boot                  |
| <code>seed_settings</code>        | <code>backend/shared/helpers/settings.py</code>                                                                         | Settings rows ( <code>alerts.cooldown_minutes</code> ,<br><code>chase.max_consecutive_errors</code> , etc.)                                                  | Yes — operator<br>edits via<br><code>/admin/settings</code> ;<br>seeder preserves<br><code>value</code> and only<br>refreshes metadata               |
| <code>seed_templates</code>       | <code>backend/api/algo/templates_seed.py</code>                                                                         | 4-default order templates ( <code>default-bull</code> , <code>default-<br/>long-option</code> , <code>default-bear</code> , <code>default-short-vol</code> ) | Partial — system<br>templates are<br>overwritten on<br>every restart, but<br>operator's saved<br>copies (different<br><code>user_id</code> ) survive |
| <code>seed_global_pinned</code>   | <code>backend/api/routes/watchlist.py</code>                                                                            | Pinned global watchlists (Markets, Default)                                                                                                                  | Editable per-user;<br>global rows refresh                                                                                                            |
| <code>seed_hedge_proxies</code>   | <code>backend/api/routes/hedge_proxies.py</code><br>(runs from <code>app.on_startup</code> , not <code>init_db</code> ) | Six default pairs (GOLDBEES → GOLD/GOLDM, etc.)                                                                                                              | Editable via<br><code>/admin/settings</code><br>→ Hedge proxies                                                                                      |

The recurring tension in seeders: **how much to overwrite on every boot vs preserve operator changes?** The pattern:

- **Description / schema / metadata** — always refreshed (code is the source of truth)
- **value / conditions / actions / events** — preserved on existing rows (operator's overrides)
- **status** — preserved EXCEPT for built-in summary agents which force-reset to `inactive` (see `seed_agents` )
- **New rows added** with whatever default the SEEDS const ships

If you're adding a new seeder, follow this pattern. The audit-friendly path is `INSERT ... ON CONFLICT DO UPDATE SET <metadata only>` so existing values can't be clobbered.

#### 4.5.8 Settings cache layer

`backend/shared/helpers/settings.py` exposes `get_int`, `get_float`, `get_bool`, `get_string` readers backed by an in-process dict cache:

```
_SETTINGS_CACHE: dict[str, Any] = {}
_CACHE_GENERATION = 0

def get_int(key: str, default: int) -> int:
    val = _SETTINGS_CACHE.get(key)
    if val is None:
        return default
    return int(val)
```

The cache loads on first read + invalidates on every PATCH (the route handler bumps `_CACHE_GENERATION` and clears `_SETTINGS_CACHE`). This means:

- ✔ Hot-path reads are O(1) dict lookup — settings are checked thousands of times per minute on background tasks.
- ✔ Operator edits take effect on the next read, no service restart.
- ✗ Multi-worker would need per-worker invalidation (we're single-worker — see §4.1 — so this isn't an issue).

When adding code that reads a setting, **always go through `get_*` helpers**. Never `SELECT ... FROM settings WHERE key = ...` in a hot path.

#### 4.5.9 `attached_gtts_json` — the small-state JSON blob pattern

`AlgoOrder.attached_gtts_json` deserves its own subsection because it's the load-bearing column for template attach (§9):

```
# Persisted shape (string-encoded JSON)
[
  {
    "kind": "gtt",
    "label": "TP",
    "id": "abc123",           # broker GTT id
    "current_trigger": 250.0,
    "sl_trail_pct": null,
    "tp_trigger": 250.0,
    "highest_ltp": 200.0,    # set if sl_trail_pct is non-null
    "lowest_ltp": 200.0,
    "sibling_id": null      # set for Groww emulated OCO pairs
  },
  {
    "kind": "wing",
    "label": "Wing BUY",
    "id": "def456",         # broker order id (not GTT – wings are real positions)
    "qty": 50,
    "symbol": "NIFTY24APR21500PE"
  },
  ...
]
```

**Why a blob and not a child table?** Three reasons:

- Atomic write per parent — readers never see "half-attached" state.
- Easy to refactor the spec shape (just version the JSON inside, no migration).
- GTT inspection is rare — we don't JOIN against it. When we read it, we read the whole bracket anyway.

**Idempotency rule:** the `_fire_template_attach_on_fill` function checks `attached_gtts_json IS NULL` before writing. Once populated, it's never re-attached (see §9 for the full guard chain). To force a re-attach, the operator hits `POST /orders/algo/<id>/retry-template`.

#### 4.5.10 Pool sizing + connection accounting

Defaults:

- `pool_size=5` connections kept warm
- `max_overflow=10` extra one-off connections under burst
- `pool_pre_ping=True` checks the connection is alive before checkout

We've never seen pool exhaustion in prod because:

- Single worker (§4.1) caps concurrent request handlers at the asyncio scheduler's natural limit.
- Background tasks use SHORT sessions (`async with` inside the loop body, not around the loop).

Symptoms of pool exhaustion (if you ever see them):

- Slow request handlers despite low DB CPU.
- `TimeoutError: QueuePool limit of size 20 overflow 10 reached` in logs.

Fix path: investigate which handler is holding sessions across `await` to a slow external call. **Don't bump `pool_size` first** — it's almost always a code-shape problem, not a sizing one.

#### 4.5.11 Demo masking — at the data layer boundary

Read paths for demo sessions mask account values via `mask_column` (§22). The masking happens **at the route layer**, not at the data layer:

```
# Wrong – masking inside the ORM
class AlgoOrder(Base):
    account: Mapped[str] = mapped_column(String(16))
    @property
    def account_display(self):
        return mask_column(self.account)

# Right – masking at the route handler
@get("/orders")
async def list_orders(self, request: Request) -> list[OrderInfo]:
    rows = await self._fetch()
    is_demo = request.state.is_demo
    return [
        OrderInfo(account=mask_column(r.account) if is_demo else r.account, ...)
        for r in rows
    ]
```

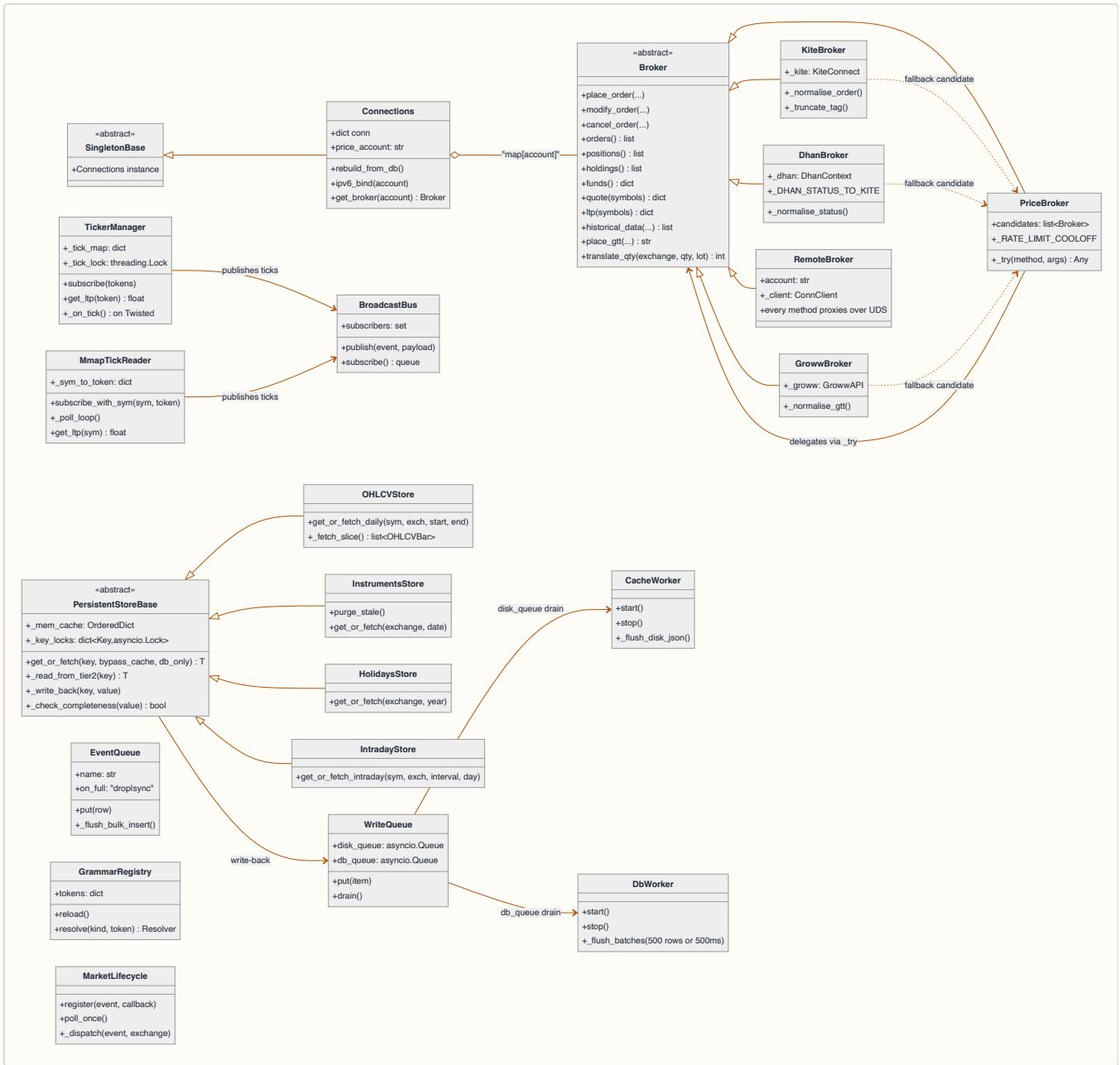
This keeps the data layer free of presentation concerns and avoids subtle bugs (e.g. ORM expressions seeing masked values during a JOIN).

#### 4.5.12 Backend runtime object graph

The class diagram below is the high-level shape of the backend runtime. Every box is a real class you can grep. Method sets are the operationally-important surface (not every method on every class — those live in the source).

Two clusters:

- **Broker layer** ( `backend/brokers/` ) — `Connections` singleton owns all broker sessions; adapters implement the Broker ABC; local vs remote proxy split lets the same route code run against in-process brokers OR a UDS `conn_service` (see §14.5).
- **Persistence layer** ( `backend/api/persistence/` ) — `PersistentStoreBase` is the three-tier read/write scaffold; four concrete stores share it. Two write-back workers ( `cache_worker` + `db_worker` ) drain the in-process write queues.



**Grep map** — every class name above resolves to a single file:

| Class                                                         | Module                                                                 |
|---------------------------------------------------------------|------------------------------------------------------------------------|
| Connections (+ _IPv6SourceAdapter)                            | backend/brokers/connections.py                                         |
| Broker ABC                                                    | backend/brokers/base.py                                                |
| KiteBroker / DhanBroker / GrowwBroker                         | backend/brokers/adapters/{kite,dhan,groww}.py                          |
| PriceBroker + get_broker + get_market_data_broker             | backend/brokers/registry.py                                            |
| RemoteBroker + ConnClient                                     | backend/brokers/client/remote_broker.py + client.py                    |
| TickerManager + BroadcastBus                                  | backend/brokers/kite_ticker.py                                         |
| MmapTickReader                                                | backend/brokers/mmap_ticker.py                                         |
| PersistentStoreBase                                           | backend/api/persistence/store_base.py                                  |
| OHLCVStore / InstrumentsStore / HolidaysStore / IntradayStore | backend/api/persistence/{ohlcv,instruments,holidays,intraday}_store.py |
| WriteQueue / CacheWorker / DbWorker                           | backend/api/persistence/{write_queue,cache_worker,db_worker}.py        |
| EventQueue                                                    | backend/api/persistence/event_queue.py                                 |
| GrammarRegistry                                               | backend/api/algo/grammar_registry.py                                   |
| MarketLifecycle                                               | backend/api/algo/market_lifecycle.py                                   |

## 4.6 Database schema overview

RamboQuant's data model spans 35+ SQLAlchemy tables, split into logical domains. Two PostgreSQL databases: `ramboq` (prod, main branch) and `ramboq_dev` (dev branches). All tables live in the branch-local DB except `broker_accounts`, which is shared.

### Table categories

**Auth + User Management** — User identity, email verification, roles, compliance:

| Table                             | Purpose                                                                                                     | Key columns                                                                                                                                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>users</code>                | Operator profiles + LP investor records. Unified table for internal staff + external LPs.                   | id (PK), account_id (unique), username, role (designated/trader/risk/admin/partner), email, pan, kyc_verified, contribution_date, share_pct, assigned_accounts, assigned_strategies, token_version (for force-logout) |
| <code>auth_tokens</code>          | One-time email verification + password-reset tokens. Single table with <code>purpose</code> discriminator.  | id (PK), user_id (FK), purpose (verify reset), token (unique), expires_at, used_at                                                                                                                                    |
| <code>impersonation_events</code> | Audit of admin/designated impersonating a partner. Tracks session start + end.                              | id (PK), actor_username, target_username, started_at, ended_at, end_reason                                                                                                                                            |
| <code>investor_tokens</code>      | Long-lived URL tokens for LP-facing portal access. Token IS the credential.                                 | id (PK), user_id (FK), token (unique), expires_at, revoked_at, last_visit_at, visit_count                                                                                                                             |
| <code>investor_events</code>      | LP capital ledger — subscriptions, redemptions, bootstrap events. Source of truth for units-based NAV math. | id (PK), user_id (FK), event_type (subscription redemption bootstrap), event_date, amount, nav_per_unit, units_delta                                                                                                  |
| <code>monthly_statements</code>   | Audit of auto-emailed LP statements. One row per (user, period_year, period_month).                         | id (PK), user_id (FK), period_year, period_month, generated_at, sent_at, recipients_json, pdf_size_bytes, error                                                                                                       |

**Broker Connection** — Account credentials + market metadata:

| Table                                | Purpose                                                                              | Key columns                                                                                                                                                                                  |
|--------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>broker_accounts</code>         | Shared table (ramboq DB only) storing encrypted broker credentials for all branches. | id (PK), account (unique, e.g. ZG0790), broker_id (kite dhan groww), user_id (FK), api_key_enc, access_token_enc, source_ip, priority, poll_priority, circuit_breaker_enabled, display_order |
| <code>market_holidays</code>         | Exchange holiday calendar (NSE/MCX/CDS). Seeded from broker API, cached.             | id (PK), exchange, holiday_date (unique per exchange)                                                                                                                                        |
| <code>market_special_sessions</code> | Special trading sessions (e.g. Muhurat trading). Operator-editable overrides.        | id (PK), exchange, date, start_time, end_time, reason                                                                                                                                        |

**Watchlists** — User-defined symbol groups:



| Table           | Purpose                                                                                    | Key columns                                                                             |
|-----------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| watchlists      | Named list containers. Global rows (is_global=True, user_id=NULL) shared across all users. | id (PK), user_id (FK nullable), name, is_global, is_default, is_pinned, sort_order      |
| watchlist_items | Individual symbols in a watchlist. Includes operator-supplied alias.                       | id (PK), watchlist_id (FK), tradingsymbol, exchange, alias (optional label), sort_order |

#### Orders + Execution — Core order lifecycle and attribution:

| Table             | Purpose                                                                                                                     | Key columns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| algo_orders       | Master order row. Spans manual + agent-fired + template-attached + chase iterations. Mode: sim/paper/live/replay/shadow.    | id (PK), account, symbol, exchange, transaction_type (BUY/SELL), quantity, filled_quantity (cumulative across partials), initial_price, current_limit (re-quoted during chase), fill_price, status (OPEN/FILLED/REJECTED/CANCELLED), mode, engine (manual/sim/paper/live/replay/shadow/expiry), agent_id (FK nullable), broker_order_id, template_id (FK), attached_gtts_json (JSON list of {kind, label, id, ...}), basket_tag, parent_order_id (for TP/SL children), strategy_id (FK), request_id (for audit drill-through) |
| algo_order_events | Append-only timeline per order. One row per state transition (placed, chase_modify, fill, unfill, reject, cancel).          | id (PK), order_id (FK), ts, kind (placed chase_modify fill unfill reject cancel postback ...), message, payload_json                                                                                                                                                                                                                                                                                                                                                                                                          |
| algo_events       | Legacy event log (pre-AgentEvent era). Deprecated; kept for compatibility.                                                  | id (PK), algo_order_id (FK nullable), event_type, detail, timestamp                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| strategies        | Named bucket for order attribution. Owns lots + provides per-strategy P&L rollup.                                           | id (PK), slug (unique), name, description, owner_user_id (FK nullable), capacity_cap_inr, target_volatility, is_active                                                                                                                                                                                                                                                                                                                                                                                                        |
| strategy_lots     | Per-strategy FIFO lot ledger. Opens on fill, closes when counter-direction consumes it. Authoritative for per-strategy P&L. | id (PK), strategy_id (FK), open_order_id (FK nullable), account, symbol, exchange, side (B/S), qty, remaining_qty, open_price, close_price, realized_pnl, opened_at, closed_at                                                                                                                                                                                                                                                                                                                                                |

#### Agents + Conditions — Rule engine and alert infrastructure:

| Table          | Purpose                                                                                                               | Key columns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agents         | Declarative rules: condition tree → alert → actions. Includes loss alerts, expiry alerts, user-defined custom agents. | id (PK), slug (unique), name, long_name (3-part descriptor), description, conditions (JSONB tree), events (alert channels), actions (order/cancel/close side-effects), scope (per_account all_accounts), schedule (market_hours continuous ...), cooldown_minutes, fire_at_time (gate to HH:MM window), trade_mode (paper live per-agent override), status (active inactive cooldown completed), lifespan_type (persistent one_shot n_fires until_date), tier (critical high medium low for noise reduction), topic (agent grouping tag), digest_window_sec (buffer outgoing alerts), debounce_minutes, condition_first_true_at, tags (JSONB list), blackout_windows (quiet hours) |
| agent_events   | Alert events fired by agents. Persisted for operator inspection + MCP queries.                                        | id (PK), agent_id (FK), event_type (fired suppressed error ...), trigger_condition (the condition leaf that fired), detail, sim_mode (simulator flag), timestamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| grammar_tokens | Token catalog (metrics, scopes, operators, channels, actions). Extensible alphabet for condition/alert/action trees.  | id (PK), grammar_kind (condition notify action), token_kind, token (name), value_type, resolver (dotted path to function), params_schema, enum_values (JSONB), is_active, is_system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

#### NAV + Performance — Investor slicing and daily metrics:

| Table              | Purpose                                                                                                       | Key columns                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| nav_daily          | Daily firm-level NAV snapshot. Written after broker positions settle. Authoritative for all investor slicing. | id (PK), as_of_date (unique), nav, cash_total, positions_mtm, holdings_mtm, accounts_snapshot (JSONB list), note      |
| strategy_snapshots | Daily per-strategy P&L rollup. Charts the strategy performance curve.                                         | id (PK), strategy_id (FK), as_of_date, open_lots_count, open_notional, realised_pnl, unrealised_pnl, margin_allocated |
| perf_snapshots     | Nightly static + runtime metrics per page. Used for trend analysis + performance budgets.                     | id (PK), page_or_route, metrics_json (JSONB), static_metrics_json, captured_at                                        |

#### Data Snapshots — Intraday market state + persistent cache:

| Table                | Purpose                                                                                                                                                                                                                                                                 | Key columns                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| daily_book           | Intraday snapshot of positions, holdings, or funds per account per symbol. Captured at market close, useful when markets closed. Schema includes <code>previous_close</code> (frozen yesterday's settlement price) to stabilize day P&L math during closed-hours reads. | id (PK), kind (positions holdings funds), account, symbol, exchange, qty, avg_price, ltp, pnl, pnl_pct, date, captured_at, segment (NSE MCX ...), <b>previous_close</b> |
| ohlcv_daily          | 5-year OHLCV history. Persistence tier 2 fallback for chart data.                                                                                                                                                                                                       | symbol, exchange (PK), date (PK), open, high, low, close, volume                                                                                                        |
| instruments_snapshot | Per-exchange symbol→token map. Refreshed daily.                                                                                                                                                                                                                         | id (PK), exchange, date, payload (JSONB full instruments dict), row_count                                                                                               |
| holidays_snapshot    | Exchange holiday sets per year. Immutable once year closes.                                                                                                                                                                                                             | id (PK), exchange, year, dates_json (JSONB list)                                                                                                                        |
| intraday_bars        | 5/15/30/60-minute bars. 90-day rolling retention.                                                                                                                                                                                                                       | id (PK), symbol, exchange, date, interval (5min 15min 30min 60min), bar_ts, open, high, low, close, volume                                                              |
| movers_snapshots     | Nightly snapshot of top movers (NIFTY, NIFTYNXT50, etc). Pre-computed so /pulse doesn't timeout.                                                                                                                                                                        | id (PK), index_symbol, snapshot_date, movers_json (JSONB list)                                                                                                          |

**Audit + Compliance** — Forensic trails:

| Table              | Purpose                                                                                      | Key columns                                                                                                                                                                                                                                         |
|--------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| audit_log          | HTTP request + mutation audit trail. Every write captured by middleware + explicit handlers. | id (PK), actor_user_id, username, role, action, category (order.place order.fill agent.action system.nav ...), method, path, target_type, target_id, status_code, summary, request_id (FK to al go_orders for drill-through), client_ip, created_at |
| mcp_audit          | Mutations initiated via MCP (Claude Code research mode). Tracks tool calls + results.        | id (PK), user_id (FK), tool_name, args_redacted, result_status, result_summary, request_id, created_at                                                                                                                                              |
| admin_email_events | Audit of admin-triggered alert sends (e.g. manual test notifications).                       | id (PK), admin_user_id, recipient_email, subject, body_preview, sent_at, error                                                                                                                                                                      |
| visitor_log        | Minimal analytics — timestamps of /auth/login + visitor count.                               | id (PK), visitor_ip, last_seen_at, visitor_count                                                                                                                                                                                                    |

**Market Lifecycle + Background Jobs** — System state:

| Table                   | Purpose                                                                                   | Key columns                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| market_lifecycle_events | Audit log of market open/close transitions per exchange. Indexed for operator drill-down. | id (PK), exchange, event_type (nse:open nse:close nse:close_settled ...), fired_at, captured_at |
| code_metrics_snapshots  | Captured per release (commit SHA). Query count, test counts, response times.              | id (PK), release_version, static_metrics_json, runtime_metrics_json, captured_at                |

**Configuration + Extensibility:**

| Table            | Purpose                                                                                                                    | Key columns                                                                                                                                                                                     |
|------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| settings         | DB-backed tunables: alert cooldown, retry counts, market hours, etc. Operator-editable via <code>/admin/settings</code> .  | id (PK), category, key (unique), value_type (int float bool string), value (operator-editable), default_value, description, schema (JSON validation), units                                     |
| order_templates  | Reusable bracket recipes. System templates (default-bull, default-bear, etc) seeded at boot; operator saves custom copies. | id (PK), user_id (FK nullable), name, slug, owner_user_id (FK nullable), tp_pct, sl_pct, wing_premium_pct, wing_strike_offset, wing_qty, tp_scales_json, tp_order_type, sl_trail_pct, is_system |
| hedge_proxies    | Cross-reference between holdings (GOLDBEES) and option roots they hedge (GOLD). Includes $\beta$ regression.               | id (PK), proxy_symbol, target_root, is_active, note, beta, correlation, regression_at                                                                                                           |
| research_threads | Persistent Chat threads in <code>/admin/research</code> (MCP Lab). One row per thread; messages stored as JSON.            | id (PK), created_by_user_id (FK), title, slug, messages_json (JSONB), summary, created_at, updated_at                                                                                           |
| sim_recordings   | Deterministic event logs for replay. Captures every state mutation so operator can re-run identical scenarios.             | id (PK), label, scenario, seed_mode, started_at, ended_at, duration_sec, tick_count, event_count, payload (JSONB event stream), owner_user_id (FK)                                              |
| sim_iterations   | Multi-run coordinator. First iteration references itself; others reference iteration 1 via <code>parent_run_id</code> .    | id (PK), slug (unique), parent_run_id (FK self-ref nullable), iteration_index, iterations_total, regime, seed, started_at, ended_at, end_reason, params_json, summary_json                      |

**News + Market Reports:**

|  |
|--|
|  |
|--|

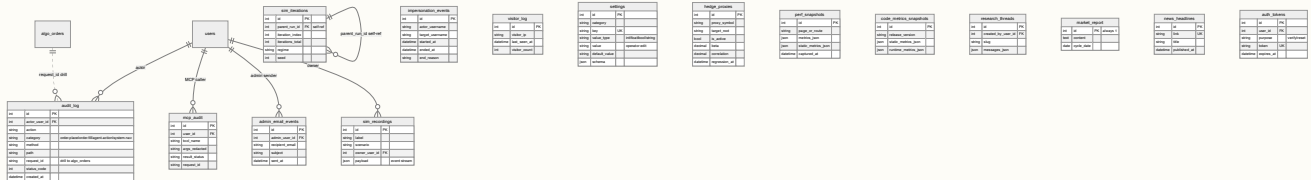
| Table          | Purpose                                                           | Key columns                                                            |
|----------------|-------------------------------------------------------------------|------------------------------------------------------------------------|
| market_report  | Single-row cache (id=1) for daily market summary from Gemini API. | id (PK, always 1), content, cycle_date, refreshed_at, generated_at     |
| news_headlines | RSS headlines cache. Pre-filtered on /market page.                | id (PK), link (unique), title, summary, published_at, source, category |

### 4.7.3 Panel C — Persistence · market data cache

[illegible]

Panel C has NO foreign keys (persistence stores are self-contained market-data cache). `daily_book` is the closed-hours snapshot anchor for every operator-visible surface (positions, holdings, NAV, sparklines); the latest-batch CTE in `snapshot_gate.latest_snapshot_ltp_map` reads from here. `market_lifecycle_events` audits every open / close / close\_settled transition dispatched by the singleton `MarketLifecycle` bus.

#### 4.7.4 Panel D — Audit · configuration · extensibility



### Cross-panel foreign keys:

- `algo_orders.agent_id` (Panel A) → `agents.id` (Panel B) — ON DELETE SET NULL
- `algo_orders.request_id` (Panel A) drill-through from `audit_log.request_id` (Panel D)
- `agent_events.agent_id` (Panel B) → `agents.id` (Panel B) — ON DELETE CASCADE

### Key foreign-key patterns:

- `algo_orders.agent_id` → `agents.id`: `ON DELETE SET NULL` (preserve order history if agent deleted)
- `agent_events.agent_id` → `agents.id`: `ON DELETE CASCADE` (retire agent history when agent deleted)
- `algo_orders.parent_order_id` → `algo_orders.id`: `ON DELETE SET NULL` (self-ref for TP/SL children)
- `algo_order_events.order_id` → `algo_orders.id`: `ON DELETE CASCADE` (timeline only meaningful with parent)
- `investor_events.user_id` → `users.id`: `ON DELETE RESTRICT` (capital ledger is audit trail; explicit operator action required to delete LP)
- `User.id`: multiple targets use `ON DELETE CASCADE` (`auth_tokens`, `investor_tokens`, `monthly_statements`) or `SET NULL` (impersonation, strategy owner)

## 4.8 Retention policies

Data retention is staggered — critical audit trails kept longer than ephemeral cache. Configured via `settings` table; operator can adjust retention via `/admin/settings` → Retention.

| Table                | Retention           | Config key                       | Rationale                                                                          |
|----------------------|---------------------|----------------------------------|------------------------------------------------------------------------------------|
| ohlcv_daily          | 5 years             | (hardcoded)                      | SEBI Cat-III backtest horizon                                                      |
| instruments_snapshot | 7 days              | (hardcoded)                      | Symbol token changes rarely; old snapshots are cheap to drop                       |
| holidays_snapshot    | Forever             | (hardcoded)                      | Reference data — rarely used but occasionally queried for audits                   |
| intraday_bars        | 90 days             | (hardcoded)                      | Intraday charts old after 3 months; 90-day window covers quarterly strategy review |
| algo_events          | 30 days             | retention.algo_events_days       | Operational log; older entries rarely queried                                      |
| algo_order_events    | 90 days             | retention.algo_order_events_days | Order timeline — longer window for compliance disputes                             |
| auth_tokens          | 7 days after expiry | retention.auth_tokens_days       | One-time verify/reset tokens; ephemeral by design                                  |
| mcp_audit            | 90 days             | mcp.audit_retention_days         | MCP tool calls (research mode) — compliance audit trail                            |
| audit_log            | 365 days            | retention.audit_log_days         | User action audit trail — 1-year window for disputes                               |
| nav_daily            | Forever             | (hardcoded)                      | Investor reporting — Cat-III requires 8-year retention                             |
| daily_book           | Forever             | (hardcoded)                      | Intraday snapshots become the P&L source after market close                        |
| investor_events      | Forever             | (hardcoded)                      | Capital ledger — units-based NAV depends on full history                           |
| monthly_statements   | Forever             | (hardcoded)                      | Investor statements are permanent records                                          |
| strategy_snapshots   | Forever             | (hardcoded)                      | Strategy performance is historical record                                          |
| All others           | Forever             | —                                | Configuration, metadata, non-critical operational logs kept for full history       |

**Cleanup mechanism:** nightly cron ( 03:10 IST and staggered) runs `DELETE FROM <table> WHERE created_at < now() - interval .` Per-table cleanup is idempotent — multiple runs don't corrupt state.

**Operator override:** Set retention to 0 in `/admin/settings` to disable deletion for that table (useful during active investigation).

## 4.9 Metrics + Performance tracking

RamboQuant tracks two orthogonal signals: **static code health** (per-file metrics) and **runtime Web Vitals** (per-page response times, JavaScript heap pressure). Both are persisted nightly to `perf_snapshots` and visualized in `/admin/perf`.

🔗 **TECH — Why centralized metrics** — **WHY** Operator feedback (Jun 2026): "dropdown lag", "refresh button stuck", "frontend heap keeps growing". A single dashboard surfacing cyclomatic complexity + LCP + JS heap reveals systemic issues (subscription leaks, long tasks, state bloat) before they degrade UX. **WHAT** Nightly snapshot of static + runtime metrics, persisted with ISO8601 timestamp + `page_or_route` tag. **HOW** `scripts/perf_baseline.py` computes static via `radon cc + line-count`; `scripts/perf_capture_run.sh` runs Playwright against dev to harvest LCP/TBT/heap via `performance.timing` + Chrome DevTools Protocol. Cron orchestrator ( `_task_perf_snapshot` at 04:00 IST) executes both; inserts row via `POST /api/admin/perf/snapshot` (internal, no auth required from cron). **WHERE** `backend/api/models.py::PerfSnapshot` , `backend/api/routes/admin.py` , `scripts/perf_*.py` .

### Static metrics (per file)

| Metric          | Ideal                           | How                                                                              | Notes                                                         |
|-----------------|---------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------|
| LOC             | Components < 1500, Pages < 3000 | <code>wc -l</code> (excludes blank + comment-only)                               | Baseline for cognitive load                                   |
| cc_max          | < 20                            | Cyclomatic complexity of most complex function in file via <code>radon cc</code> | Red flag at > 50: hard to maintain, high defect risk          |
| cc_avg          | < 8                             | Mean complexity across all functions                                             | Safe zone; > 15 indicates refactor candidate                  |
| \$effect count  | < 20 per Svelte file            | Regex boundary match <code>\$effect(</code> declarations                         | Too many effects = hard to trace reactivity                   |
| \$state count   | < 50 per Svelte file            | Regex boundary match <code>\$state(</code> declarations                          | Large state trees risk stale-cache bugs                       |
| \$derived count | Monitor trend                   | Regex boundary match <code>\$derived</code> declarations                         | No hard limit; watch for explosion as sign of over-reactivity |

**Cyclomatic complexity thresholds** (colour coding):

- Green: `cc < 10` — safe, easy to read
- Yellow: `cc 10–20` — moderately complex, watch carefully
- Red: `cc > 20` — hard to maintain, high defect risk
- Critical: `cc > 50` — refactor mandatory before merge

## Runtime metrics (Web Vitals)

| Metric           | Ideal        | How measured                                                      | Notes                                                        |
|------------------|--------------|-------------------------------------------------------------------|--------------------------------------------------------------|
| <b>LCP</b>       | < 2500 ms    | Largest Contentful Paint — largest above-fold element render time | Chrome DevTools via Playwright                               |
| <b>TBT</b>       | < 200 ms     | Total Blocking Time — sum of long-task durations during page load | Chrome DevTools via Playwright                               |
| <b>JS Heap</b>   | < 50 MB      | V8 heap size after page settles (Chrome only)                     | Leak detector; baseline should not grow session-over-session |
| <b>Route p50</b> | < 200 ms     | Median backend request latency (per endpoint)                     | Sampled from access logs                                     |
| <b>Route p95</b> | < 500 ms     | 95th-percentile tail latency                                      | Catches outlier slow requests                                |
| <b>QPS</b>       | < 20 typical | Requests per second per route                                     | Expected load during market hours                            |

## Data flow

**Static compute** — `scripts/perf_baseline.py --with-cyclomatic --no-build`:

- Walks `backend/` + `frontend/src` recursively
- For each `.py` file: LOC count + `radon cc` function list
- For each `.svelte` file: LOC count + regex boundary match for `$effect(, $state(, $derived` declarations (regex patterns in the script handle `$derived.by(...)` and `$props({...})`)
- Outputs JSON with per-file metrics

**Runtime capture** — `scripts/perf_capture_run.sh`:

- Spins up Playwright against `dev.ramboq.com`
- For each route in a curated list (positions, holdings, orders, dashboard, etc):
  - Navigate → wait for settle (3s idle) → capture `window.performance.timing.navigationStart`, `performance.memory.usedJSHeapSize` (Chrome only)
  - Extract LCP via `PerformanceObserver` + `'largest-contentful-paint'` entripoint
  - Sum `performance.getEntriesByType('longtask')` duration for TBT
- Outputs JSON per route

**Nightly persist** — `_task_perf_snapshot` (04:00 IST):

1. Calls `perf_baseline.py --with-cyclomatic --no-build`
2. Calls `perf_capture_run.sh`
3. Merges static + runtime JSONs
4. POST `/api/admin/perf/snapshot` with `{page_or_route, static_metrics, runtime_metrics}`
5. Inserts `PerfSnapshot(captured_at=now(), page_or_route=..., metrics_json=..., static_metrics_json=...)` row

## DB schema

```
class PerfSnapshot(Base):
    __tablename__ = "perf_snapshots"

    id                : int (PK)
    page_or_route      : str (e.g. "GET /api/positions", "/orders", "/dashboard")
    metrics_json       : dict (JSONB — runtime metrics: lcp_ms, tbt_ms, heap_mb, qps, ...)
    static_metrics_json : dict (JSONB — static metrics: loc, cc_max, cc_avg, ...)
    captured_at        : datetime (UTC, unique per (page_or_route, day))

    __table_args__ = (
        Index("ix_perf_snapshots_route_ts", page_or_route, captured_at.desc()),
    )
```

**Retention:** 365 days (config: `retention.perf_snapshots_days`, default 365).

## Admin endpoints

- GET `/api/admin/perf/latest` — latest snapshot per page/route (no history).
- GET `/api/admin/perf/history?page=<route>&days=30` — time series for a single `page_or_route` over N days. Returns `[{captured_at, metrics_json, static_metrics_json}]`.

- GET `/api/admin/perf/regressions?days=7&threshold_pct=10` — detect pages where any metric exceeded 7-day median by > 10%. Returns list with regression alert.
- POST `/api/admin/perf/snapshot` — internal cron endpoint. No auth; HMAC signature ( `X-PERF-SIGNATURE` header, signed with `api_secret` ) required.

All endpoints are `admin`-guarded except `/snapshot`.

## Frontend surface

`/admin/perf dashboard` — card grid layout:

- **Code Health Card** — time-series chart of `cc_max` + LOC trend (30-day window)
- **Runtime Card** — time-series chart of LCP + TBT + heap (30-day window)
- **Regression Alert** — sticky callout at top if regressions detected in last 7 days
- **Metrics Glossary** — info tooltips (hover on metric label → show WHAT/IDEAL/IMPACT/FIX)

**Metric metadata SSOT** — `frontend/src/lib/data/metricMetadata.js` exports `METRIC_META`:

```
export const METRIC_META = {
  cc_max: {
    WHAT: "Highest cyclomatic complexity of any function in the file",
    IDEAL: "< 20 for maintainability",
    IMPACT: "High cc means more defects, longer review cycles",
    FIX: "Extract complex functions into helpers, simplify boolean logic"
  },
  lcp_ms: {
    WHAT: "Largest Contentful Paint — largest above-fold element render time",
    IDEAL: "< 2500 ms",
    IMPACT: "Poor LCP = operator sees blank page for seconds",
    FIX: "Lazy-load off-viewport components, defer non-critical JS"
  },
  // ... per metric
};
```

**InfoHint component** (reused everywhere) displays this tooltip on hover.

## #major tag workflow

When operator adds `#major` prefix to any ask, the concurrent refactor flow is triggered:

1. Audit agent fetches top-3 cyclomatic/LOC hotspots from last 30 days of `perf_snapshots`
2. Identify regressions flagged in last 7 days
3. Parallel refactor tasks per hotspot
4. Recompute metrics after changes + assert no regression vs baseline

Example:

```
#major add a new route

→ audit sees cc_max creeping from 18→22 in chase.py over 2w
→ refactor: extract _chase_place_order, _chase_poll, _chase_status into focused
  functions (break one 60-line function into 3 functions × ~20 lines each)
→ recompute: cc_max now 14, tests pass
→ merge
```

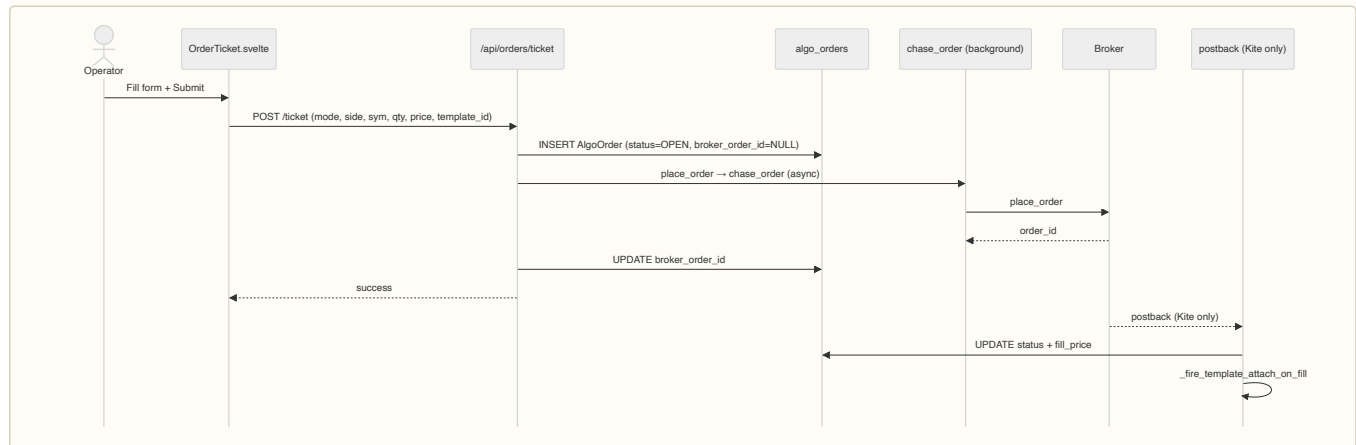
## Key files

- `backend/api/models.py::PerfSnapshot` — schema
- `backend/api/routes/admin.py::perf_snapshot` — cron endpoint
- `scripts/perf_baseline.py` — static metrics compute
- `scripts/perf_capture_run.sh` — runtime capture via Playwright
- `backend/api/background.py::_task_perf_snapshot` — orchestrator (04:00 IST)
- `frontend/src/lib/data/metricMetadata.js` — `METRIC_META` glossary
- `frontend/src/routes/(algo)/admin/perf/+page.svelte` — dashboard



## Part II — Order lifecycle

### 5. Order placement — single ticket (Ticket tab)



**Files:** `frontend/src/lib/order/OrderTicket.svelte` (submit) → `backend/api/routes/orders.py::ticket_order` → `backend/api/algo/chase.py::chase_order` → postback (Kite) or polling.

**Race-window guard:** `AlgoOrder` commits with `broker_order_id=NULL` first. Fast IOC fill in this window is caught by postback-fallback matching (account, symbol, side, qty, status=OPEN) within 60s.

#### 5.1 F&O order quantity convention (lots-first API)

**P0 commitment:** all F&O order APIs ( `/ticket` , `/basket` , `/preview-ticket-template` ) now accept **LOTS** as input for instruments with `lot_size > 1` (futures + options). The API converts lots → contracts at the request boundary using `contracts = lots × lot_size`.

**Request shape:**

- **F&O** ( `lot_size > 1` ): send `quantity: <lots>` (frontend builds `_lots` key)
- **Equity** ( `lot_size = 1` ): send `quantity: <raw_qty>` (no change)

**Guard chain** (G1, G2 — critical):

| Guard                    | Check                                                                   | Level                                                                        | Applies to                     |
|--------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------|
| G1 LOT_MULTIPLE          | <code>qty % lot_size == 0</code>                                        | Removed from <code>_ticket_enforce_lot_and_fat_finger</code>                 | Dead (correct by construction) |
|                          |                                                                         | Kept in <code>_arm_take_profit</code> (live path, inline before broker call) | Live F&O close orders          |
|                          |                                                                         | Kept in <code>apply_plan_live</code> GTT layer (top of function, sync check) | Template attach paths          |
| G2 FAT_FINGER_5_LOT_CAP  | <code>qty ≤ 5 lots</code> (NFO/CDS/BFO) or <code>≤ 20 lots</code> (MCX) | Main path, checked at <code>_ticket_validate_input</code>                    | All F&O places                 |
|                          | <code>CLOSE</code> intent bypass                                        | G2 skipped if <code>intent='close'</code> (operator kill)                    | Closes only                    |
| Adapter ceiling (50-lot) | Hard block at <code>kite.py:place_order</code>                          | Applies to all quantities                                                    | Last-line defense (no bypass)  |

**Files:**

- `backend/api/routes/orders_place.py::_ticket_validate_input` — lots → contracts conversion
- `backend/api/routes/orders_place.py::_ticket_enforce_lot_and_fat_finger` — G2 check (G1 removed)
- `backend/api/routes/orders_basket.py` — same conversion + guards for batch legs
- `backend/api/algo/template_attach.py::apply_plan_live` — GTT layer G1 + G2 sync checks before `broker.translate_qty`
- `backend/brokers/adapters/kite.py::translate_qty` , `place_order` — adapter-level qty handling
- `frontend/src/lib/order/orderTicketSubmit.js::buildPlacePayload` — sends `_lots` for F&O

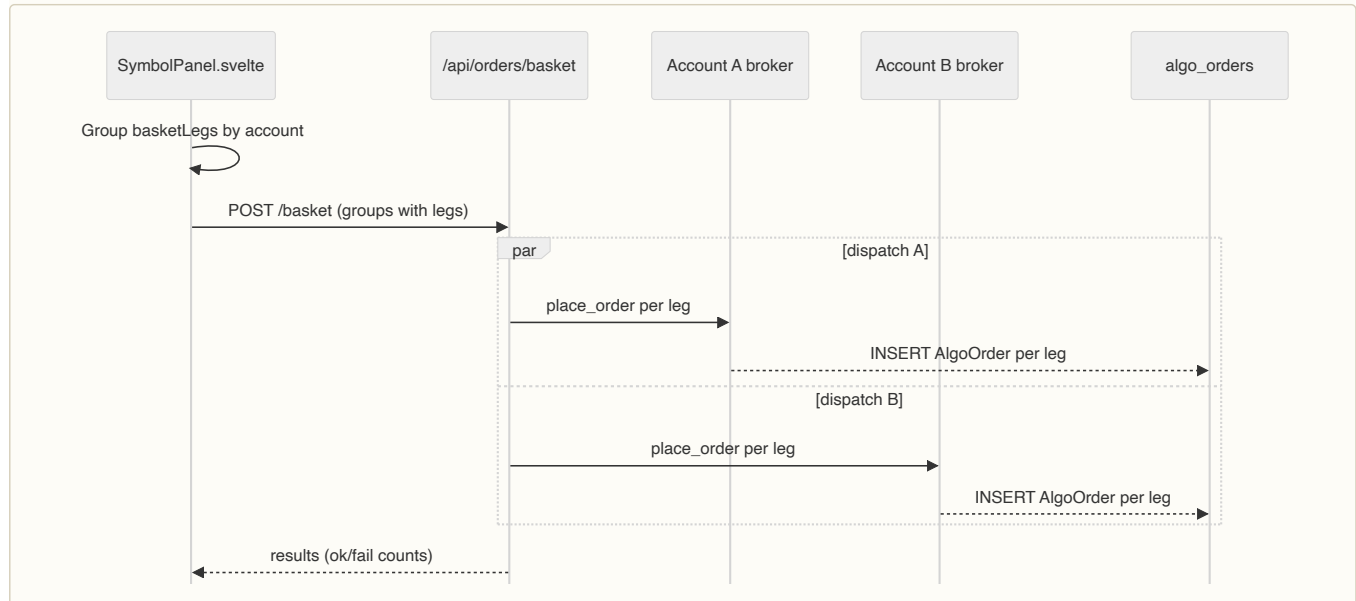
**Cold-cache 503 guard** — extended from MCX-only to ALL F&O exchanges (NFO, CDS, BFO, BCD). When instruments cache is empty at startup or post-refresh, `/api/orders/ticket` returns 503 with a "cold start" message if it can't resolve the `lot_size`. The operator retries after 5-10s once warm.

**Schema update** — `BasketLeg msgspec.Struct` adds optional `strategy_id: Optional[int] = None` field for future per-leg strategy attribution.



## 6. Order placement — basket (Chain tab)

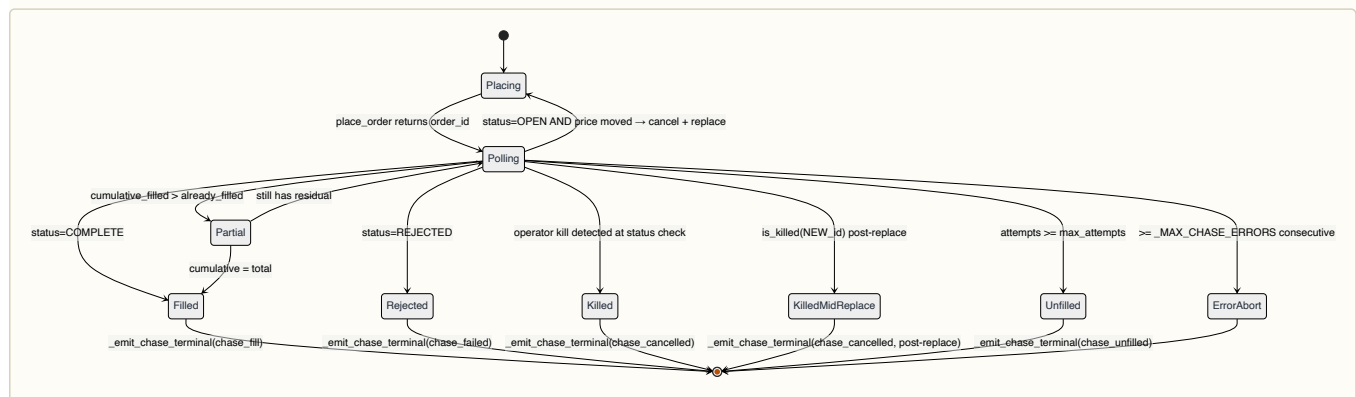
Multi-leg submission grouped per-account, dispatched in parallel via `asyncio.gather` :



**Files:** `frontend/src/lib/SymbolPanel.svelte::submitBasket` (grouping) → `backend/api/routes/orders.py::place_basket` (parallel dispatch).

**Template isolation:** legs with explicit `template_id` ignore shell overrides. Legs with no `template_id` inherit shell defaults.

## 7. Chase loop lifecycle



**Key files:**

- `backend/api/algo/chase.py::chase_order` — main loop
- `backend/api/algo/chase.py` — partial-fill branch (search `_record_partial_fill`)
- `backend/api/algo/chase.py` — kill-race post-replace check (search `is_killed(current_order_id)` after `_sync_algo_order_id`)
- `backend/api/algo/chase.py::_emit_chase_terminal` — snapshot + downstream attach
- `backend/api/algo/chase.py::_sync_algo_order_id` — writes `broker_order_id` + `current_limit`

⚡ **TECH — sync polling vs WebSocket order updates** — WHY Postback delivery is unreliable for non-Kite brokers (Dhan + Groww are poll-only). Sync polling is the lowest-common-denominator that works everywhere. WHAT `chase_order` calls `_order_status` every 20s (configurable per chase). HOW Each iteration: depth quote → adjusted limit → cancel old + place new → sync ID → wait → poll status. WHERE `backend/api/algo/chase.py`.

**Partial-fill math (post C-1 fix):**

```

already_filled = quantity - remaining_qty
new_delta = cumulative_filled - already_filled
fire partial branch when: cumulative_filled > 0 AND new_delta > 0 AND cumulative_filled < quantity
  
```

## 8. The order/chase/template tripod

This is the most complex part of the codebase. Three subsystems with overlapping responsibilities:

### 8.1 What each one owns

|                                                               | Owns                                                                                            | Reads                                               |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Order routing</b> ( <code>orders.py</code> )               | The broker-facing entry path. Single ticket + basket.                                           | Settings, templates                                 |
| <b>Chase loop</b> ( <code>chase.py</code> )                   | The per-order placement lifecycle. Cancel + replace + status polling + partial fill accounting. | Broker, AlgoOrder, kill signal                      |
| <b>Template attach</b><br>( <code>template_attach.py</code> ) | Post-fill exit-rule wiring. TP/SL/Wing/Scale/Trail GTTs at the broker.                          | OrderTemplate, AlgoOrder (read-only at attach time) |

### 8.2 Why three? Why not one big "manage this order" function?

History: the chase loop existed first (single ticket → place + chase to fill). Templates were bolted on later (Phase 0–3 + Sprints A–E). The current shape is intentional — each subsystem can be tested in isolation:

- Chase tests use a mock `_order_status` that returns a scripted sequence.
- Template tests build a `TemplatePlan` directly and assert the GTT spec shape.
- Routes are integration-tested with real broker mocks ( `backend/tests/` ).

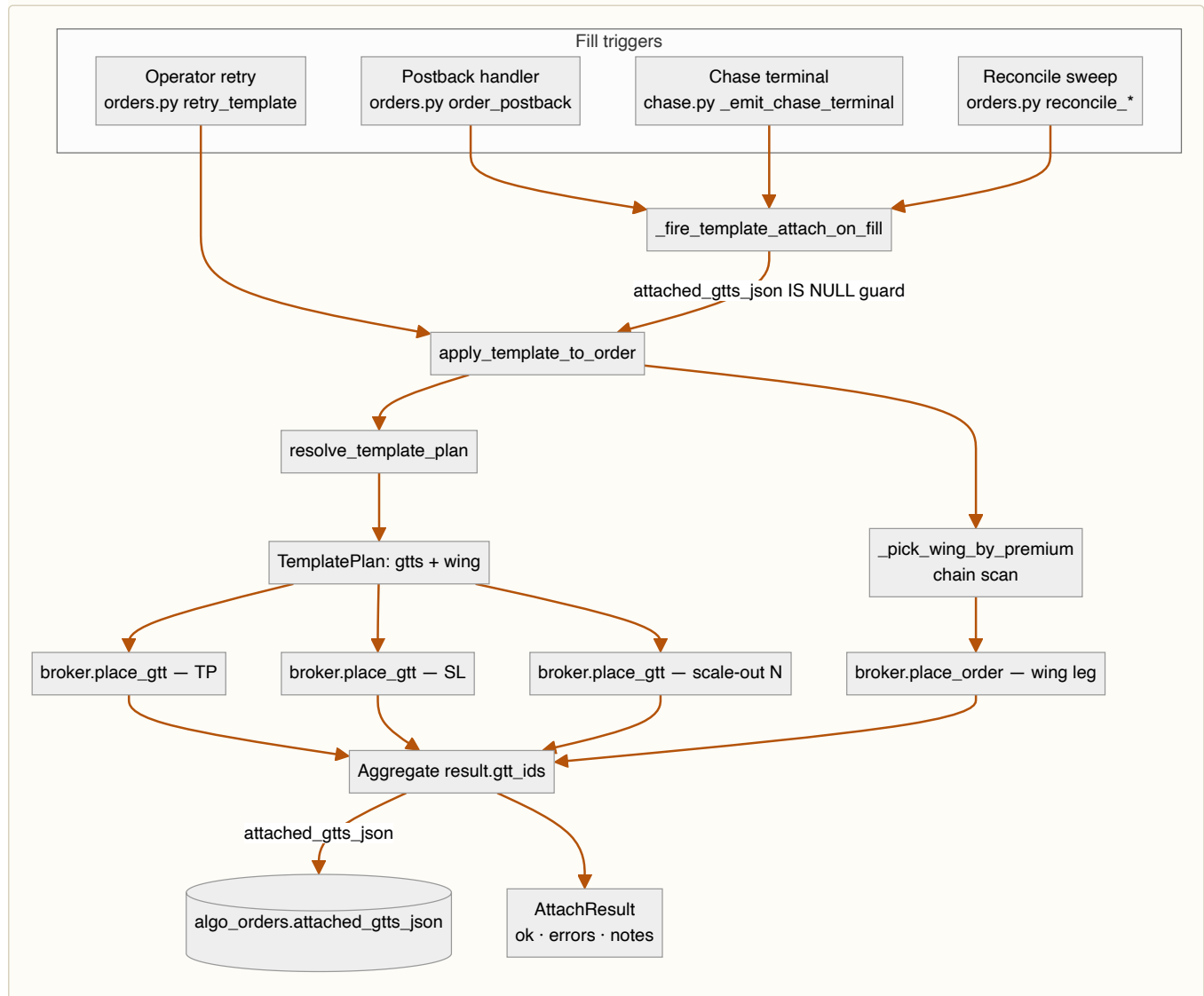
### 8.3 The mode pivot

`mode ∈ {sim, paper, live, shadow}` decides which adapter the order actually hits. The pivot happens at submit time ( `_resolve_mode` in `backend/api/algo/actions.py` ) and is **persisted on the AlgoOrder row** — every downstream branch (chase terminal, postback, reconcile, template attach) reads `row.mode` to decide whether to call a real broker or the paper engine.

**Gotcha:** the chase loop runs the same code regardless of mode. Paper mode is achieved by injecting the paper engine's `place_order` adapter at the broker registry boundary. Don't add `if mode == 'live'` branches inside chase — the abstraction is the broker registry, not the chase.

## Part III — Templates + exits

### 9. Template attach pipeline



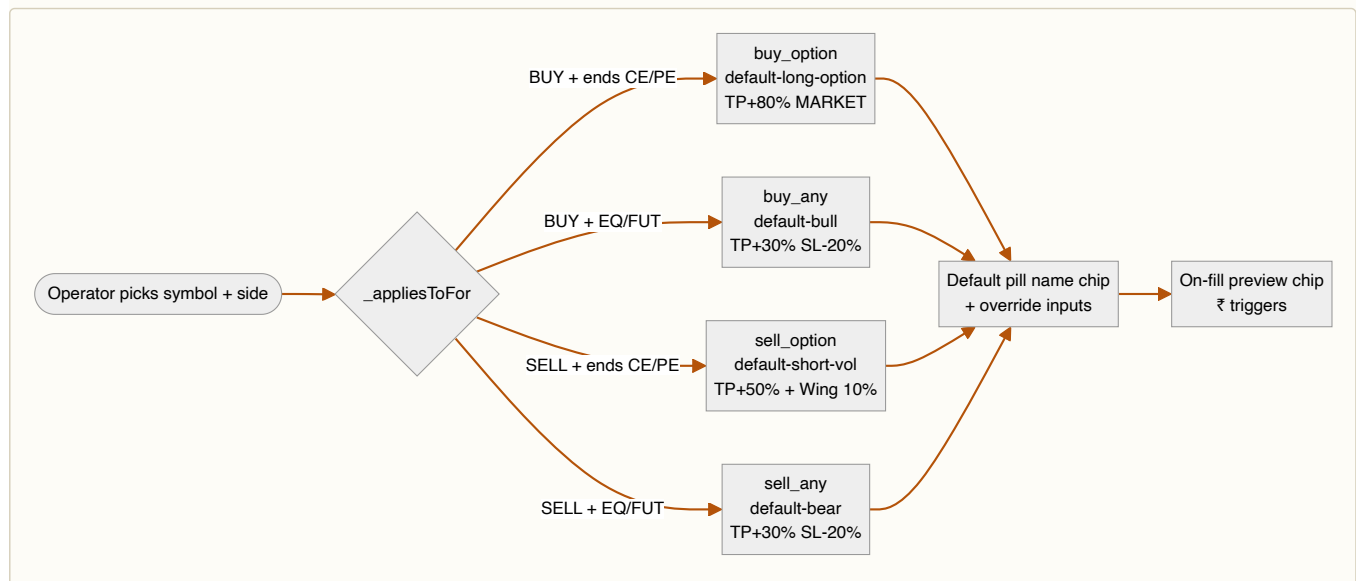
#### Key files:

- `backend/api/algo/template_attach.py::resolve_template_plan` — override merge + scope resolution
- `backend/api/algo/template_attach.py::_pick_wing_by_premium` — OI + spread filters
- `backend/api/algo/template_attach.py::AttachResult` — return type (NOT `TemplateAttachResult` — the docs previously had this wrong)
- `backend/api/routes/orders.py::_fire_template_attach_on_fill` — idempotency guard + persistence
- `backend/api/routes/orders.py::retry_template` — manual re-fire path. Persists `attached_gtts_json` per H-7 + trail-stop scaffolding per Sc.5

**Idempotency:** `_get_template_attach_lock(parent_row_id) + attached_gtts_json IS NULL` check. Strong dict with 1h TTL after M-5 fix replaces the prior `WeakValueDictionary`.

⦿ **TECH — JSON blob vs normalized table for `attached_gtts_json`** — WHY Each parent has 1-5 GTTs + maybe a wing. A child table would mean a JOIN on every order grid render. The blob lets us read the whole bracket in one column-fetch. WHAT Stored as a JSON array of entries ( `{kind: "gtt", label: "TP", id: "...", current_trigger: ..., sl_trail_pct: ...}` ). HOW Always write atomically (single column update); read+parse on every access (cheap because rows are small). WHERE `backend/api/models.py::AlgoOrder.attached_gtts_json + _fire_template_attach_on_fill`.

## 10. 4-default template matrix



### Key files:

- `backend/api/algo/templates_seed.py::SYSTEM_TEMPLATES` — seeded defaults + rebalance logic
- `frontend/src/lib/order/OrderTicket.svelte::_appliesToFor`
- `frontend/src/lib/SymbolPanel.svelte::_appliesToFor` — same helper at shell level
- `frontend/src/lib/SymbolPanel.svelte::_sideAwareDefault` — with fallback to focused-leg symbol
- `frontend/src/routes/(algo)/automation/templates/+page.svelte` — coverage view (note: `/automation/templates` is the actual route; older docs reference `/admin/templates` which is stale)

## 11. Template override merge

Operator overrides flow through multiple layers; understanding the merge order saves hours of debugging.

```
Request:
  template_id=T1, tp_pct_override=20, sl_pct_override=None

Backend persist (orders.py::_build_overrides_json ~line 541):
  AlgoOrder.template_id = T1
  AlgoOrder.template_overrides_json = '{"tp_pct": 20}' # only NON-None overrides

At fill (template_attach.py::_pick):
  tp_pct = _ov.get("tp_pct") or template.get("tp_pct")
  → 20 (override wins)
  sl_pct = _ov.get("sl_pct") or template.get("sl_pct")
  → template's saved sl_pct (no override)
```

**Per-leg vs shell:** when a basket leg has `template_id` set explicitly (not inherited from `_sharedTemplateId`), the SHELL overrides DO NOT flow through. This is the audit-Sc.12 fix — pre-fix the shell's `tp_pct_override` silently contaminated a leg that the operator had retargeted to a different template.

```
submitBasket logic:
  effective_template = leg.template_id ?? shell_template
  if leg has its own template_id:
    tp_override = leg.tp_pct_override (do NOT fall through to shell)
  else:
    tp_override = leg.tp_pct_override ?? shell.tp_pct_override
```

## 12. Chase loop invariants

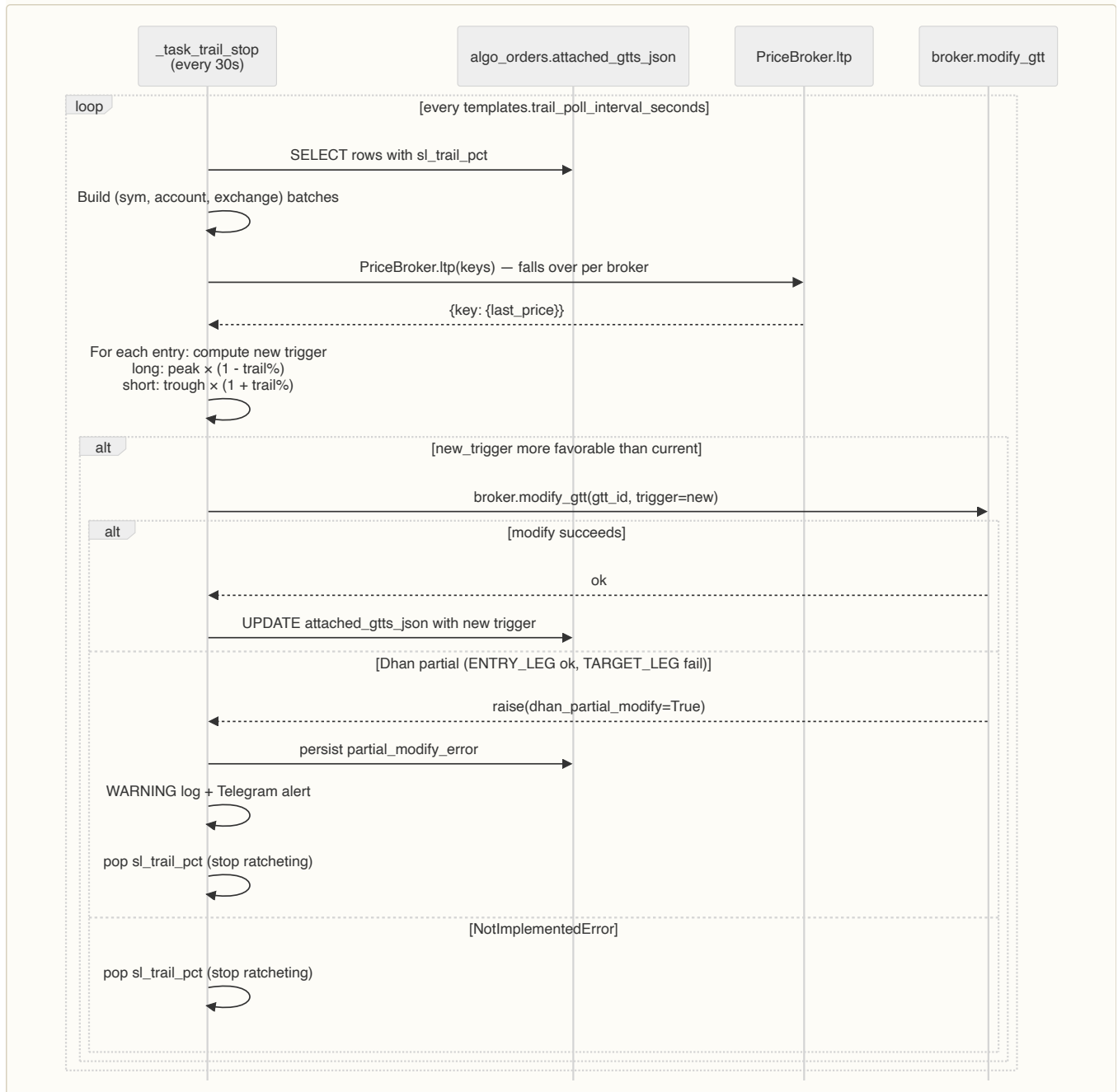
Six things the chase loop MUST guarantee:

1. **AlgoOrder.broker\_order\_id always matches the LATEST broker order.** Cancel-and-replace updates this via `_sync_algo_order_id`. Without it the postback handler can't resolve a row by `broker_order_id`.

2. **AlgoOrder.current\_limit reflects the latest re-quoted limit.** Added in M-6. ChaseCard renders this when present so the operator sees the live limit, not entry.
3. **AlgoOrder.filled\_quantity is monotonic and never exceeds AlgoOrder.quantity.** The `MAX(prior, cumulative)` clamp in `_record_partial_fill` enforces this post C-1 fix. Template attach reads `filled_quantity` to size exit GTTs.
4. **Operator kills take effect on the very next loop iteration.** `mark_killed(broker_order_id)` is synchronous + the loop checks `is_killed(current_order_id)` (a) at status-check time AND (b) immediately after replace (C-2 fix). The dict has a 60-min TTL so a stale kill flag can't survive across days.
5. **Partial fills get persisted on every NEW delta, not just the first.** The branch fires when `cumulative > already_filled` post-C-1 fix.
6. **A chase that hits `>= _MAX_CHASE_ERRORS` consecutive exceptions aborts.** Prevents infinite re-trying against a broker that's down.

Break any of these and template attach sizes wrong, kills get ignored, or zombie chases burn rate limit.

## 13. Trail-stop subsystem



### Key files:

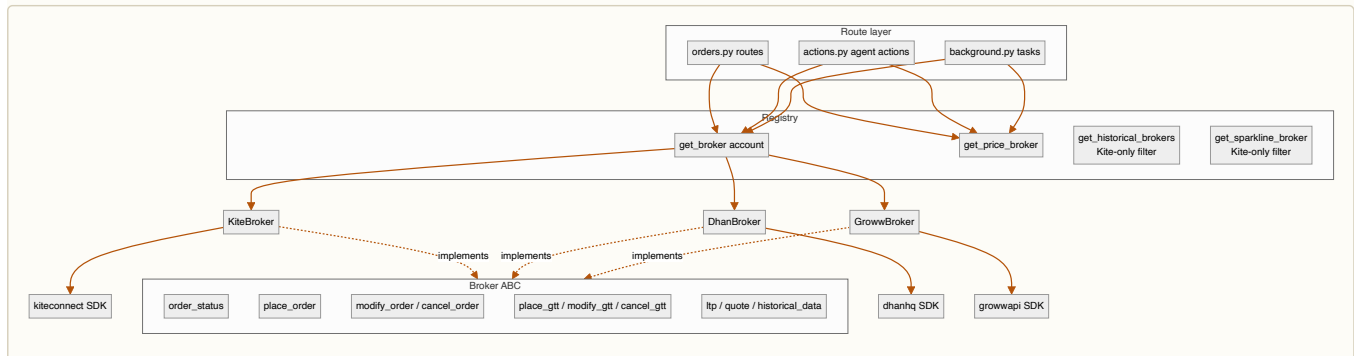
- `backend/api/background.py::_task_trail_stop`

- `backend/api/background.py` — Dhan partial-modify detect + alert (M-2 fix, search `dhan_partial_modify`)
- `backend/brokers/adapters/dhan.py::modify_gtt` — two-leg dispatch (Sprint C)
- `backend/brokers/adapters/groww.py` — emulated OCO trail (currently `NotImplementedError` -skip)
- `backend/brokers/adapters/dhan.py::ltp` — wired via instruments cache (B-2 fix)

**Asymmetric SELL guard note:** the poller's SELL ratchet check is `current_trigger > 0 AND proposed < current_trigger`. If `current_trigger=0` (entry never persisted), the guard short-circuits → trail silently dead. This is why **every persistence path that writes a trail entry MUST seed `current_trigger`** (see Sc.5a fix in `retry_template`).

## Part IV — Brokers

### 14. Broker abstraction



#### Capability matrix surface:

- `backend/brokers/capabilities.py::BrokerCapabilities` — dataclass with every capability flag
- `backend/brokers/registry.py::get_historical_brokers` — Kite-only filter
- `frontend/src/lib/data/brokerCapWarnings.js` — single source of truth for warning strings (H-5)
- `frontend/src/lib/order/OrderTicket.svelte::capWarningFor` — single-account
- `frontend/src/lib/SymbolPanel.svelte::aggregateCapWarnings` — cross-account (H-5)

🔗 **TECH — PriceBroker fallback chain** — **WHY** Some brokers can answer quote/ltp/historical (Kite), some can't (Dhan returns `{}` by design for quote). Walking the chain lets the operator's chart still render even when their primary account is throttled. **WHAT** `PriceBroker._try(method_name, *args)` iterates eligible brokers, calls method, checks predicates (`_quote_has_data` / `_ltp_has_data` / `_historical_has_data`), returns first successful response. **HOW** Add a new method by name in the predicate map. Rate-limit cool-off (`_RATE_LIMIT_COOLOFF`) excludes throttled accounts for 30s. **WHERE** `backend/brokers/registry.py::PriceBroker`.

#### 14.1 Kite account flipping — market-data broker resolution

`get_market_data_broker()` is the **single** resolution path for every quote / ltp / instruments / historical\_data call in route handlers and background tasks. A `contextvars.ContextVar ( _MDB_CTX )` caches the `PriceBroker` instance for the lifetime of one HTTP request so every callsite within the same handler picks the **same** broker session. On `PriceBroker._try()` failover: try broker A, on transient error / empty predicate, log `[MARKET-DATA-FALLBACK]`, try broker B, etc.

Selection order:

1. `connections.price_account` operator pin (setting)
2. `broker_accounts.priority` ASC
3. Insertion order



3. **IPv6 source-binding** — Kite + Dhan enforce one-session-per-IP rules. Each account binds to a unique IPv6 via `_IPv6SourceAdapter` (Kite/Dhan) or `ContextVar` proxy (Groww).
4. **Token cache** — each broker persists tokens to `.log/<broker>_tokens.json`. On startup, skips login if fresh token cached; fires full login only on miss/expiry/manual delete.
5. **Registry factories** — use `get_broker(account)` for operator actions, `get_price_broker()` for shared market data, `get_historical_brokers()` for OHLCV + regression, `get_sparkline_broker()` for KiteTicker.
6. **Capabilities immutable** — frozen dataclass with every field explicit per broker (no defaults). Used to render warning chips on `OrderTicket` when template asks for unsupported GTT shape.

**PriceBroker fallback chain** — when a broker returns empty data (Dhan returns `{}` for MCX quotes by design), walk to the next broker. Rate-limit cool-off excludes throttled accounts for 30s.

**Capability matrix — verbatim from `backend/brokers/capabilities.py`:**

```
KITE_CAPS = BrokerCapabilities(
    broker_id="kite", display_name="Kite (Zerodha)",
    gtt_single=True, gtt_oco=True, gtt_modify=True,
    gtt_cap_per_account=50, gtt_validity_days=365, gtt_supports_mcx=True,
    bracket_order=False, cover_order=True, atomic_basket=True,
    order_tag=True, margin_preview=True,
    postback_gtt="reliable", rate_limit_orders_sec=10,
)
DHAN_CAPS = BrokerCapabilities(
    broker_id="dhan", display_name="Dhan",
    gtt_single=True, gtt_oco=True, gtt_modify=True,
    gtt_cap_per_account=50, gtt_validity_days=365, gtt_supports_mcx=False,
    bracket_order=True, cover_order=True, atomic_basket=True,
    order_tag=True, margin_preview=True,
    # Audit fix — no Dhan WebSocket / GTT-fire postback handler is wired
    # in the codebase today; detection is the poll-based _task_oco_pair_watcher
    postback_gtt="poll_only",
    rate_limit_orders_sec=20,
)
GROWW_CAPS = BrokerCapabilities(
    broker_id="groww", display_name="Groww",
    gtt_single=True, gtt_oco=False, gtt_modify=True,
    gtt_cap_per_account=25, gtt_validity_days=90, gtt_supports_mcx=False,
    bracket_order=False, cover_order=False, atomic_basket=False,
    order_tag=False, # No native tag; broker_order_link sidecar covers it
    margin_preview=False,
    postback_gtt="poll_only",
    rate_limit_orders_sec=5,
)
```

OCO emulation for Groww (no `gtt_oco`) is implemented in `groww.py::place_gtt` via two single-trigger GTTs + the `_task_oco_pair_watcher` background task that cancels the surviving sibling when one fires. There's no `gtt_emulated_oco` flag; emulation is an implementation detail behind the adapter.

`CAPS_BY_BROKER_ID` maps both the canonical `"zerodha_kite"` and the legacy `"kite"` alias to `KITE_CAPS`, so older YAML-seeded rows keep resolving without a column rewrite.

Frontend reads via `GET /api/admin/brokers/{account}/capabilities` (in-memory, no broker call). UI helper `brokerCapWarnings.js` consults the matrix to warn the operator at submit time when a template requests a feature the broker can't provide natively.

### 14.5.9 Per-broker quirks worth knowing

These are documented inline in code, but listed here for orientation:



| Broker | Quirk                                                                   | Where it's handled                                                                                      |
|--------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Kite   | tag is 20-char max                                                      | <code>_truncate_tag</code> in <code>backend/brokers/adapters/kite.py</code>                             |
| Kite   | Postback HMAC validation                                                | <code>order_postback</code> route checksum check                                                        |
| Kite   | Rate-limited <code>historical_data</code> quota (low per-second budget) | <code>_RATE_LIMIT_COOLOFF</code> 30s window in registry — <code>_RATE_LIMIT_COOLOFF_SECONDS = 30</code> |
| Kite   | One-IP-per-app rule                                                     | <code>_IPv6SourceAdapter</code> mount                                                                   |
| Dhan   | Token dashboard validity defaults to 5min                               | Operator must extend to 24h in Dhan dashboard                                                           |
| Dhan   | <code>ltp()</code> returns <code>{}</code> for MCX commodity by design  | PriceBroker fallback + B-2 fix logs the empty response                                                  |
| Dhan   | <code>modify_gtt</code> needs TWO calls (ENTRY_LEG + TARGET_LEG)        | Sprint C dispatch in <code>dhan.py::modify_gtt</code>                                                   |
| Dhan   | One-active-token-per-app-per-IP rule                                    | <code>_IPv6SourceAdapter</code> + multi-account stabilizer in <code>Connections</code>                  |
| Groww  | Module-level <code>requests</code> calls with no session hook           | ContextVar monkey-patch (see §14.5.5)                                                                   |
| Groww  | No native OCO                                                           | Emulated via two single GTTs + <code>_task_oco_pair_watcher</code> cancellation of survivor             |
| Groww  | <code>cancel_gtt</code> needs exchange (numeric id collision risk)      | M-4 fix raises if exchange missing                                                                      |
| Groww  | No <code>historical_data</code> support                                 | <code>historical_data=False</code> cap; sparkline + chart endpoints fall over to Kite                   |

#### 14.5.9.5 BSE ticker subscription and NSE→BSE equity token fallback

**50-cap fix** — `_perf_subscribe_book_symbols` used a hard `[:50]` slice that truncated the unresolved-symbols list. BSE holdings appearing after the 50th NSE/NFO entry were never subscribed. Fix: replaced with chunked loop ( `CHUNK=50` ) that covers ALL unresolved symbols.

**NSE→BSE equity companion** — for equity exchanges (NSE/BSE), `quote.py:_resolve_token_for_sym` now pairs companions immediately instead of walking derivatives first. Walk order:

| Exchange        | Token walk order                             |
|-----------------|----------------------------------------------|
| NSE             | <code>[NSE, BSE, MCX, CDS, NFO, BFO]</code>  |
| BSE             | <code>[BSE, NSE, MCX, CDS, NFO, BFO]</code>  |
| MCX/CDS/NFO/BFO | Check that exchange first, no companion pair |

Same fix applied at three sites:

- `quote.py:_resolve_token_for_sym` (main ticker lookup)
- `batch_sparkline` token resolution
- `_task_sparkline_warm` token lookup

#### BFO (BSE F&O) verification:

- ✓ In `_SPARKLINE_EXCHANGES` — subscribed
- ✓ Maps to "NSE" gate (09:15-15:30 IST)
- ✓ Route lookup via `_symbol_exchange_open` → `ctx['nse_open']`
- ✓ Kite segment mapping: BFO → "derivatives"
- ✓ Token found at index 0 in walk order

#### Files:

- `backend/api/background.py::_perf_subscribe_book_symbols` — chunked loop
- `backend/api/routes/quote.py::_resolve_token_for_sym` — companion pairing
- `backend/api/routes/sparkline.py` — batch + warm token lookup

#### 14.5.10 Modifying the broker layer — guard rails

If you're touching anything in `backend/brokers/`:

- **Never branch in callers by broker\_id**. The whole point of the abstraction is that callers don't know which vendor they're talking to. If you find yourself writing `if isinstance(broker, KiteBroker)` in a route, the right fix is a new capability flag.
- **Always re-shape vendor responses to Kite shape**. Frontend + chase + template all expect Kite shape. Skipping the normalize step silently breaks things downstream.
- **Status maps are non-negotiable**. Every vendor status must map to a Kite-canonical status ( `COMPLETE`, `OPEN`, `CANCELLED`, `REJECTED`, `EXPIRED`, `TRIGGER_PENDING` ). A missing entry breaks chase fill detection.
- **Wrap sync SDK calls in `asyncio.to_thread`**. Adapters are sync internally; callers MUST go through `asyncio.to_thread(broker.method, ...)` in async handlers to avoid blocking the event loop.

- **Log silent failures.** B-4 audit fix: when a broker SDK returns empty data instead of raising, log `WARNING` with method + symbol + account so the failure surfaces in `api_log_file`.
- **Update `capabilities.py` first.** If you discover a vendor supports something we'd marked `False`, update the matrix BEFORE writing code that uses the feature. Operator-visible warnings flow from the matrix.

## 15. How to add a new broker

If you're integrating a new vendor (e.g. "Upstox"):

### Backend

1. **Implement the adapter** in `backend/brokers/upstox.py`. Subclass `Broker` (ABC at `base.py`). Implement EVERY method — there's no "partial" mode. If a method genuinely doesn't apply, raise `NotImplementedError` with a clear message rather than returning empty.
2. **Translate to Kite shape.** Every method that returns operator-facing data (orders, positions, ltp, GTTs) must shape its return to match Kite's structure. Frontend renders are Kite-shape; downstream chase + template code expects Kite-shape. Build a `_normalise_*` helper per category. Mirror the patterns in `ghan.py` and `groww.py`.
3. **Status-string normalization.** Add `_UPSTOX_STATUS_TO_KITE = {...}`. Every Kite-canonical status (`COMPLETE`, `OPEN`, `CANCELLED`, `REJECTED`, `EXPIRED`) must map from one Upstox string. The B-1 audit lesson: a single missing entry silently breaks chase fill detection for an entire broker.
4. **Capabilities.** Add `UPSTOX_CAPS` in `capabilities.py` with EVERY field set explicitly. Don't rely on dataclass defaults — being explicit makes capability gaps visible at code review.
5. **Register in `registry.py`.** Add to `_ADAPTERS` map + `CAPS_BY_BROKER_ID`.
6. **Token caching.** Each broker has its own `.log/<broker>_tokens.json`. Follow the connection wrapper pattern in `connections.py`.
7. **Tests.** Add `backend/tests/test_upstox_broker.py`. Mock the vendor SDK at the boundary; assert your `_normalise_*` outputs.

### Frontend

8. **No frontend code change needed.** The `BrokerCapabilities` dataclass is the contract; the cap warning helper at `brokerCapWarnings.js` reads it generically. Operator-visible capabilities surface automatically.

## 16. Broker gotchas

Documented so you don't relearn them the hard way:

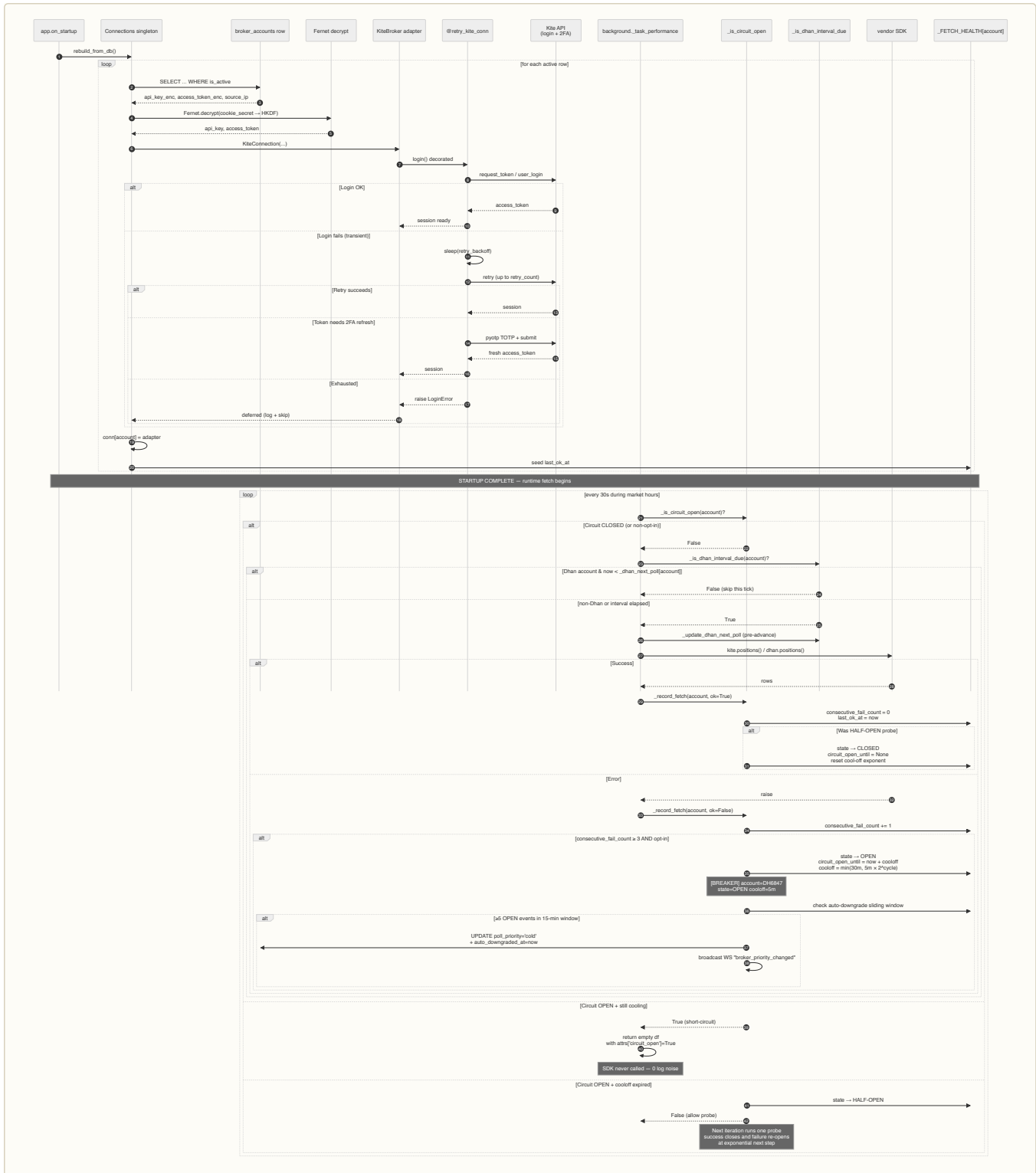
| Gotcha                                                                                                                                         | Bit us in                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Postback arrives before broker_order_id is committed</b>                                                                                    | Race window between <code>AlgoOrder</code> pre-persist + seed-broker_id second commit. Fix: fallback recent-NULL-id match (C-3)                                                                  |
| <b>Cumulative vs delta in status polls</b>                                                                                                     | Every broker reports <code>filled_quantity</code> cumulatively. Pre-fix we added the cumulative value each call → inflation across restarts (C-1)                                                |
| <b>Kill recorded against old broker_order_id</b>                                                                                               | Cancel-and-replace creates a new id; kill was only checked against old. Operator's kill silently ignored (C-2)                                                                                   |
| <b>WeakValueDictionary GC during await</b>                                                                                                     | <code>_TEMPLATE_ATTACH_LOCKS</code> could be GC'd between mint and acquire. Fix: strong dict with TTL (M-5). 1h chosen because longest realistic live-chase window is ~30 min; 1h is 2x headroom |
| <b>Reconcile attach BEFORE commit</b>                                                                                                          | Attach pipeline opened its own session and read pre-commit state. Fix: defer to after commit (C-4 single + bulk)                                                                                 |
| <b>Empty <code>_normalise_orders</code> status map</b>                                                                                         | Groww's "EXECUTED" passed through verbatim; chase loop never saw "COMPLETE" → no fill detection (B-1)                                                                                            |
| <b>Dhan <code>ltp()</code> returned <code>{}</code></b> by design until B-2. Trail stop silently dead — no log, no Telegram, just zero ratchet |                                                                                                                                                                                                  |
| <b>Groww <code>cancel_gtt</code> blind segment fallback</b>                                                                                    | Could cancel wrong GTT on numeric id collision. Now raises if exchange missing (M-4)                                                                                                             |
| <b>Naive <code>datetime.now()</code> in DB writes</b>                                                                                          | Mix with tz-aware columns → "AT TIME ZONE" errors. Always <code>datetime.now(timezone.utc)</code>                                                                                                |
| <b>Kite's tag is 20-char max</b>                                                                                                               | We truncate via <code>_truncate_tag</code> in <code>chase.py</code>                                                                                                                              |
| <b>Trail-stop persistence missing <code>current_trigger</code></b>                                                                             | Asymmetric SELL guard short-circuits with <code>0</code> . Every trail-write path must seed (Sc.5a)                                                                                              |
| <b>OCO double-fire 15s window</b>                                                                                                              | <code>oco_pair_poll_seconds</code> default. Matches the <code>poll_only</code> GTT detection lag operator sees in the cap warning chip. Telegram alert fires on detection (H-8)                  |
| <b>60-second postback fallback window</b>                                                                                                      | Long enough to cover the slowest IOC fill + DB commit race; short enough to avoid cross-pollination with new orders (C-3)                                                                        |

## 16.1 Connection retries per account — login + circuit breaker

Startup `Connections.rebuild_from_db()` iterates `broker_accounts` rows. Per account: decrypt secrets (Fernet), login via adapter, on failure the `@retry_kite_conn` decorator retries with backoff. If token refresh is needed, the 2FA flow re-mints. Post-startup, the runtime fetch path is guarded by two independent gates on `broker_apis._fetch_*_local`:

1. **Circuit breaker** ( `_is_circuit_open` , opt-in per row via `circuit_breaker_enabled` ) — CLOSED → normal fetch (increments `consecutive_fail_count` on error) → OPEN after 3 consecutive fails (cool-off 5m → 10m → 20m → 30m exponential) → HALF-OPEN probe → back to CLOSED on success.
2. **Dhan interval gate** ( `_dhan_next_poll[account]` ) — advances the next-poll timestamp BEFORE the fetch runs, so a crash mid-fetch doesn't cause a tight-retry loop. Non-Dhan accounts always pass.

Auto-downgrade: when a Dhan account with both `circuit_breaker_enabled=True` AND `auto_downgrade_enabled=True` hits ≥5 breaker-OPEN events inside a 15-min sliding window, `poll_priority` flips `hot` → `cold` and a 5-min cooloff prevents re-firing.



Key gates & timings:

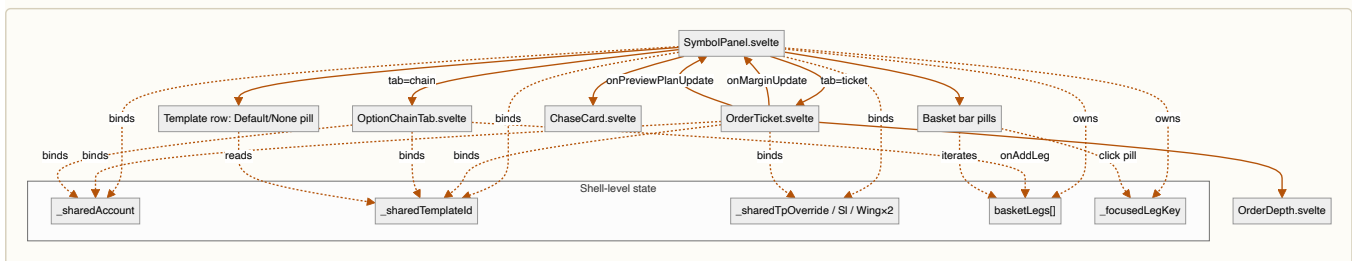
| Gate                                                 | File                                                       | Behaviour                                                                                                     |
|------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>@retry_kite_conn</code>                        | <code>backend/shared/helpers/decorators.py</code>          | Login retries with backoff; falls into 2FA flow on token expiry                                               |
| <code>_is_circuit_open(account)</code>               | <code>backend/brokers/broker_apis.py</code>                | Fast in-process dict lookup; O(1); non-opt-in accounts always return False                                    |
| <code>_record_fetch(account, ok, error)</code>       | same file                                                  | Increments/resets <code>consecutive_fail_count</code> , opens/closes circuit, evaluates auto-downgrade window |
| <code>_is_dhan_interval_due(account)</code>          | same file                                                  | Dhan-only; checks <code>now &gt;= _dhan_next_poll[account]</code>                                             |
| <code>_update_dhan_next_poll(account, broker)</code> | same file                                                  | Pre-advances timestamp BEFORE fetch (crash-safe)                                                              |
| Cool-off schedule                                    | same file                                                  | 5m → 10m → 20m → 30m exponential (cap 30m); resets on successful HALF-OPEN probe                              |
| Auto-downgrade sliding window                        | same file                                                  | 5 OPEN events in 15 min → <code>poll_priority='cold'</code> ; 5-min re-fire cooloff                           |
| Restore endpoint                                     | <code>POST /api/admin/brokers/{id}/restore-priority</code> | Operator resets to 'hot', clears stamps                                                                       |

#### Surfaces:

- `/api/admin/broker-health` returns `circuit_state`, `consecutive_fail_count`, `circuit_open_until`, `circuit_breaker_enabled`, `poll_priority`, `auto_downgraded_at`, `auto_downgrade_reason` per account.
- `BrokerHealthBadge.svelte` shows OPEN chip + "circuit open until HH:MM" tooltip for opt-in accounts; non-opt-in red badges show "retrying every poll".

## Part V — Frontend

### 17. Frontend modal state



#### Key files:

- `frontend/src/lib/SymbolPanel.svelte` — shell + Template row + basket bar + chase card mount
- `frontend/src/lib/order/OrderTicket.svelte` — Ticket form + depth ladder + margin preview
- `frontend/src/lib/order/OptionChainTab.svelte` — strike grid + futures + chain quotes
- `frontend/src/lib/order/OrderDepth.svelte` — bid/ask depth (visibility-gated polling)

#### Preview chip swap rule (Chain tab):

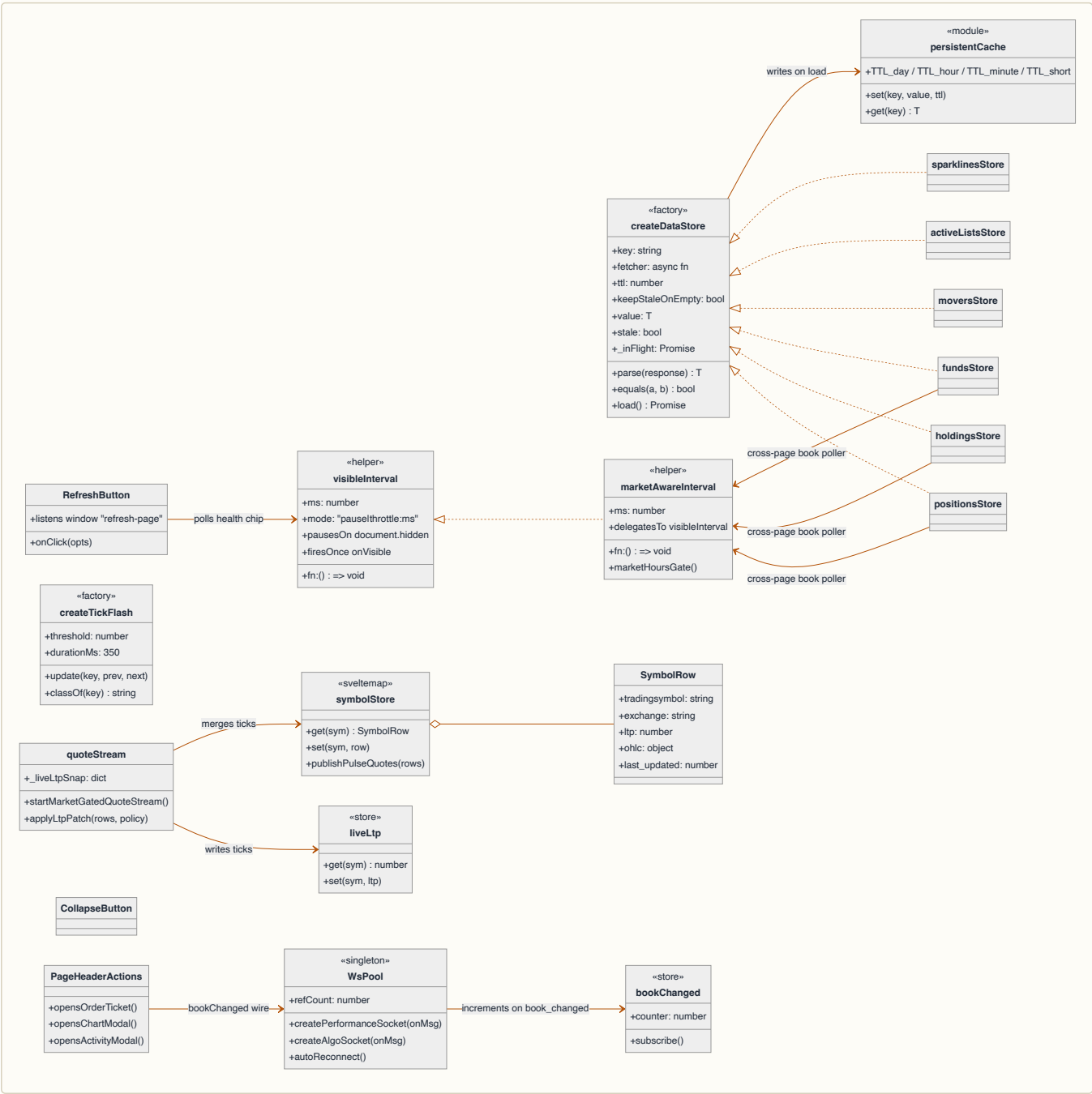
- `basketLegs.length === 0` → Ticket-form preview
- `basketLegs.length > 0` + no focus → last-leg preview
- `_focusedLegKey !== null` → that specific leg's preview, badge shows `LEG N/M` ●
- Click any basket pill → set `_focusedLegKey`
- Click chip itself → cycle to next leg
- Operator `x` on focused leg → key clears, falls back to last-leg

⚙️ **TECH — Svelte 5 `$bindable()` props** — **WHY** Two-way sync without prop-drilling or a global store. **WHAT** Child component declares `let { templateId = $bindable(null) } = $props()`; parent writes `bind:templateId={_sharedTemplateId}`. **HOW** Mutations on either side propagate. Avoid `bind:` for derived values (use a `$derived` instead). **WHERE** `SymbolPanel.svelte` ↔ `OrderTicket.svelte` ↔ `OptionChainTab.svelte` template + account props.

# 18. Frontend state architecture

## 18.0 Frontend runtime object graph

The class diagram below covers the load-bearing frontend runtime — the data-store factory, the singleton WebSocket pool, the polling helpers that hibernate on tab-hidden, and the tick-flash / symbol-store primitives that surface live prices in ag-Grid cells.



**Grep map** — every symbol above resolves to a single file:

| Symbol                                                                                         | Module                                           |
|------------------------------------------------------------------------------------------------|--------------------------------------------------|
| createDataStore factory                                                                        | frontend/src/lib/data/dataStore.svelte.js        |
| positionsStore / holdingsStore / fundsStore / moversStore / activeListsStore / sparklinesStore | frontend/src/lib/data/marketDataStores.svelte.js |
| persistentCache                                                                                | frontend/src/lib/data/persistentCache.js         |
| symbolStore + publishPulseQuotes                                                               | frontend/src/lib/data/symbolStore.svelte.js      |
| liveLtp + quoteStream                                                                          | frontend/src/lib/data/quoteStream.js             |
| visibleInterval + marketAwareInterval                                                          | frontend/src/lib/stores.js                       |
| createTickFlash                                                                                | frontend/src/lib/data/tickFlash.svelte.js        |
| createPerformanceSocket / createAlgoSocket (WsPool)                                            | frontend/src/lib/ws.js                           |
| bookChanged counter                                                                            | frontend/src/lib/stores.js                       |
| RefreshButton / CollapseButton / PageHeaderActions                                             | frontend/src/lib/*.svelte                        |

## 18.1 Why no global store for order state?

Svelte stores would be the obvious pattern but we don't use them for order modal state. Reasons:

- **Modals are short-lived.** The operator opens, fills, submits, closes. State outlives a single modal mount maybe 5% of the time (basket persists across tab flips).
- **Component-local state with bindable props is enough.** `bind:value` on Svelte 5 runes provides bidirectional sync without the boilerplate.
- **One modal at a time.** We don't need a global "current order context" — the modal owns its context.

The exceptions: `executionMode` (navbar drives every page), `authStore` (every page), `dataCache` (PositionStrip + dashboards share), `orderTemplatesStore` (template CRUD broadcast). These are all narrow — they don't carry order-specific state.

## 18.2 The "shell" pattern

`SymbolPanel.svelte` is a shell. It owns:

- Header (account + symbol pickers)
- Tabs (Ticket / Chain)
- Template row (Default/None pill + override inputs + preview chip)
- Basket bar (when basket has legs)
- Common action footer (margin chip + Submit)

The actual tab content ( `OrderTicket.svelte` , `OptionChainTab.svelte` ) is mounted as a child. State pipes through:

- **Down via props:** shell → tab (e.g. `_sharedAccount` → `account` prop)
- **Up via callbacks:** tab → shell (e.g. `onMarginUpdate` , `onPreviewPlanUpdate` )
- **Two-way via `bind`:** for shared mutable state (e.g. `bind:templateId={_sharedTemplateId}` )

When you add a new piece of shell-visible state, decide once:

- Is it tab-specific? → Stay in the tab component.
- Should it survive tab flips? → Lift to shell.
- Does any tab need to READ it? → Pipe down via prop.
- Does any tab need to WRITE it? → `bind`: it.

## 18.3 Frontend column factory SSOT — `pulseColumns.js`

`frontend/src/lib/data/pulseColumns.js` is the **single source of truth** for ag-Grid column definitions across the platform. Factory functions replace inline column specs, enabling consistent styling and behaviour across surfaces. New factories added this session:

| Factory                                   | Returns                              | Used by                     |
|-------------------------------------------|--------------------------------------|-----------------------------|
| <code>mkWeightPctCol()</code>             | Column spec for portfolio weight %   | PerformancePage, Dashboard  |
| <code>mkDeltaCol()</code>                 | Column spec for P&L delta            | PerformancePage             |
| <code>mkThetaCol()</code>                 | Column spec for option theta         | /admin/derivatives          |
| <code>mkNavBreakdownCols(segments)</code> | Array of NAV breakdown columns       | Dashboard NavBreakdown card |
| <code>mkUtilPctCol()</code>               | Column spec for margin utilization % | /admin/funds                |
| <code>mkFundsDetailCols()</code>          | Array of funds detail columns        | /admin/history Funds tab    |

⚡ **TECH — Centralized column specs vs inline definitions** — **WHY** Operator brief: "weight % looks different on three pages". Inline column specs (valueFormatter, width, headerName, tooltips) scatter across components; a missed update breaks visual consistency. **WHAT** Every column that appears on multiple pages gets a factory in `pulseColumns.js`. Pages call the factory instead of defining columns inline. **HOW** Factories export immutable column objects; pages `.map()` them into the `ag-Grid` `columnDefs` prop. **WHERE** `frontend/src/lib/data/pulseColumns.js`; used by `PerformancePage`, `Dashboard`, `/admin/derivatives`, `/admin/history`, `/admin/funds`.

**Usage pattern:**

```
// Before: inline spec on every page
const positionCols = [
  { headerName: "Weight %", valueFormatter: pctFmt, width: 90, ... },
  { headerName: "Delta", valueFormatter: aggFmt, width: 80, ... },
];

// After: factory-based
const positionCols = [
  mkWeightPctCol(),
  mkDeltaCol(),
];
```

**Shared formatters** — `frontend/src/lib/format.js` now exports `aggFmtGrid({ value })` and `pctFmtGrid({ value })` — `ag-Grid`-compatible wrapper functions over the existing `aggCompact` and `pctFmt` formatters. These return styled strings with the correct precision; used by column factories for `MarketPulse` and `PerformancePage` grid definitions.

**Files:**

- `frontend/src/lib/data/pulseColumns.js` — factory SSOT
- `frontend/src/lib/format.js` — `aggFmtGrid()` and `pctFmtGrid()` `ag-Grid` wrappers
- `frontend/src/lib/data/nav.js` — `aggregateDayPnlForPositions()` helper (see §19.1)
- Callers: `PerformancePage`, `Dashboard`, `/admin/derivatives`, `/admin/history`

### 18.4 Pure module extractions — Phase 1

Four stateless JavaScript modules ship core logic isolated from component tree:

| Module                         | Exports                                                                                                                                                   | Used by                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <code>riskMath.js</code>       | <code>normCdf</code> , <code>probAbove</code> , <code>expectedValueOnCurve</code> , <code>multilegPopOnCurve</code> , <code>RISK_FREE_R=0.07</code>       | <code>Derivatives page</code> , <code>/admin/derivatives</code> |
| <code>templateScope.js</code>  | <code>appliesToFor(side, sym) → 'sell_option' 'sell_any' 'buy_option' 'buy_any' 'both'</code>                                                             | <code>OrderTicket</code> , <code>SymbolPanel</code>             |
| <code>pulseGridSetup.js</code> | <code>PULSE_DEFAULT_COL_DEF</code> , <code>PULSE_SORTING_ORDER</code> , <code>pulseRowId</code> , <code>summaryRowId</code> , <code>postSortGroups</code> | <code>MarketPulse (PositionStrip)</code>                        |
| <code>chart/paths.js</code>    | <code>smaPath</code> , <code>emaPath</code> , <code>vwapPath</code> , <code>bbPaths</code> , <code>rsiSeries</code> , <code>macdSeries</code>             | <code>ChartWorkspace (overlays + indicators)</code>             |

**Design pattern:** each module is a zero-dependency utility collection; no props, no `svelte` imports, no stores. Enables unit-test coverage + reuse across unrelated surfaces.

**Files:**

- `frontend/src/lib/data/riskMath.js` — Black-Scholes, normal CDF, payoff integration
- `frontend/src/lib/data/templateScope.js` — symbol-side decision tree for default-template pickup
- `frontend/src/lib/data/pulseGridSetup.js` — `ag-Grid` sorting + row ID scheme + column definitions
- `frontend/src/lib/chart/paths.js` — SVG path builders for overlays + indicators

### 18.5 Svelte component extractions — Phase 2 + Phase 3

**Phase 2** (small, light):

| Component                          | Props                                                                                                     | Used by                                                                                      |
|------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <code>ChaseAggPicker.svelte</code> | <code>variant='ticket' 'panel' + value/onChange</code>                                                    | <code>OrderTicket</code> (variant=ticket, default), <code>SymbolPanel</code> (variant=panel) |
| <code>OhlcvTooltip.svelte</code>   | <code>bar</code> , <code>pxLeft</code> , <code>pxTop</code> , <code>pinned</code> , <code>onClose</code>  | <code>ChartWorkspace</code> (candlestick hover)                                              |
| <code>TickTooltip.svelte</code>    | <code>tick</code> , <code>pxLeft</code> , <code>pxTop</code> , <code>pinned</code> , <code>onClose</code> | <code>ChartWorkspace</code> (live-tick overlay hover)                                        |

**Phase 3** (large, Playwright-gated):



| Component                            | Lines | Props                                                                                                                                        | Used by                                                            |
|--------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| OrderKnobsRow.svelte                 | 320   | order_type, product, variety, validity (all <code>\$bindable()</code> )                                                                      | OrderTicket (modal)                                                |
| TemplateBar.svelte                   | 280   | tp_pct_override, sl_pct_override, wing_qty_override, wing_leg_override (all <code>\$bindable()</code> ) + <code>sideAwareDefault</code> prop | SymbolPanel (shell Template row)                                   |
| CandidateLegRow.svelte (derivatives) | 841   | <code>c</code> , <code>legsTab</code> , <code>legAnalytics</code> + 6 callback props                                                         | /admin/derivatives page, legs grid                                 |
| AddToPulseModal.svelte               | 380   | 13 <code>\$bindable()</code> state vars + 7 async callbacks                                                                                  | MarketPulse (add-to-watchlist)                                     |
| CardHeader.svelte                    | 200   | <code>title</code> , <code>showControls</code> , <code>onDownload</code> , <code>variant</code> props + CSS custom property theming          | 9 card header sites (NavBreakdown, performance, derivatives, etc.) |

#### Svelte 5 patterns:

- **`$bindable()` two-way:** Child declares `let { fieldName = $bindable(default) } = $props()`; parent writes `bind:fieldName={parent_var}`. Mutations sync both directions atomically.
- **Callback forwarding:** Large components expose callback props for state changes; parent wires them to stores or sibling updates. Example: `CandidateLegRow` → `onLegAnalyticsChange(newVal)` → parent `legsTab.analytics[idx] = newVal`.

#### Files:

- `frontend/src/lib/order/ChaseAggPicker.svelte` — variant-driven L/M/H aggression UI
- `frontend/src/lib/chart/OhlcvTooltip.svelte` + `TickTooltip.svelte` — anchored chart tooltips
- `frontend/src/lib/order/OrderKnobsRow.svelte` — extracted from `OrderTicket`, unified knob layout
- `frontend/src/lib/TemplateBar.svelte` — extracted from `SymbolPanel`, override row + Default button state
- `frontend/src/routes/(algo)/admin/derivatives/CandidateLegRow.svelte` — 841-line leg table row, internally isolated
- `frontend/src/lib/AddToPulseModal.svelte` — watchlist add-to modal, async symbol search + confirm
- `frontend/src/lib/CardHeader.svelte` — unified card title bar, CSS var theming, embedded controls

**CardHeader theming:** Uses CSS custom properties ( `--ch-*` ) set at layout level ( `algo-dark` in `+layout.svelte` , `public-light` in `public` viewport). `showControls` prop gates the right-side icon tray; `onDownload` callback wires to `NavBreakdown.downloadCsv()` via a named method exposed on the component. Embeds `CardControls` component internally.

#### Bug fixes bundled:

- `GridDownloadButton.svelte` — added `autoMargin` prop (default `true`); `CardControls` passes `autoMargin={false}` so fullscreen layout spacer handles alignment instead.
- `NavBreakdown.svelte` — exposes `downloadCsv()` method; dashboard wires it through `CardHeader.onDownload`.

## 19. The preview pipeline

The on-fill preview chip ( `on fill` → `TP ₹250 / SL ₹180 / + Wing BUY ...` ) is the single most useful piece of context at submit time. It's computed via two independent pipelines:

- **OrderTicket's `_previewPlan`** ← computed against the Ticket form's symbol/side/qty/price/template.
- **SymbolPanel's `_lastLegPlan`** ← computed against the last basket leg (or operator-focused leg).

The chip render switches between them based on `_activeTab === 'chain' && basketLegs.length > 0`. **Why two pipelines?** Because the inputs are different:

- Ticket: form state, not yet a "leg"
- Last-leg: a fully-formed leg with its own account + symbol + overrides

Both call the same backend endpoint ( `previewTicketTemplate` ) with the same payload shape. The frontend just feeds them differently.

🔗 **TECH — Backend preview endpoint vs frontend simulation** — **WHY** Operators trust ₹ values that come from the same code path that will ACTUALLY fire on fill. Computing them in the frontend would risk drift; computing them in the backend guarantees the chip reflects reality. **WHAT** `POST /api/orders/preview-ticket-template` returns `{plan: {gtts: [...], wing: {...}, notes: [...]}}`. **HOW** Frontend debounces 200ms after any override change; backend runs `resolve_template_plan` with `apply_path="preview"` so no broker calls fire. **WHERE** `backend/api/routes/orders.py::preview_ticket_template`.

### 19.1 aggregateDayPnlForPositions helper

`frontend/src/lib/data/nav.js::aggregateDayPnlForPositions(rows)` reduces position rows using `baseDayPnlForPosition(r)` per row. Used wherever total Day P&L across all positions must be summed (e.g. PerformancePage TOTAL row, Dashboard hero, NavStrip P pill).

```
export function aggregateDayPnlForPositions(rows) {
  if (!rows || rows.length === 0) return 0;
  return rows.reduce((sum, r) => sum + baseDayPnlForPosition(r), 0);
}
```

**Why not `sum(day_change_val)` ?** The `day_change_val` field omits Kite's edge cases (new positions, fully-flat intraday closes).

`baseDayPnlForPosition(r)` applies the override: when `r.overnight_quantity === 0` && `r.pnl !== 0`, use `r.pnl` instead. Aggregating with the helper ensures the total matches the sum of individual per-cell Day P&L values.

#### Callers:

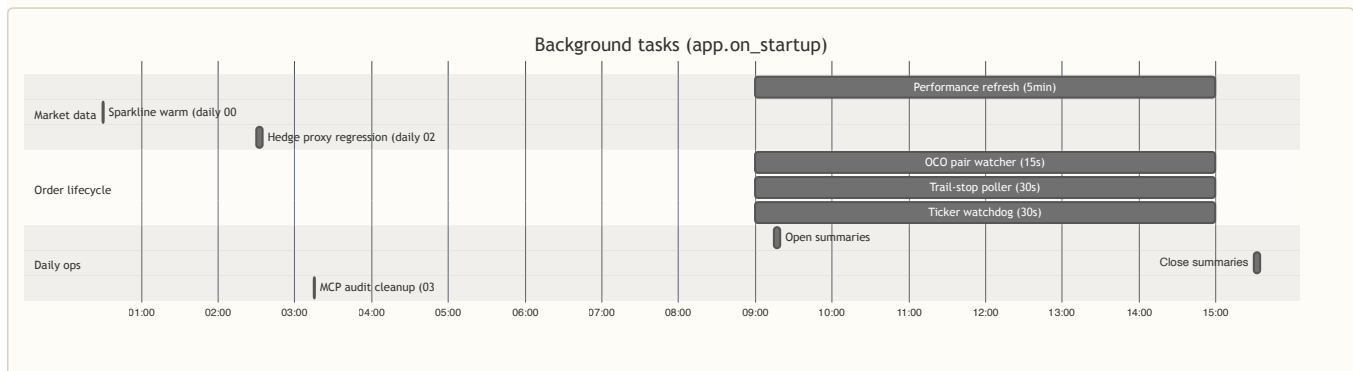
- PerformancePage TOTAL row aggregation
- Dashboard hero P&L + position summary
- NavStrip P pill slot 1

#### Files:

- `frontend/src/lib/data/nav.js::aggregateDayPnlForPositions` + `baseDayPnlForPosition`

## Part VI — Runtime

### 20. Background task topology



#### Key files:

- `backend/api/background.py` — all task definitions
- `backend/api/app.py::on_startup` — spawn list

#### Tasks that touch operator orders:

- `_task_performance` (5min) — fetches positions/holdings/funds; runs `agent_engine.run_cycle`
- `_task_oco_pair_watcher` (15s) — Groww emulated OCO sibling cancel
- `_task_trail_stop` (30s) — Dhan + Kite trail SL ratchet
- `_task_ticker_watchdog` (30s) — KiteTicker reconnect on disconnect

🔗 **TECH — Why poll-based + not event-based** — **WHY** Vendor postbacks are unreliable (Dhan + Groww have no inbound webhook; Kite drops 0.5-2% in our experience). Polling is the conservative floor. **WHAT** Each task runs on its own `asyncio` cadence; no scheduler library. **HOW** Pick interval based on operator latency tolerance: trail-stop = 30s (slow ratchet OK), OCO watcher = 15s (faster because both legs settling within window means double-fire). **WHERE** `backend/api/background.py`.

#### 20.1 Sparkline refresh pipeline

`_task_sparkline_warm` (startup + 00:30 IST + segment opens) populates KiteTicker subscription universe from three sources:

1. **Daily book union** — all positions + holdings from past 7 days (backstop, survives `conn_service` restart)
2. **Watchlist symbols** — from `WatchListItem` table (TRADES, key: `trading_symbol` + `exchange`)
3. **Virtual root resolution** — MCX/CDS symbols resolved to active contract (e.g. `CRUDE0IL` → `CRUDE0IL26JULFUT`)

#### Three-part defect fix:

- **MCX/CDS watchlist symbols get correct exchange** — `_sparkline_universe_symbols` now queries `(trading_symbol, exchange)` pair instead of loose `trading_symbol` lookup

- **Virtual roots resolved before OHLCV fetch** — `snapshot_sparkline` resolves MCX/CDS virtual roots via `symbol_resolver.resolve_symbol()` before calling `ohlcv_store.get_or_fetch_daily`
- **Tier 4 fallback** — `batch_sparkline` added fallback to read from `daily_book WHERE kind='sparkline'` when `ohlcv_store` is cold (`db_only` mode)

**Stale-better merge** — frontend `_mergeSparkSeries` keeps cached real curve over fresh flat/degenerate series to prevent chart collapse when broker feed lags.

**Mover grace-window** — `loadSparklines()` in `MarketPulse` carries the previous mover rotation's pairs into each new call via `_prevMoverSparkPairs`. Symbols that just left the winners/losers list get a one-rotation (30s) grace period before sparklines are pruned from cache. Symbols re-entering the top 10 show instantly from cache rather than fetching fresh.

**Sparkline gradient fill** — sparkline SVG cells now render a gradient area fill beneath each curve. Color is trend-aware: up-trend = teal `rgba(91,142,149,0.3)` tapering to 0% at bottom, down-trend = amber `rgba(196,122,61,0.3)`, flat = slate `rgba(126,151,184,0.3)`. SVG uses `<defs><linearGradient>` with 30% opacity at the curve tapering to 0% at the cell bottom; `<polygon>` fills the area, `<polyline>` draws the line on top. Implemented in `frontend/src/lib/components/SparklineCell.svelte`.

#### Files:

- `backend/api/background.py::_task_sparkline_warm`
- `backend/api/routes/sparkline.py::snapshot_sparkline`, `batch_sparkline`
- `backend/api/algo/symbol_resolver.py::resolve_symbol`
- `frontend/src/lib/PerformancePage.svelte::_mergeSparkSeries`
- `frontend/src/lib/MarketPulse.svelte::loadSparklines` — grace-window logic + `_prevMoverSparkPairs`
- `frontend/src/lib/components/SparklineCell.svelte` — gradient fill rendering

## 20.2 Background task async safety — broker API calls in `to_thread`

`backend/api/background.py::_task_warm_backfill` and other background tasks invoke broker SDK methods (`fetch_holdings()`, `fetch_positions()`) which are synchronous. These are now wrapped with `await asyncio.to_thread(...)` to prevent blocking the event loop during backfill warm or other background operations.

Pre-fix: sync broker calls in background tasks could block asyncio schedulers, causing delayed response handling on concurrent HTTP requests.

#### Files:

- `backend/api/background.py::_task_warm_backfill` — `to_thread` wrappers on broker calls

## 20.3 OCO pair watcher — mutual sibling pointer cleanup

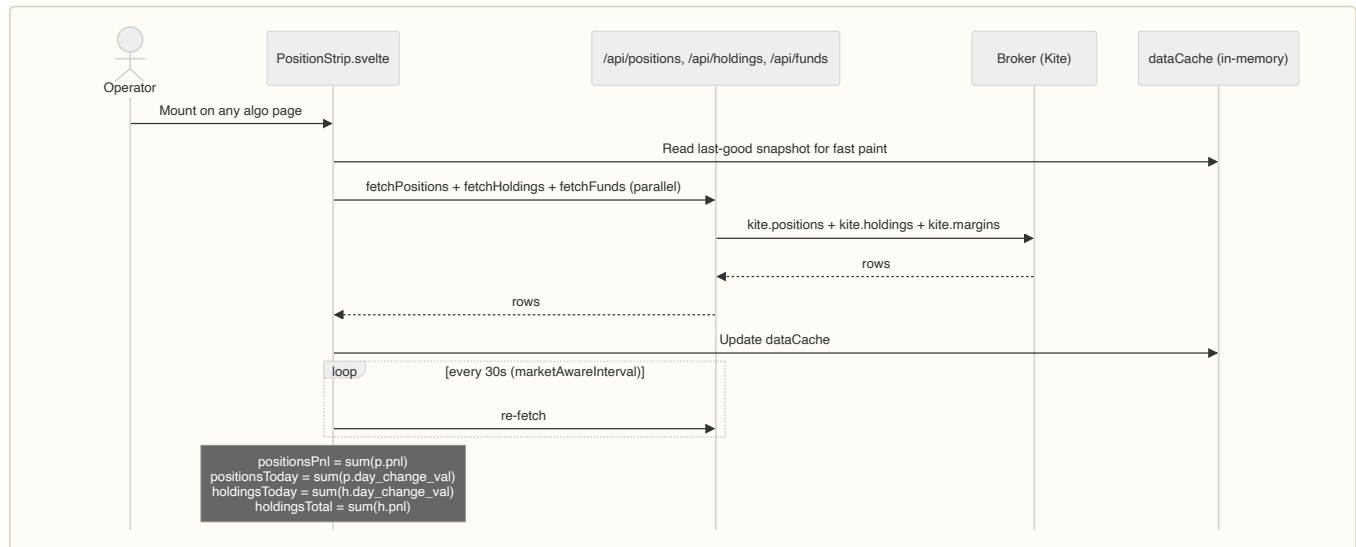
`backend/api/background.py::_task_oco_pair_watcher` now clears mutual `sibling_id` pointers when both OCO entries settle simultaneously. When entry A and entry B both reach a terminal status in the same poll cycle, `sib_entry.pop("sibling_id", None)` is called for the sibling entry (in addition to the primary), so the bidirectional reference is fully cleared in the both-settled branch.

Pre-fix: only the primary entry's sibling pointer was cleared, leaving the sibling entry with a dangling reference to a potentially stale ID.

#### Files:

- `backend/api/background.py::_task_oco_pair_watcher` — mutual pointer cleanup

## 21. Data refresh — PositionStrip + Dashboard



### Key files:

- `frontend/src/lib/PositionStrip.svelte` — navbar strip aggregations
- `backend/api/routes/positions.py`, `holdings.py`, `funds.py` — REST endpoints
- `backend/brokers/broker_apis.py::fetch_positions / fetch_holdings / fetch_margins`
- `backend/api/cache.py` — server-side cache (per-key locking + TTL)

`/admin/derivatives` Snapshot **TOTAL reconciles to PositionStrip** by adding back the rows the page filters out (equity intraday positions + derivative-looking holdings) via `_excludedByAccount`. See `frontend/src/routes/(algo)/admin/derivatives/+page.svelte` (search `_byUnderlyingTotal`).

⚙️ **TECH — marketAwareInterval polling vs WebSocket** — **WHY** Position state changes when fills happen; we already get fills via KiteTicker, but positions are aggregated server-side. Polling is cheaper than rebuilding aggregations client-side. **WHAT** `marketAwareInterval(fn, 30000)` polls every 30s during market hours, pauses on `document.hidden`. **HOW** Use the helper from `$lib/stores`; never raw `setInterval`. **WHERE** `frontend/src/lib/stores.js::marketAwareInterval`.

### 21.1 PositionStrip tick-border animation gate

`frontend/src/lib/PositionStrip.svelte` `tickBus.subscribe()` callback has a market-hours gate. If `!isNseOpen()` && `!isMcxOpen()`, the bottom-border refresh animation is skipped immediately. This prevents the tick animation from firing during closed hours and giving a false "live data refreshing" impression to the operator.

Pre-fix: animation ran continuously even when markets were closed and the WebSocket was sending stale ticks (reduced-cadence poll during quiet hours).

### Files:

- `frontend/src/lib/PositionStrip.svelte::tickBus.subscribe` — market-hours gate

## 21.5 Frontend → broker API — full round-trip

Detailed sequence of a live `/api/positions` request when `RAMB0Q_USE_CONN_SERVICE=1`. Shows the `_MDB_CTX` contextvar reset, raw-broker-DataFrame cache (`_RAW_CACHE`, 30s TTL) on the return path, the closed-hours snapshot branch (`closed_hours_or_broker` gate), and the two-process split (main API ↔ `conn_service` over UDS).



```
day_delta = pnl - overnight_quantity * (close_price - average_price)
```

- When `prev_settlement_pnl` is null, compute yesterday's unrealized P&L synthetically
- Gives `pnl - 0 = pnl` for new intraday positions ✓

#### Backend:

- `backend/api/routes/positions.py::_override_stale_close_from_snapshot` — calls `_backfill_prev_settlement_pnl(rows)` to populate `PositionRow.prev_settlement_pnl` from `daily_book.total_pnl` (same timing guard as `close_price` override)
- `backend/api/models.py::PositionRow` — added `prev_settlement_pnl: Optional[float]`

#### Frontend SSOT:

- `baseDayPnlForPosition(r)` in `frontend/src/lib/data/nav.js` — applies two-tier formula above
- `livePositionDayPnl(r, liveLtp, pollLtp)` wraps `baseDayPnlForPosition` + adds live SSE tick adjustment  $(liveLtp - pollLtp) \times qty$
- Used by:
  - PerformancePage TOTAL row (sum of daily P&L)
  - Derivatives page `_byUnderlyingTotal` loop (F&O aggregate)
  - Dashboard hero P&L + position summary
  - NavStrip P pill slot 1 (intraday P&L)
  - MarketPulse position card
  - Snapshot rows + Legs grid + Payoff overlay

## 21.5.6 previous\_close column in daily\_book — stable reference price

Kite overwrites `positions.close_price` to today's settlement price after market close, breaking the day P&L fallback formula during closed-hours snapshot reads. The `daily_book` table now includes a `previous_close` column that freezes yesterday's official settlement price at the first intraday snapshot of each trading day, providing a stable reference.

#### Problem it solves:

- Between MCX close (23:30 IST) and next market open (09:00 IST), Kite's `positions.close_price` has already been overwritten to today's settlement (known at 00:00 IST approx).
- The closed-hours snapshot reader calls `baseDayPnlForPosition` fallback: `pnl - overnight_qty * (close_price - avg_price)`
- Using today's settlement price here → fallback computes wrong, collapses to 0 for overnight positions
- Result: NavStrip P pill + Dashboard + Performance page show zero day P&L during overnight window

#### Solution:

- `DailyBook.previous_close: Optional[float]` column (DOUBLE PRECISION)
- Written once per (date, account, kind, symbol) at first snapshot via COALESCE UPSERT
- COALESCE ensures subsequent snapshots don't overwrite the frozen value
- `_positions_snapshot` reads and passes this to `build_snapshot_position_row`, which uses it as `PositionRow.close_price` when available (`> 0`)

#### Implementation details:

- **Capture point:** `backend/api/algo/daily_snapshot.py::_positions_rows()` writes `previous_close = position.close_price` at first snapshot of each IST date
- **Persistence:** `backend/api/database.py::init_db()` creates column via idempotent `ALTER TABLE daily_book ADD COLUMN IF NOT EXISTS previous_close DOUBLE PRECISION`
- **Read point:** `backend/api/routes/positions.py::_positions_snapshot()` SELECTs `db.previous_close` and passes to builder
- **Use:** `backend/api/routes/positions_helpers.py::build_snapshot_position_row()` uses `previous_close` as `close_price` in the `PositionRow` response when `previous_close > 0`

#### Frontend visibility:

- Closed-hours positions now show the stable overnight `close_price`
- `baseDayPnlForPosition` fallback uses the correct price, preserving overnight P&L during MCX window
- NavStrip P pill, Dashboard hero, Performance TOTAL all stay in sync during overnight hours

#### Files:

- `backend/api/models.py::DailyBook` — new column definition
- `backend/api/database.py::init_db` — migration via ALTER TABLE
- `backend/api/algo/daily_snapshot.py::_positions_rows` — write at first snapshot
- `backend/api/routes/positions.py::_positions_snapshot` — read + pass to builder

- `backend/api/routes/positions_helpers.py::build_snapshot_position_row` — use as `close_price`
- Frontend: no changes (uses existing `PositionRow.close_price`)

**Key rule:** never read `day_change_val` directly; always use `baseDayPnlForPosition(r)`.

#### Files:

- `backend/api/routes/positions.py::_backfill_prev_settlement_pnl`
- `backend/api/models.py::PositionRow.prev_settlement_pnl`
- `frontend/src/lib/data/nav.js::baseDayPnlForPosition`
- All callers listed above

### 21.5.6 MCX snapshot multiplier fix — closed-hours position undercount

`_positions_snapshot` in `backend/api/routes/positions.py` reads closed-hours snapshots from `daily_book` and rebuilds position rows. Pre-fix, raw Kite `quantity` fields (in lots for MCX) were passed directly to the row builder, while the live path (`broker_apis.fetch_positions`) multiplies by Kite's `multiplier` field to return contracts. This asymmetry meant closed-hours MCX options showed 100× undercount (qty in lots instead of contracts).

**Fix:** `backend/api/routes/positions_helpers.py` exports `extract_snapshot_multiplier(snapshot_payload)` which reads the `multiplier` field from the snapshot JSON. `_positions_snapshot` applies it before building each row, ensuring closed and open hours show identical contract quantities.

#### Files:

- `backend/api/routes/positions_helpers.py::extract_snapshot_multiplier` — extracts + applies multiplier
- `backend/api/routes/positions.py::_positions_snapshot` — caller
- `backend/tests/test_positions_helpers.py` — 5 tests covering MCX extraction + conversion

### 21.5.7 Holdings snapshot `close_price` fix — preserves prior-session close

`_overlay_snapshot_for_closed_exchanges` in `backend/api/routes/holdings.py` was overwriting `close_price` with the snapshot LTP value. During closed hours, this zeroed the day P&L formula:  $(\text{snapLtp} - \text{close\_price}) \times \text{qty} = 0$  when both fields held the same snapshot value, masking actual overnight price changes.

**Fix:** `close_price` is no longer overwritten. It preserves the broker's prior-session close, so the formula remains correct:  $(\text{snapLtp} - \text{prior\_close}) \times \text{qty}$  computes legitimate overnight P&L.

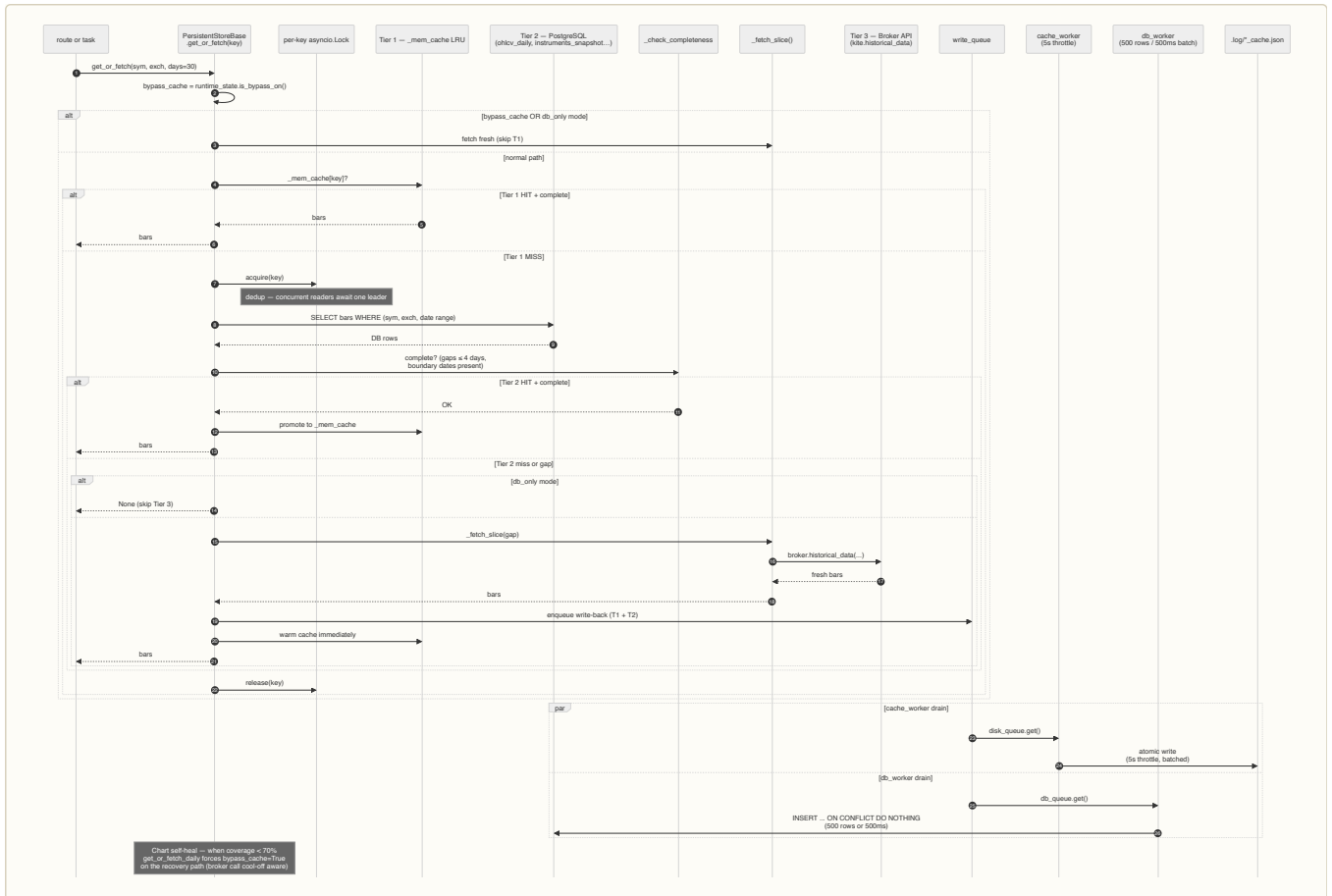
#### Files:

- `backend/api/routes/holdings.py::_overlay_snapshot_for_closed_exchanges` — preserves `close_price` field

## 21.6 Persistence three-tier — cache → DB → broker

Every OHLCV / instruments / holidays / intraday read walks Tier 1 (in-memory LRU) → Tier 2 (PostgreSQL row) → Tier 3 (broker API). A per-key `asyncio.Lock` deduplicates concurrent in-flight fetches. Broker writes return immediately to the caller; persistence runs off-path via two parallel worker coroutines (`cache_worker`, `db_worker`). Refresh modes (off / soft / hard) let the operator force Tier 3 refetches.





**Refresh modes** (operator toggles via `P05T /api/admin/persistence/mode/{off|soft|hard}`):

- `off` — normal hierarchy (default, safe)
- `soft` — Tier 1+2 bypass, fetch from broker, write-back heals both tiers
- `hard` — soft + ticker recycle (unsubscribe → reconnect → resubscribe)

**Write-back workers:**

- `cache_worker` — drains `disk_queue`, flushes `.log/sparkline_cache.json` etc. Throttled to 5s; last-write-wins on duplicate keys.
- `db_worker` — drains `db_queue`, batched SQL upserts per kind (500-row / 500ms boundary).
- On queue full: warn + drop; next read re-fetches from broker.

**Chart self-heal** (\$4.5 companion) — `/api/options/historical` detects <70% coverage in DB (threshold `chart_self_heal_coverage_threshold` in settings, default 0.70) and auto-fetches from broker when ≥1 broker available. Response carries `partial: bool` for the frontend "partial data" hint.

**Files:**

- `backend/api/persistence/store_base.py` — `PersistentStoreBase.get_or_fetch`, per-key lock, completeness checks
- `backend/api/persistence/ohlcv_store.py` — concrete OHLCV implementation
- `backend/api/persistence/write_queue.py` — `disk_queue` + `db_queue`
- `backend/api/persistence/cache_worker.py` / `db_worker.py` — background drainers
- `backend/api/persistence/runtime_state.py` — `is_bypass_on()`, mode toggles

## 21.7 Stale data semantics — `keepStateOnEmpty` and error recovery

Frontend data stores (`positionsStore`, `holdingsStore`, `fundsStore`, `sparklinesStore`) use `keepStateOnEmpty` semantics: when a broker fetch returns empty or errors, the prior snapshot is retained client-side so grids don't flicker blank. No red banner appears on transient HTTP errors.

**PositionStrip stale-data tint:**

- Track consecutive fetch failures per store with `_staleFailCount`
- When `positionsStore.error` || `holdingsStore.error` persists for 2+ consecutive polls (`_staleFailCount` ≥ 2), apply CSS class `ps-stale` to the `PositionStrip`
- Visual: amber gradient background tint + orange border (subtle, no modal or text)



- Resets `_staleFailCount` to 0 on the next successful poll

#### Orders page banner gating:

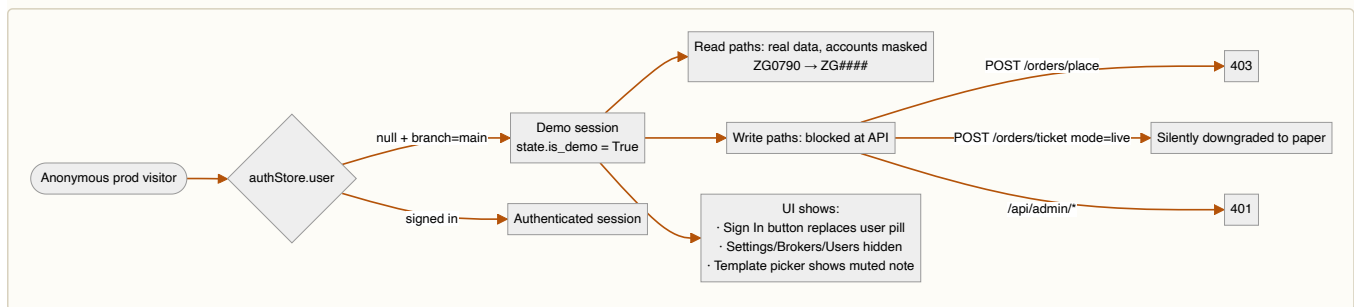
- `/orders` page tracks `_orderLoadFails` (incremented per failed fetch)
- Banner "Orders feed unavailable" only renders when `_orderLoadFails >= 3`
- Suppresses noise from single transient 502s; alerts operator on sustained feed loss
- Resets counter on successful reload

**Rationale:** HTTP errors are often transient (network blip, broker throttle for 1-2 seconds). Keeping the stale snapshot visible prevents jarring blanks while the backend retries automatically. The PositionStrip amber tint provides subtle visual feedback that data may be stale (for 2+ polls, real signal) without alarming on every hiccup.

#### Files:

- `frontend/src/lib/PositionStrip.svelte` — `_staleFailCount` tracking + `ps-stale` CSS class application
- `frontend/src/routes/(algo)/orders/+page.svelte` — `_orderLoadFails` counter + 3-failure banner gate
- `frontend/src/lib/data/marketDataStores.svelte.js` — `keepStaleOnEmpty` store config

## 22. Demo mode



#### Key files:

- `backend/api/auth_guard.py::is_demo_request` + `auth_or_demo_guard`
- `frontend/src/routes/(algo)/+layout.svelte` — demo nav-link gating
- `frontend/src/lib/SymbolPanel.svelte` — template row demo gate (L-3)
- `backend/brokers/broker_apis.py::mask_column` — for demo + public

## 22.5. Investor portal — token-as-credential

Public read-only NAV surface for LPs. The URL `/investor/<token>` IS the credential — no login, no password. Operator mints from `/admin` per-user Portal button, copies the URL, forwards through their own channel (WhatsApp / email).

#### Why token-as-credential

The boutique fund has 1–5 LPs. Asking each LP to manage a password for a quarterly NAV check is friction nobody wants. Carta, SS&C/GP-Link, and Yieldstreet all converge on the same pattern for LP-facing statements: a long-lived per-LP URL that's revocable on suspicion of leakage. Same threat model as a long-lived API key — if you trust the recipient with the URL, the URL is fine.

⚡ **TECH — Long-lived URL token vs JWT magic-link** — **WHY** JWT magic-links are short-lived (5–15 min); they're great for a one-time "log in to this session" handshake but useless for "bookmark this URL and re-check the value every Friday." The investor portal is a recurring read-only surface, not a session. **WHAT** 32-byte `secrets.token_hex` → 64-char string, 90-day default expiry, revocable. **HOW** Stored raw in `investor_tokens.token` (same convention as `AuthToken` — the token IS the URL slug; hashing adds no security since possession of the URL == access). **WHERE** `backend/api/models.py::InvestorToken`, `backend/api/routes/investor.py`.

#### Schema

```

class InvestorToken(Base):
    __tablename__ = "investor_tokens"
    id, user_id (FK users.id), token (64-char unique),
    expires_at, revoked_at (nullable),
    last_visit_at, visit_count (operator visibility),
    note (admin label), created_by (FK users.id), created_at
  
```

`Base.metadata.create_all` picks it up on next deploy — no migration.

## Active check

`_is_active(row, now) := row.revoked_at is None AND row.expires_at > now`. Three terminal states surfaced in the admin UI: ACTIVE (green) / REVOKED (red) / EXPIRED (slate).

## Endpoints

**Admin** ( `manage_investor_tokens` cap, admin-only):

- GET `/api/admin/users/{id}/investor-tokens` — list (preview only, never full token)
- POST `/api/admin/users/{id}/investor-tokens` — mint (returns full token + portal URL ONCE)
- DELETE `/api/admin/users/{id}/investor-tokens/{tid}` — revoke

**Public** (no auth — token in URL is the credential):

- GET `/api/investor/{token}/slice` → `InvestorSliceResponse` (same math as `/api/nav/me`)
- GET `/api/investor/{token}/history?days=180` → curve

## Visit tracking

`_resolve_token()` bumps `last_visit_at` + `visit_count` on every successful resolve via a best-effort `UPDATE` (try/except, rollback on failure). The operator's admin modal surfaces the timestamp + count so they know "this LP last looked 3 weeks ago" without leaving the page. The counter increments per endpoint hit, so a single page load bumps it by 2 (slice + history).

## Frontend separation

The portal page lives at `frontend/src/routes/investor/[token]/+page.svelte` — sibling of `(public)` and `(algo)` route groups. It inherits only the root `+layout.svelte` (which is empty — just `{@render children() }`), so it gets none of the algo navbar or the public marketing nav. Cream + champagne palette matching the marketing site so LPs land on a "professional statement" page, not a Bloomberg-style trading desk.

Robots `noindex,nofollow` in `<svelte:head>` so leaked URLs don't end up in search engines.

## Revocation model

Revoke is destructive but trivially reversible — admin clicks Revoke (confirms via `ConfirmModal`), `revoked_at` flips to `now()`, next visit 401s. Re-mint creates a new row; the old row stays for the audit trail.

Idempotent: revoking an already-revoked row is a no-op (the original `revoked_at` timestamp is preserved so "when did we revoke this?" remains accurate).

## Source files

- `backend/api/models.py::InvestorToken`
- `backend/api/routes/investor.py` — both controllers
- `backend/api/rbac.py::CAPS["manage_investor_tokens"]`
- `frontend/src/routes/investor/[token]/+page.svelte` — LP-facing page
- `frontend/src/routes/(algo)/admin/+page.svelte` — `openPortal()` modal + mint flow
- `frontend/src/lib/api.js::{fetchInvestorTokens, mintInvestorToken, revokeInvestorToken}`

## 22.6. Investor portal — units-based NAV math

The v1 model (`slice = share_pct × firm_nav`) is **retired**. Every LP slice / cost-basis / P&L computation now flows through the standard fund-accounting units model:

```
units_held(user, t)  = Σ units_delta for user's events <= t
total_units(t)       = Σ units_held across every LP
nav_per_unit(t)      = firm_nav(t) / total_units(t)
slice(user, t)       = units_held × nav_per_unit
cost_basis(user, t)  = Σ amount (sub+bootstrap) - Σ amount (redemption)
pnl(user, t)         = slice - cost_basis
```

☞ **TECH — Units vs static share\_pct** — WHY static `share_pct` breaks when an LP joins mid-period (their cost basis is the day they bought in, not "since fund inception") or partially redeems (the remaining slice's basis must shrink in proportion). WHAT Each LP holds a count of partnership units; the fund publishes a per-unit value daily; slices are products. HOW Subscription buys units at the day's `nav_per_unit`, redemption sells at the day's `nav_per_unit`, gains accrue automatically through `firm_nav` growth. WHERE `backend/api/algo/investor_units.py` is the single math source; four callsites consume it.

## Single source of math

`backend/api/algo/investor_units.py` exposes:

- `ensure_user_bootstrap(s, user)` — idempotent synthetic-event seed for v1 → units migration
- `ensure_all_bootstrapped(s)` — covers every eligible LP; called at the start of every units compute
- `units_held(events, as_of) / cost_basis(events, as_of)` — pure-function primitives
- `slice_value(user_events, all_events, firm_nav, as_of)` — returns `(slice, nav_per_unit)`
- `compute_slice(s, user, firm_nav, as_of)` — DB-aware wrapper
- `compute_slice_history(user_events, all_events, firm_curve)` — pre-fetched, walks dates in pure Python

Switched callsites (all four):

| Surface                                                             | Old                                     | New                                                |
|---------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------|
| <code>/api/nav/me</code> (authenticated LP)                         | <code>share_pct × firm_nav / 100</code> | <code>compute_slice()</code>                       |
| <code>/api/nav/me/history</code>                                    | scaled curve                            | <code>compute_slice_history()</code>               |
| <code>/api/investor/{token}/slice + /history</code> (public portal) | scaled curve                            | <code>compute_slice / compute_slice_history</code> |
| <code>compute_statement()</code> (monthly PDF)                      | period-anchored scale                   | event-walked slice                                 |

## Auto-bootstrap rule

On every units compute, `ensure_all_bootstrapped(s)` scans every eligible LP (`is_active=True AND share_pct > 0`) and inserts a synthetic event for any without one. The bootstrap row encodes the v1 state:

```
InvestorEvent(
    event_type = "bootstrap",
    units_delta = user.share_pct,
    amount = user.contribution,
    nav_per_unit = contribution / share_pct # 1.0 fallback when contribution=0
    event_date = contribution_date or created_at.date() or today,
)
```

This guarantees that **the first request after this code lands reproduces v1 numbers identically** when `share_pcts` sum to 100 across all eligible LPs. When the sum != 100 (operator-residual case with implicit ownership), the units model redistributes proportionally and slices sum to `firm_nav` by construction — slightly different numbers, but internally consistent + correct going forward.

## Day-delta semantics under units

`day_delta_share` on `/api/nav/me` is computed as `slice(today) - slice(prior)`, both via the SAME event set. This means:

- A pure market gain shows up as positive day-delta ✓
- A subscription between the two snapshots inflates `slice(today)` but ALSO appears in `cost_basis`, so it doesn't read as P&L on the LP's portal
- A redemption deflates `slice(today)` symmetrically

The portal UI labels this as "Day Δ" but the operator should read it as "change in your slice's market value since yesterday's snapshot."

## Smoke-test invariants

Math is verified end-to-end against a fixture (see commit `322f0c22`):

| Scenario                                                       | Invariant                                                  | Verified |
|----------------------------------------------------------------|------------------------------------------------------------|----------|
| All LPs at bootstrap, <code>share_pcts</code> sum to 100       | slices match v1 exactly                                    | ✓        |
| Fund grows N%                                                  | each LP's slice grows N% on their basis                    | ✓        |
| LP_A subscribes mid-period at higher <code>nav_per_unit</code> | extra subscription doesn't double-count as P&L             | ✓        |
| Any state                                                      | $\sum \text{slices} == \text{firm\_nav}$ (modulo rounding) | ✓        |

## Bootstrap edit / correction path

Operator wants to fix a bootstrap row (wrong contribution, wrong `share_pct` in `users` row, missing LP):

1. Edit `User.contribution / share_pct` in `/admin` (existing flow)
2. Delete the bootstrap event in `/admin` → Portal → Events tab
3. Next compute auto-bootstraps with the corrected User columns

The delete-then-recompute cycle preserves history (bootstrap event has the old timestamp) without manual reconciliation.

Source files

- `backend/api/algo/investor_units.py` — math + bootstrap
- `backend/api/algo/investor_statement.py::compute_statement` — PDF math via units
- `backend/api/routes/nav.py::my_slice + my_history` — authenticated LP endpoints
- `backend/api/routes/investor.py::slice + history` — public portal endpoints
- `backend/api/models.py::InvestorEvent` — events journal table

22.7. Audit log — forensic trail

Single `audit_log` table catches every mutating event the platform produces — HTTP mutations (via middleware), broker fills (via postback handler), agent-initiated actions (via the action dispatcher), and background-task events (NAV compute, monthly statement send). Read surface at `/admin/audit` is cap-gated to `view_audit` (designated `/admin / risk`).

⚙️ **TECH — ASGI middleware + fire-and-forget writes** — WHY SEBI Cat-III's "every mutating event" requirement can't be satisfied with per-route decorators (easy to forget; impossible to enforce). A middleware catches everything by default. WHAT `AuditMiddleware` wraps every HTTP response; on a mutating 2xx/3xx it schedules a `_write_audit` coroutine via `asyncio.create_task`. Response leaves the server immediately; the DB write lands shortly after. HOW Failed audit writes log a warning and drop — the user's request never blocks on the audit pipeline. WHERE `backend/api/audit.py`.

Schema

```
class AuditLog(Base):
    id, actor_user_id (FK users.id), actor_username, actor_role,
    action, category, method, path,
    target_type, target_id,
    status_code, summary,
    request_id (UUID, mirrored in X-Request-ID),
    client_ip, user_agent, created_at
```

- `actor_*` fields are SNAPSHOTTED — a later role demotion doesn't rewrite history.
- `category` is a coarse tag (`order.fill`, `agent.action`, `system.nav`, ...) populated by `_derive_category_from_path` for HTTP rows and explicitly by `write_audit_event` callers.
- `request_id` correlates each audit row with the API log line for the same request.

Two write paths

**1. HTTP middleware (default)** — `AuditMiddleware.handle` watches every response. Skips non-mutating methods + `_SUPPRESS_PREFIXES`. Captures actor from JWT, status code from the wrapped `send`, body summary from the first 1 KB of response. Path-derived category via `_derive_category_from_path`.

**2. Non-HTTP helper (added Jun 2026)** — `write_audit_event(category, action, ...)` is the public API for any code path that mutates state without going through HTTP:

| Caller                                                       | Category                                              | Actor                           |
|--------------------------------------------------------------|-------------------------------------------------------|---------------------------------|
| Broker postback handler ( <code>orders.py</code> )           | <code>order.fill / order.cancel / order.reject</code> | <code>broker</code>             |
| Agent action dispatcher ( <code>actions.py::execute</code> ) | <code>agent.action</code>                             | <code>agent:&lt;slug&gt;</code> |
| Monthly statement send ( <code>background.py</code> )        | <code>system.statement</code>                         | <code>system</code>             |
| NAV compute ( <code>background.py</code> )                   | <code>system.nav</code>                               | <code>system</code>             |

The helper is fire-and-forget ( `asyncio.get_running_loop().create_task(...)` ); failed writes log a warning and drop. Sim-mode actions are intentionally NOT audited — they're already isolated in the sim event log and don't touch real state.

Failed mutations toggle

`audit.log_failed_mutations` setting (default `False`). When ON, the middleware also writes audit rows for 4xx/5xx mutating responses. Use for defect tracking ("operator hit SUBMIT and saw 422 — what blocked?"); toggle off otherwise to avoid volume spikes from validation errors.

Category routing

`_PATH_CATEGORY_RULES` in `audit.py` is the prefix-match table. Adding a new business surface is one tuple. Example:

```

_PATH_CATEGORY_RULES: tuple[tuple[str, str], ...] = (
    ("/api/orders/ticket", "order.place"),
    ("/api/orders/postback", "order.fill"),
    ...
)

```

For `PUT/PATCH /api/orders/{id}` and `DELETE /api/orders/{id}`, the middleware narrows the generic `order` category to `order.modify` / `order.cancel` based on HTTP method.

## UI — filter pills

`/admin/audit` carries a row of category pills (All / Orders / Agents / Users / Config / System) above the existing column filters. Each pill passes a comma-separated category list to the backend; `SQL IN (...)` does the rest. The Category column in the table carries per-bucket tints (green orders / cyan agents / amber users / violet config / slate system).

## Performance contract

- Every audit write is `asyncio.create_task(_write_audit(...))` — no `await`. The middleware's `handle()` returns immediately after the wrapped response.
- The DB insert pays no transaction beyond its own session; failures swallow into a logger warning.
- Helper invocations from background tasks pay the same fire-and-forget cost.
- Read path is one paginated query with `LIMIT/OFFSET`; the `(category, created_at)` and `(actor_user_id, created_at)` indexes cover the common UI queries.

## Source files

- `backend/api/audit.py` — middleware + `write_audit_event` helper
- `backend/api/models.py::AuditLog` — table
- `backend/api/routes/audit.py` — read surface + filters
- `frontend/src/routes/(algo)/admin/audit/+page.svelte` — viewer + pills

## 22.8. Postback fan-out — book\_changed bus

Single coordinated refresh trigger for every position-derived surface after a broker postback. Replaces the prior pattern where each surface polled its own cadence and downstream aggregates (Snapshot grid totals, strategy analytics, payoff curve) lagged the per-cell qty patch by 5–15 s.

🔗 **TECH — Coordinated invalidation vs per-page polling** — **WHY** the postback handler historically invalidated only the `orders` cache. `positions` and `holdings` had their own 30 s TTL, and the strategy endpoint memoised its own analytics — so the Snapshot grid showed patched per-cell qty (via `position_filled` optimistic patch) but stale aggregates until the next per-page poll cycle. Operator's report: "snapshot grid updated two iterations." **WHAT** Terminal postbacks now invalidate every dependent cache atomically + broadcast a single `book_changed` event. A frontend singleton subscriber re-emits via a Svelte store; every position-derived page subscribes once and refetches its primary loader. **HOW** 200 ms upstream debounce coalesces basket-order bursts. Monotonic counter store lets `$effect` re-run on every increment without payload comparison. **WHERE** `backend/api/routes/orders.py::order_postback` + `frontend/src/lib/data/bookChanged.js`.

## Backend chain

`POST /api/orders/postback` on any terminal status (COMPLETE / CANCELLED / REJECTED / EXPIRED):

```

invalidate("orders")
if _terminal:
    invalidate("positions")
    invalidate("holdings")
    broadcast({
        "event": "book_changed",
        "account": masked, "exchange": ..., "trading_symbol": ...,
        "reason": status, "ts": int(time() * 1000),
    })
if status == "COMPLETE":
    broadcast({"event": "position_filled", "qty": signed_delta, ...})

```

`position_filled` is preserved alongside the new event — it carries the qty delta the per-cell optimistic-patch path on Pulse + Performance reads. `book_changed` is the broader coordination signal that also covers CANCELLED / REJECTED paths where there's no qty to patch.

## Frontend bus

`$lib/data/bookChanged.js`:

- Singleton subscriber via `createPerformanceSocket`. Started from `(algo)/+layout.svelte::onMount` so every algo page sees it. Idempotent — multiple `startBookChangedBus()` calls share one WS.
- Listens for `book_changed`, debounces 200 ms, increments `bookChanged` (a monotonic counter writable store) + sets `lastBookEvent` (latest payload).
- Pages subscribe with a `$effect` that watches the counter and calls their primary loader once per increment. Counter pattern (vs payload comparison) lets the effect re-run trivially on every change.

## Subscription map

| Page                            | Loaders called on increment                                |
|---------------------------------|------------------------------------------------------------|
| <code>/admin/derivatives</code> | <code>loadPositions()</code> + <code>loadStrategy()</code> |
| <code>/dashboard</code>         | <code>loadHero()</code>                                    |
| <code>/pulse</code>             | <code>loadPulse()</code>                                   |
| <code>/orders</code>            | <code>_debouncedLoadOrders()</code>                        |
| <code>/performance</code>       | <code>loadAll({ fresh: true })</code>                      |

## Recipe — wire a new page to the bus

```
<script>
  import { bookChanged } from '$lib/data/bookChanged';

  async function loadXxx() { /* page's primary loader */ }

  let _bookCounter = 0;
  $effect(() => {
    const n = $bookChanged;
    if (n <= _bookCounter) return;
    _bookCounter = n;
    loadXxx();
  });
</script>
```

That's it. The counter guard prevents re-entry; the upstream debounce handles burst coalescing.

## Performance contract

- Backend fan-out runs inside the postback handler's existing `_asyncio.create_task` block — zero added latency on the broker ack path.
- One extra JSON broadcast per terminal status (~150 bytes wire). At 10 fills/sec that's 1.5 KB/sec across all WS clients combined.
- Frontend debounce keeps loader calls to one per 200 ms window per page.
- Pages without the wiring fall back to their existing pollers — additive, never breaks the prior path.

## Source files

- `backend/api/routes/orders.py::order_postback` — invalidation chain + broadcasts
- `frontend/src/lib/data/bookChanged.js` — singleton subscriber + stores
- `frontend/src/routes/(algo)/+layout.svelte` — `startBookChangedBus()` callsite
- Wired pages: [admin/derivatives](#), [dashboard](#), [MarketPulse](#), [orders](#), [PerformancePage](#)

## 22.9. History — multi-day orders / trades / funds

`/admin/history` is the row-level forensic surface — three tabs over the platform's append-only datasets: Orders (`algo_orders`), Trades (`daily_book.kind='trades'`), Funds (`daily_book.kind='funds'`). Companion to `/admin/audit` (event-level log); same `view_audit` cap, different storage shape.

🔗 **TECH — Append-only daily\_book vs broker SDK** — **WHY** Kite Connect (and most Indian broker SDKs) expose ONLY today's orders + trades; historical data must be scraped from the Console UI. RamboQuant snapshots every loaded account at 15:35 IST into `daily_book` so the platform owns the multi-day record-of-truth without depending on the broker's UI export. **WHAT** One row per (date, account, kind, symbol) with a unique constraint that lets re-runs upsert idempotently. **HOW** `_task_daily_snapshot` background task fires at 15:35 IST + on startup. **WHERE** `backend/api/algo/daily_snapshot.py` + `backend/api/models.py::DailyBook`.

## Endpoint surface

`backend/api/routes/history.py::HistoryController` — `view_audit` cap, three reads:

| Endpoint                      | Source                    | Default range | Pagination       |
|-------------------------------|---------------------------|---------------|------------------|
| GET /api/admin/history/orders | algo_orders               | 30 days       | 50/page, cap 500 |
| GET /api/admin/history/trades | daily_book[kind='trades'] | 30 days       | 50/page, cap 500 |
| GET /api/admin/history/funds  | daily_book[kind='funds']  | 90 days       | unpaged          |

Shared params: from\_date / to\_date / accounts / symbols (comma-separated lists for accounts + symbols). Orders adds status / mode . Funds drops symbols .

Response shape highlights

- Orders:** counts field is a SQL-side GROUP BY status histogram. UI renders as summary pills without paginating.
- Trades:** summary.total\_notional is  $\sum qty \times avg\_cost$  across the FILTERED set, computed via `_func.sum()` so pagination doesn't degrade accuracy.
- Funds:** earliest\_date is `MIN(daily_book.date) WHERE kind='funds'` — the UI's "tracking started X" chip uses it to set expectations while historical backfill catches up.

Funds capture (new — Jun 2026)

`_funds_rows` — runs alongside the existing holdings / positions / trades capture inside `_task_daily_snapshot` . Per account, per segment (equity / commodity), one row per day. Idempotent via the existing `daily_book` ON CONFLICT clause.

Column mapping (re-using the generic `daily_book` schema to avoid a new table):

| daily_book column | Funds semantic                                           |
|-------------------|----------------------------------------------------------|
| qty               | utilised.debits (₹ debited today)                        |
| avg_cost          | available.cash                                           |
| ltp               | available.opening_balance                                |
| day_pnl           | utilised.realised_m2m                                    |
| total_pnl         | net (segment net worth)                                  |
| symbol            | '__seg__' sentinel (unique constraint requires non-null) |
| exchange          | segment label uppercased                                 |

The mapping is intentionally pragmatic — `daily_book` is denormalised by design, and adding a separate `funds_book` table would duplicate the schema without adding value. The semantics are clear from `kind='funds'` .

Drill, delta, backfill — closed limits

**Per-row audit drill** — closed. `algo_orders.request_id` (nullable VARCHAR(36), indexed) captured on `POST /api/orders/ticket` from `request.scope.state.request_id` stamped by `AuditMiddleware` . `GET /api/admin/audit` accepts a `request_id` filter param; the audit page reads `?request_id=...` URL param on mount + widens `since_hours` to 90 days. History Orders tab grows an `Audit` > column per row that opens `/admin/audit?request_id=<uuid>` pre-filtered.

**Cashbook Δ on Funds tab** — closed. `FundsRow.cash_delta` computed server-side: `HistoryController.list_funds` walks rows in O(N), groups by (account, segment) , sorts ASC by date, sets `prior_cash` to the previous row's `cash_available` each step. Response keeps DESC order (newest first) for the UI; per-row delta carries the move within the (account, segment) series. UI tints positive green / negative red / em-dash for the first row in a series.

**Funds backfill** — endpoint + Dhan adapter both wired.

Adapter contract:

```
def funds_ledger(self, from_date: str, to_date: str) -> list[dict]:
    """Return a list of normalised per-(date, segment) rows:
    [{date, segment, cash_available, opening_balance,
      debits, realised_m2m, net, payload}, ...]
    """
```

Endpoint flow:



```
@post("/funds/backfill", guards=[admin_guard])
async def backfill_funds(...) -> FundsBackfillResponse:
    broker = get_broker(account)
    if not hasattr(broker, "funds_ledger"):
        raise HTTPException(status_code=501, detail=...)
    entries = await loop.run_in_executor(
        None, lambda: broker.funds_ledger(from_iso, to_iso))
    # INSERT ... ON CONFLICT DO UPDATE per entry — same column
    # mapping as _funds_rows in the live snapshot path.
```

Broker support matrix:

- **Kite (zerodha\_kite)** — no programmatic ledger. Always 501.
- **Dhan** — wired (`DhanBroker.funds_ledger`). SDK method discovery probes `get_ledger_report (v2) / get_funds_ledger / ledger_report` (fork variants) with `kwargs`→positional fallback. Aggregates voucher-level entries per `(voucherdate, segment)`; `_DHAN_SEGMENT_MAP` collapses Dhan exchange codes to our 2-segment vocabulary.
- **Groww** — pending. Same single-file pattern: add `funds_ledger(from, to)` to `GrowwBroker` returning the normalised shape.

## ⚙️ TECH — Voucher-level aggregation vs daily snapshot

Dhan's `/v2/statement/ledger` returns voucher-level entries (one per transaction: a trade settlement, a brokerage debit, an MTM credit), not daily summaries. The adapter aggregates because `daily_book[kind='funds']` is intentionally per-day per-segment — re-using the existing snapshot schema instead of adding a `funds_ledger_voucher` table.

Aggregation logic:

- Group entries by `(voucherdate, segment)`.
- Sum `debit + credit` separately per group.
- Track first + last `runbal` as SOD / EOD proxies (Dhan returns entries in chronological order within a day).
- Output `cash_available = close_runbal`, `opening_balance = close_runbal - (credits - debits)`, `realised_m2m = credits - debits` (semantically "net daily cash move", not pure MTM — voucher entries include brokerage / STT / exchange charges that operator should not interpret as P&L).

## Idempotency on backfill

The backfill loop uses `INSERT ... ON CONFLICT (date, account, kind, symbol) DO UPDATE SET ...` — same clause as `_upsert_rows` in the daily snapshot path. Re-running a backfill with a wider date range overwrites existing rows with the canonical Dhan numbers (intentional — if both the live snapshot AND a backfill cover the same day, the backfill's voucher-aggregated numbers are more accurate than a single `broker.margins()` snapshot taken at 15:35 IST).

Per-row try/except + single bulk commit at the end: a single bad voucher entry doesn't lose a multi-month pull. Failed rows log to debug + increment the `skipped` counter; the response surfaces both counts.

## Remaining limit

- **Cashbook view as a separate tab** — running-balance walk that reconciles trade-leg deltas against funds snapshots row by row. The  $\Delta$  column gives the daily move; a dedicated tab could enumerate the trade contributions that produced it. Not in scope for this slice; a follow-up SQL view + 4th tab.

## Source files

- `backend/api/routes/history.py` — controller + 3 endpoints
- `backend/api/algo/daily_snapshot.py` — `_funds_rows` + pipeline wiring
- `backend/api/models.py::DailyBook` — table (unchanged; just a new `kind` value)
- `frontend/src/routes/(algo)/admin/history/+page.svelte` — viewer
- `frontend/src/lib/api.js` — `fetchHistoryOrders/Trades/Funds` wrappers

## 22.10. Order placement latency — preflight + tick cache + paper-skip

Closes the order-placement deterioration the operator flagged ("placement feels slow now"). Three orthogonal fixes ship together because they target the same hot path:

⚙️ **TECH — Sequential awaits vs `asyncio.gather`** — **WHY** `await` on a `run_in_executor` blocks the route until the broker SDK returns; four sequential awaits = ~800-1200ms on Kite's typical 200-300ms round-trip. Even though each call is itself `async`-scheduled, the `await` chain serializes them. **WHAT** Wrap each independent broker call in its own helper coroutine with self-contained exception handling, then fire all four with `asyncio.gather`. Total wall-time becomes `max(individual call)` instead of `sum(individual calls)`. **HOW** Each helper returns a plain Python value on success or a sentinel (`None` / tuple-with-error) on failure, so the consumer can branch on result type rather than handle exceptions



across the gather boundary. WHERE `backend/api/algo/actions.py::run_preflight` — fan-out helpers `_fetch_profile` / `_fetch_instruments` / `_fetch_basket_margin` / `_fetch_account_margins`.

## Fix 1 — preflight parallelization

`run_preflight` previously ran four broker calls in strict sequence:

```
profile = await loop.run_in_executor(None, broker.profile)           # ~300ms
instruments = await loop.run_in_executor(None, broker.instruments, exchange) # ~250ms
bm_result = await loop.run_in_executor(None, broker.basket_order_margins, ...) # ~300ms
m = await loop.run_in_executor(None, broker.margins)                 # ~300ms
# Total: ~1150ms sequential
```

The new structure:

```
# Stage 1 (synchronous) — build basket_orders (uses cached get_lot_size).
# Stage 2 (parallel) — gather the 4 independent broker calls:
profile_res, instruments_res, bm_res, margins_res = await asyncio.gather(
    _fetch_profile(),           # None for non-Kite brokers
    _fetch_instruments(),       # None when exchange not in F&O / qty<=0
    _fetch_basket_margin(),     # Exception on failure (handled below)
    _fetch_account_margins(),   # (seg_dict, error_str) tuple
)
# Total: ~max(300ms) parallel
```

Each helper handles its own exceptions so a single broker failure surfaces as a logged warning + None result rather than tearing down the gather. `_fetch_basket_margin` returns the exception object (not raising) so the consumer can re-raise into the existing `MARGIN_SHORTFALL` block's `try/except` — minimal change to the downstream handler.

## Fix 2 — tick-size index

`_align_price_to_tick` looked up the contract's `tick_size` via a linear scan:

```
for inst in items:           # items = 10-50k rows
    if inst.s == sym_u and inst.e == ex_u:
        tick = float(inst.ts or 0)
        break
```

Ticket route called this twice per order (price + trigger), so a single ticket paid ~100k linear iterations.

Now a module-level `_TICK_INDEX: dict[tuple[str, str], float]` is built lazily from the instruments cache. `_TICK_INDEX_STAMP` holds the cached `InstrumentsResponse` object; identity comparison (`resp is not _TICK_INDEX_STAMP`) detects cache refresh and triggers a rebuild. Subsequent ticket calls are O(1) dict lookups.

Trade-off: the rebuild itself is still O(N) — one scan per cache refresh (typical TTL ~10 minutes). Hot-path savings dominate; ~50ms per ticket recovered.

## Fix 3 — PAPER skips route-level preflight

`PaperTradeEngine.register_open_order` already runs `basket_order_margins` internally — it's the gate that decides REJECTED vs OPEN on an open order, and writes the broker's exact error string into `AlgoOrder.detail` when the basket margin check fails. The route-level preflight before that was running the SAME `basket_margin` call (plus three others) for the SAME order, costing ~800ms with zero additional correctness.

The PAPER branch of `ticket_order` no longer calls `run_preflight()`. LIVE preflight stays — it's the only chance to block before `kite.place_order` actually fires.

## Combined ticket-path savings

| Path         | Before                                                           | After                                                        |
|--------------|------------------------------------------------------------------|--------------------------------------------------------------|
| LIVE ticket  | ~1200ms preflight + ~150ms route + ~300ms place_order ≈ 1.65s    | ~300ms preflight + ~150ms route + ~300ms place_order ≈ 0.75s |
| PAPER ticket | ~1200ms preflight + ~150ms route + ~50ms engine register ≈ 1.40s | ~150ms route + ~50ms engine register ≈ 0.20s                 |

PAPER is the bigger win because the entire preflight goes away; LIVE saves roughly half the latency.

## Source files

- `backend/api/algo/actions.py::run_preflight` — parallel gather
- `backend/api/routes/orders.py::_align_price_to_tick` + `_TICK_INDEX` — O(1) tick lookup
- `backend/api/routes/orders.py::ticket_order` — PAPER preflight skip block

## 22.11. Navbar audit — rename + resequence

Operator-requested audit of the algo navbar.

### Renames:

- `modes` group → `explore`. The old name was vestigial from the `sim/paper/live/shadow/replay` terminology before the mode toggles moved to the navbar dropdown (Wave C). Group now contains just `Sandbox`; can grow when `Replay` gets its own dedicated entry.
- `Lab` label → `Sandbox`. Industry-standard term across `QuantConnect / Streak / Sensibull`; reads faster to first-time visitors than the prior internal jargon. URL `/admin/execution` unchanged — bookmarks + deep links preserved.

**Monitor resequence** (rationale = daily-trader workflow frequency):

```
old: Tour Pulse Dashboard Derivatives Strategies NAV Orders Charts Automation
new: Tour Pulse Dashboard Orders      Derivatives Charts Automation Strategies NAV
```

Orders moved ahead of analysis surfaces (Derivatives / Charts) since active trading is the trader's primary entry point. Strategies + NAV move to the end — attribution + LP-facing views are weekly, not minute-by-minute.

### Implementation:

```
// frontend/src/routes/(algo)/+layout.svelte
const GROUP_LABELS = {
  monitor: 'Monitor',
  analyze: 'Analyze',
  explore: 'Explore',
  build:   'Build',
  config:  'Config',
};
const INLINE_GROUPS = new Set(['monitor', 'analyze', 'explore']);
```

`INLINE_GROUPS` controls which groups render inline in the desktop nav; the rest collapse to dropdown triggers. Mobile drawer shows every group with a `GROUP_LABELS` caption for scan-by-intent navigation.

## 22.12. #audit workflow + Dhan / Groww postback scaffold

The operator runs periodic comprehensive audits by writing the literal hashtag `#audit` in chat. The coding agent dispatches 8 parallel `audit` subagents (one per dimension: performance / defects / stale code / UX consistency / palette / broker-API parity / data layer / docs) and synthesizes findings into a severity-tagged punch list. Audit findings ship as `audit slice <letter>` commits.

🔗 **TECH — Parallel audit subagents vs single deep review** — **WHY** A single agent asked to cover 8 dimensions produces shallow work on each. Eight focused subagents each get a tight scope brief, run concurrently, and report independently. **WHAT** 8 Agent tool-calls in one message with `subagent_type=audit` (read-only), each prompt cites the canonical patterns to check against. **HOW** Each subagent returns a focused punch list with severity tags (HIGH / MED / LOW); the coordinator synthesizes into a remediation plan rather than firing fixes unilaterally. **WHERE** Memory file `~/claude/projects/-Users-ramanambore-projects-ramboq/memory/feedback_audit_tag.md` documents the workflow.

### Audit slices A, B, C (shipped Jun 2026)

**Slice A — quick wins:** doc drift in `CLAUDE.md` + `README.md` (navbar renames `modes` → `explore`, `Lab` → `Sandbox`, monitor sequence); stale code purges (`OrderDetail.svelte`, `margin_optimizer.py`, `shadow.py` + `ShadowController`, 4 `api.js` shadow stubs, `fetchPnlRange` — all confirmed unreferenced); palette fixes (LogPanel border `#10b981` → `#4ade80`; `RefreshButton` badges 600-level→400-level).

### Slice B — perf + data layer:

- `_task_performance` ticker subscribe loop, `_task_trail_stop`, `_task_oco_pair_watcher`: sequential per-account awaits → `asyncio.gather`. Each saves ~200-300ms × N accounts.
- `get_lot_size`  $O(N)$  →  $O(1)$  via `_LOT_INDEX` dict (same identity-stamp invalidation pattern as `routes/orders.py::_TICK_INDEX`).
- 3 DB indexes added in `init_db` migration block (idempotent `CREATE INDEX IF NOT EXISTS`):
  - `ix_algo_orders_trail_stop` — partial on (mode, status) WHERE attached\_gtts\_json IS NOT NULL
  - `ix_news_headlines_published_at` — DESC
  - `ix_strategy_lots_open` — composite missing from `create_all` on pre-existing tables
- Operator-visible interrupt fixes batched in:
  - `/admin/history` + `/admin/audit`: \$effect-gated load (was `onMount` checking `_canView` once at false; load never fired)
  - Algo layout: removed blanket "non-admin → /signin" redirect (vestigial from old admin/partner tier; broke tour for trader/risk/admin roles — pre-rename names were ops/observer)

- "Prev Close" → "Close" rename across PerformancePage, MarketPulse, /admin/derivatives

### Slice C — defects + Dhan/Groww postback scaffold:

- `list_active_chases` template-attach gap: live-reconcile path flipped FILLED but never called `_maybe_fire_template_attach_for_reconcile`. Now captures `_reconciled_filled` rows + fires after commit.
- `chase.py:806` partial-fill slippage formula: `quantity` → `filled_qty` or `quantity` (the old formula overstated slippage when partials happened).
- New routes `POST /api/orders/{dhan,groww}_postback` with shared `_process_broker_postback` helper that mirrors the Kite path's fan-out (AlgoOrder sync + audit log + cache invalidate + WS broadcasts). Best-effort; logs raw payload on first hit so parser can be tuned.

### `_process_broker_postback` shared helper

`backend/api/routes/orders.py::_process_broker_postback` — extracted from the Kite postback inline logic so Dhan + Groww routes call the same fan-out:

```
async def _process_broker_postback(
    *, broker_id, order_id, status, account, symbol, txn, qty, price,
    exchange="", status_message="",
):
    # 1. AlgoOrder row sync by broker_order_id (status, fill_price, filled_at)
    # 2. order_events row (broker_postback kind)
    # 3. audit_log entry tagged order.fill|cancel|reject
    # 4. invalidate orders / positions / holdings on terminal
    # 5. broadcast order_update + position_filled + book_changed
```

Status normalization uses `_DHAN_STATUS_TO_KITE` and `_GROWW_STATUS_TO_KITE` tables already in the broker adapters. Best-effort throughout; never 5xx so the broker doesn't retry.

### Source files

- `backend/api/algo/actions.py::run_preflight` — parallel gather (slice 22.10)
- `backend/api/background.py::_task_{performance, trail_stop, oco_pair_watcher}` — slice B parallel gathers
- `backend/brokers/adapters/kite.py::get_lot_size` + `_LOT_INDEX` — slice B `O(1)` lookup
- `backend/api/database.py::init_db` — slice B index migrations
- `backend/api/routes/orders.py::list_active_chases` — slice C template-attach fix
- `backend/api/algo/chase.py:806` — slice C slippage formula
- `backend/api/routes/orders.py::order_postback_{dhan,groww}` + `_process_broker_postback` — slice C postback scaffolds

## 22.13. Audit slice D — UX consistency + palette consolidation + 2 defects

Closes the remaining MED-severity items from the #audit run. Pattern: converge on canonical components + canonical CSS-custom-prop alphas rather than introducing new code.

### UX consistency — adopting canonical components

Three pages were running their own bespoke implementations of patterns the codebase already had canonical components for. Each replacement deletes the bespoke chrome + the CSS that supported it:

| Page                                  | Bespoke pattern                                                                          | Canonical replacement                                                                                    |
|---------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <code>/admin/history</code>           | <code>.hist-tabs</code> + <code>.hist-tab</code> (+ active border + count badge)         | <code>&lt;AlgoTabs tabs={[{id, label, badge}, ...]} bind:value={tab} onChange={setTab} /&gt;</code>      |
| <code>/admin/statements</code>        | native <code>&lt;select class="ms-select"&gt;</code> + <code>.ms-select</code> CSS block | <code>&lt;Select bind:value={selectedPeriod} options={_months} /&gt;</code>                              |
| PageHeaderActions amber+cyan hover bg | <code>rgba(_, 0.12)</code> (drift)                                                       | <code>rgba(_, 0.14)</code> matching <code>--algo-amber-bg</code> / <code>--algo-cyan-bg</code> canonical |

🔗 **TECH — Canonical-component adoption vs maintaining bespoke chrome** — **WHY** Every bespoke tab strip / select / pill the operator can't visually distinguish from the canonical ones is friction on muscle memory. Slice D's audit found 4 parallel pill-strip implementations across audit / statements / templates / agent-templates pages — same job, four different CSS classes. **WHAT** Replace bespoke implementations with the existing canonical component when the contract matches; the canonical component does the styling once and every page inherits. **HOW** `AlgoTabs` already exposes `tabs: [{id, label, badge, color}]` so the History tab badge logic (showing total only on the active tab) maps trivially. `Select` accepts `options: [{value, label}]` so the period dropdown's already-correct data shape is a one-line swap. **WHERE** `frontend/src/lib/AlgoTabs.svelte`, `frontend/src/lib/Select.svelte`.

## Palette alpha consolidation

The cyan-bg alpha `0.10` was drifting across 5 files (8 callsites) while the canonical `--algo-cyan-bg` is `0.14`. `CommandBar` / `OrderCard` / `OrderTicket` / `SymbolPanel` all converged. Same fix on `PageHeaderActions` amber `0.12` → `0.14`.

Net effect: a chip background on one page now visually matches the same conceptual element on another page. Pre-fix the operator's brain had to disambiguate "is this cyan-12 or cyan-14?" — now both reach `var(--algo-cyan-bg)`.

## Defects

`paper.py::reset()` **race** — pre-fix the three dict-replace operations (`_open_orders`, `_price_history`, `_underlying_history`) ran unlocked. If a concurrent `step()` snapshot or `_capture_price_history` write happened during the replace, the price-history chart data for the first tick of a new sim could silently land in the OLD dict reference while the new sim queried the empty replacement. Fix: acquire `self._lock` around the replacements.

`history.py::backfill_funds` **docstring** — was claiming "idempotent — existing rows are not overwritten (ON CONFLICT DO NOTHING)" but the SQL is `DO UPDATE SET ...` (full overwrite). Operator-surprise risk: a hand-edit to a funds row gets clobbered on the next backfill with the same date range. Docstring now accurately documents the overwrite + warns the operator to treat funds rows as read-only.

## Source files

- `frontend/src/routes/(algo)/admin/history/+page.svelte` — `<AlgoTabs>` adoption
- `frontend/src/routes/(algo)/admin/statements/+page.svelte` — `<Select>` adoption
- `frontend/src/lib/PageHeaderActions.svelte` — amber/cyan `0.12`→`0.14`
- `frontend/src/lib/{CommandBar,SymbolPanel}.svelte` + `order/{OrderCard,OrderTicket}.svelte` — cyan `0.10`→`0.14`
- `backend/api/algo/paper.py::reset` — `self._lock` acquisition
- `backend/api/routes/history.py::backfill_funds` — docstring correction

## 22.14. Market-status — broker API beats bellwether-quote probe

The agent engine's `market_hours` schedule gate and the daily snapshot pipeline both consult `probe_market_active(exchange)` to decide whether the market is currently trading. Pre-slice-E the probe used a workaround: call `kite.quote()` on bellwether symbols (NIFTY 50 + NIFTY BANK for NSE/NFO, SENSEX for BSE/BFO, or the dynamically-resolved nearest MCX commodity futures contract — not hardcoded CRUDEOIL), check `last_trade_time` freshness within a 15-minute window. Worked, but spent Kite's quote budget on a question Kite's API can't answer directly.

🔗 **TECH — Authoritative broker API vs inferred bellwether probe** — **WHY** Kite Connect has no market-status endpoint; the only signal Kite exposes is a quote with `last_trade_time`. Dhan and Groww both ship a direct market-status API (`get_market_status` and variants). When an authoritative answer is one round-trip away, prefer it over an inferred one — bellwether probes have edge cases (illiquid contracts, weekend Muhurat sessions, MCX evening sessions) where the inference disagrees with the broker. **WHAT** Extend the `Broker ABC` with an optional `market_status(exchange) → bool | None` method. Adapters that have the API override and return `True/False`. The probe layer iterates brokers, takes the first definitive answer, falls back to the bellwether path when no broker answers. **HOW** SDK-method discovery probes (`getattr(sdk, 'get_market_status', None)` etc.) handle adapter version drift. Per-exchange 60s cache absorbs the per-tick gate evaluation. **WHERE** `backend/brokers/base.py::market_status`, `backend/brokers/adapters/dhan.py::market_status`, `backend/brokers/adapters/groww.py::market_status`, `backend/shared/helpers/market_probe.py::probe_market_active`.

## Resolution order

```
probe_market_active(exchange):
    if cache hit and fresh (60s TTL): return cached
    for broker in all_brokers():
        verdict = broker.market_status(exchange)  # ← step 1
        if isinstance(verdict, bool):
            cache[exchange] = verdict
            return verdict
    # step 2: fall back to bellwether-quote probe (unchanged path)
    kite = resolve_kite_handle()
    if kite is None: return None
    bellwethers = _candidates(exchange, broker)
    q = kite.quote(bellwethers)
    active = any(row.last_trade_time >= now - 15min for row in q)
    cache[exchange] = active
    return active
```

## Adapter contract

```
class Broker(ABC):
    def market_status(self, exchange: str) -> bool | None:
        """True / False if broker exposes a market-status endpoint
        for `exchange`; None when adapter doesn't implement one or
        the call fails. Optional method, not abstract."""
        return None
```

Both Dhan and Groww implementations probe known SDK method names ( `get_market_status` / `market_status` / `get_exchange_status` ) across SDK version drift; iterate the response rows (whatever shape the SDK returns); map our exchange vocabulary (NSE / BSE / NFO / BFO / CDS / MCX) to the broker's segment codes (Dhan: `NSE_EQ` / `BSE_EQ` / `NSE_FNO` / `BSE_FNO` / `NSE_CURRENCY` / `MCX_COMM` ; Groww: similar with variants); return `True` if ANY mapped segment reports active.

## Side effects on quote budget

A typical Kite-only deployment hits the bellwether path on every cache miss → 4 symbols × 1 quote call ≈ 4 instruments off the 10-req/sec quote budget per probe. A Dhan-loaded deployment skips that entirely for any exchange Dhan covers; only the rare cache-miss-with-Dhan-down case falls through.

## Adding the API to a new adapter

```
def market_status(self, exchange: str) -> bool | None:
    sdk = self.client
    status_fn = (getattr(sdk, "get_market_status", None)
                 or getattr(sdk, "market_status", None))
    if status_fn is None:
        return None
    try:
        resp = self._safe_call(lambda c: status_fn())
    except Exception as e:
        logger.debug(f"{self.broker_id}.market_status failed: {e}")
        return None
    # map your broker's segment codes → our vocabulary
    # return True if any mapped segment reports active
    # return False if all closed
    # return None if no mapping matched (fall through to next broker)
```

## Source files

- `backend/brokers/base.py::market_status` — ABC default
- `backend/brokers/adapters/dhan.py::market_status` — Dhan implementation
- `backend/brokers/adapters/groww.py::market_status` — Groww implementation
- `backend/shared/helpers/market_probe.py::probe_market_active` — resolution chain + cache

## 22.15. Chart indicator system — pure module + overlay persistence

Technical indicators (SMA, EMA, VWAP, Bollinger Bands, RSI, MACD) live in a single pure module rather than being inlined in ChartWorkspace.

🔗 **TECH — Indicators as a pure stateless module** — **WHY** Inline math inside a 2000-line Svelte component is untestable with `node --test` (no DOM, no Svelte runtime needed). Extracting the math to a pure module means a 32-test suite can verify hand-calculated reference values, edge cases (empty arrays, N=0, constant series), and Wilder-smoothing correctness without a browser. **WHAT** `frontend/src/lib/chart/indicators.js` exports `sma`, `ema`, `vwap`, `bollinger`, `rsi`, `macd`. Each function takes an OHLCV bars array, returns a typed series array. First (n-1) entries are `{ts, value: null}` — warmup convention. **HOW** `_assertN(n)` throws `RangeError` for non-positive or non-integer periods. `MACD` throws when `fast >= slow`. All functions are pure (no side-effects, no imports). **WHERE** `frontend/src/lib/chart/indicators.js`; tests at `frontend/scripts/indicators.test.js`.

**Overlay palette** (canonical colours, do not vary):

| Overlay        | CSS class       | Colour                                         | Notes                       |
|----------------|-----------------|------------------------------------------------|-----------------------------|
| SMA 20         | overlay-sma     | #7dd3fc sky-blue                               | dashed 4-3                  |
| SMA 50         | overlay-sma     | #c084fc violet                                 | dashed 6-3                  |
| EMA 20         | overlay-ema     | #4ade80 green                                  | solid                       |
| EMA 50         | overlay-ema     | #fb923c orange                                 | dashed 6-3                  |
| VWAP           | overlay-vwap    | #7dd3fc cyan                                   | solid 1.4px                 |
| BB mid         | overlay-bb      | #7dd3fc cyan                                   | solid 1px                   |
| BB upper/lower | overlay-bb      | #7dd3fc cyan                                   | dashed 3-2                  |
| BB fill        | overlay-bb      | rgba(125,211,252,0.06)                         | no stroke                   |
| RSI line       | overlay-rsi     | #fbbf24 amber                                  | solid 1.5px                 |
| MACD line      | overlay-macd    | #fbbf24 amber                                  | solid 1.4px                 |
| MACD signal    | overlay-macd    | #f87171 red                                    | dashed 3-2                  |
| MACD histogram | (line elements) | rgba(74,222,128,0.55) / rgba(248,113,113,0.55) | green above zero, red below |

**Sub-panel geometry** (SVG user-unit constants in ChartWorkspace.svelte):

- `RSI_H` = 48 — RSI panel height
- `MACD_H` = 56 — MACD panel height
- `_bandH` = (`_showRsi` ? `RSI_H` : 0) + (`_showMacd` ? `MACD_H` : 0) — reserved bottom space
- `_innerH` = `chartH` - `CPAD_T` - `CPAD_B` - `_bandH` — usable height for the price panel

**Overlay persistence:**

- LocalStorage key: `rbq.cache.chart-overlays.v1` (JSON array of string keys)
- Init: `$state([])` — empty server-side. Never `$state(_loadPrefs())` because `$state()` init runs during SSR where `localStorage` is undefined.
- Hydration: `onMount` reads and validates the stored array against `_OVERLAY_OPTS`. Sets `_overlaysHydrated` = `true` after reading.
- Save: `$effect` watches `_overlays` but guards with `if (!_overlaysHydrated) return` to prevent overwriting stored prefs during the first render frame.

**VWAP note** — indices (NIFTY 50, NIFTY BANK etc.) carry `volume=0` on every bar. `calcVwap()` returns `null` for all points when `cumVol=0`. The `{#if _vwapPath}` block in the SVG template silently suppresses the element. This is correct: VWAP is a price/volume metric that has no meaning for non-tradeable indices.

**Buy / sell signal markers**

⚙️ **TECH — Signal detection as pure helpers, render layer pure SVG** — **WHY** Operator brief: surface buy/sell points TradingView-style for each active indicator. Detection logic must be testable ( `node --test` ) and the marker layer must respect the canonical algo palette so the visual vocabulary matches the rest of the app. **WHAT** Five exports in `indicators.js` — `emaSignals(fast, slow)`, `vwapSignals(closes, vwapArr)`, `bollingerSignals(closes, bb)`, `rsiSignals(arr, oversold=30, overbought=70)`, `macdSignals(macdLine, signalLine)`. Each returns `[{i, type:'buy'|'sell'}]`. Inputs are duck-typed — they accept raw number arrays, `{value}` arrays (real ema output), or `{close}` bars (raw OHLCV). **HOW** In `ChartWorkspace`, `_signalMarkers` is a `$derived.by` that re-runs only when `_bars` or `_overlays` change; `_signalLayout` translates events to `{x, y, type, tag, tooltip, stack}` records with same-bar stacking. SVG renders one `<g class="signal-marker signal-{type}">` per event with a triangle + 9px monospace tag. **WHERE**  
`frontend/src/lib/chart/indicators.js::emaSignals|vwapSignals|bollingerSignals|rsiSignals|macdSignals ;`  
`frontend/src/lib/ChartWorkspace.svelte::_signalMarkers|_signalLayout .`

**Marker palette + geometry** (canonical, do not vary):



| Element                    | Spec                                                                                                         |
|----------------------------|--------------------------------------------------------------------------------------------------------------|
| Buy triangle               | filled #4ade80 emerald-400, 10×8 px, anchored at bar's low + 8 px pad, tip-up                                |
| Sell triangle              | filled #f87171 red-400, 10×8 px, anchored at bar's high – 8 px pad, tip-down                                 |
| Stroke                     | #0a0a0a 0.5 px (subtle outline so triangles read against the chart background tint)                          |
| Tag font                   | 9 px monospace, weight 700, paint-order stroke-then-fill with 2.5 px dark stroke for legibility against bars |
| Tag colour                 | matches triangle ( #4ade80 buy, #f87171 sell)                                                                |
| Stack offset               | 16 px vertical between markers on the same bar (split: buys below, sells above)                              |
| Indicator tag text         | EMA↑ / EMA↓ / VWAP↑ / VWAP↓ / BB↑ (buy = lower band) / BB↓ (sell = upper band) / RSI↑ / RSI↓ / MACD↑ / MACD↓ |
| Tooltip ( <title> element) | Buy signal – RSI 14 @ 2026-04-15 (verb + indicator + bar timestamp)                                          |
| Density throttle           | per-indicator cap of 12 events on dense ranges ( _bars.length >= 180 ); most-recent events kept              |
| Bollinger throttle         | first bar of a contiguous lower / upper band run only — prevents 5-marker spam on multi-bar breaks           |

**Signal detection rules** (peer-platform standard — TradingView / Sensibull / Streak / Upstox):

| Indicator    | Buy                                                  | Sell                                                |
|--------------|------------------------------------------------------|-----------------------------------------------------|
| EMA cross    | fast > slow AND prev fast ≤ prev slow (golden cross) | fast < slow AND prev fast ≥ prev slow (death cross) |
| VWAP         | close > vwap AND prev close ≤ prev vwap              | close < vwap AND prev close ≥ prev vwap             |
| Bollinger    | close ≤ lower band (first bar)                       | close ≥ upper band (first bar)                      |
| RSI 14       | rsi > 30 AND prev rsi ≤ 30                           | rsi < 70 AND prev rsi ≥ 70                          |
| MACD 12/26/9 | macd > signal AND prev macd ≤ prev signal            | macd < signal AND prev macd ≥ prev signal           |

**Toggle UX** — Signals chip in chart toolbar renders only when `_overlays.length > 0` (no markers to show without an indicator). Default ON, persisted to `localStorage` key `rbq.cache.chart-signals.v1`. Same height + active-state palette (cyan-400) as the Intraday chip — toolbar height SSOT ( `--chart-toolbar-h` ) preserved.

Source files

- `frontend/src/lib/chart/indicators.js` — pure indicator + signal functions
- `frontend/src/lib/chart/paths.js` — SVG path builders extracted from `ChartWorkspace`: `smaPath`, `emaPath`, `vwapPath`, `bbPaths`, `rsiSeries`, `macdSeries`. Pure functions; zero dependencies.
- `frontend/src/lib/ChartWorkspace.svelte` — imports `calcEma`, `calcVwap`, `calcMacd`, `calcBollinger`, `calcRsi`, + 5 signal helpers; uses `paths.js` functions to render SVG; all overlay paths and `_signalMarkers` / `_signalLayout` are `$derived`
- `frontend/scripts/indicators.test.js` — 52-test unit suite ( `node --test` ) covering indicator math + 5 signal helpers
- `frontend/e2e/chart_overlays.spec.js` — indicator paths Playwright spec
- `frontend/e2e/chart_signals.spec.js` — buy/sell markers Playwright spec (chromium-desktop + mobile-portrait)

22.16. Derivatives page — cold-start and payoff improvements

Three improvements accelerate page load + rendering fidelity on the `/admin/derivatives` page:

**Picker ordering** — `_getCandidates()` now prioritizes symbols in descending order of relevance: options positions → futures positions → holdings → popular underlyings. On page mount, the first candidate is auto-selected, ensuring the most-likely underlying is prefilled without waiting for candidates to load.

**Cold-start seed** — Page no longer waits for `await loadInstruments()` to complete before rendering. Instead, `OptionsPayoff.svelte` is seeded immediately with `POPULAR_UNDERLYINGS[0]` (e.g. NIFTY 50), so the payoff chart renders instantly. When instruments arrive, the page resolves to the operator's preferred symbol or re-runs `loadStrategy()`.

**Stub today\_value** — `_clientPayoffStub` in `backend/api/routes/derivatives.py` now returns `today_value: null` instead of `expiry_value`. This prevents the frontend's "today" curve from rendering with an incorrect negative slope before Black-Scholes pricing arrives. `OptionsPayoff.svelte` skips the today curve entirely when `hasTodayValues=false`.

Files:

- `frontend/src/routes/(algo)/admin/derivatives/+page.svelte` — picker ordering + cold-start seed
- `frontend/src/routes/(algo)/admin/derivatives/CandidateLegRow.svelte` — extracted leg-grid row component (841 lines, 9 props, 6 callbacks)
- `frontend/src/lib/OptionsPayoff.svelte` — `hasTodayValues` check
- `backend/api/routes/derivatives.py` — stub `today_value` change

## 22.17. Derivatives page — `_throttledTick` market-close gate

`_throttledTick` is a counter that drives all `$derived.by` blocks on the derivatives page (`liveSpot`, `_clientPayoffStub`, `_legsExpPnlTotal`, `_expiryProfit`, etc.). Each increment triggers a payoff chart re-render.

**Bug:** Pre-fix, the counter incremented on every `symbolTickCount` event unconditionally — even during closed hours when `MarketPulse`'s reduced-cadence WebSocket poll still fires. Each increment caused `_clientPayoffStub` to produce a new array reference, forcing `OptionsPayoff.svelte` to re-animate continuously overnight (expensive + misleading).

**Fix:** `_throttledTick++` is now gated behind `if (isMarketOpen())` in both `derivatives/+page.svelte` and `PositionStrip.svelte`. During closed hours, the counter stays frozen. The edge-detect clock in `marketAwareInterval` fires `loadStrategy()` once on the closed→open transition.

### Files:

- `frontend/src/routes/(algo)/admin/derivatives/+page.svelte` — market-close gate
- `frontend/src/lib/PositionStrip.svelte` — same gate for consistency

## 22.18. NavStrip — lifetime slot SSOT aligned with MarketPulse TOTAL

**Problem:** P pill slot 2 (lifetime P&L) and H pill slots 2–3 (holdings value / lifetime P&L) all had an SSE tick-delta added on top of the raw broker-snapshot sum. `MarketPulse` positions and holdings grids compute their TOTAL rows as `Σ row.pnl` (raw snapshot), producing a different number for the same data.

**Root cause:** `_livePositionsPnl`, `_liveHoldingsTotal`, and `_liveHoldingsValue` in `PositionStrip.svelte` each added `_positionsDelta` / `_holdingsDelta` to the snapshot sum. Only slot 1 (today Day P&L) should carry the SSE delta; lifetime values are broker-snapshot aggregates and must not accumulate intraday ticks.

**Fix:** Removed the delta from all three lifetime derived values. Only `_liveDayPnl` (P slot 1) retains the delta so the intraday position responds immediately to LTP moves.

**SSOT invariant:** `NavStrip` lifetime values  $\equiv \Sigma \text{row.pnl}$  in `MarketPulse` TOTAL row (within `aggCompact` rounding,  $\leq 0.1\%$ ). Day P&L (slot 1) may diverge up to 2% during open hours due to SSE delta.

### Test coverage:

- `frontend/e2e/test_navstrip_consistency.spec.js` — P slot 2 matches TOTAL within 0.1%; slot 1 within 2%
- `frontend/e2e/test_navstrip_frozen.spec.js` — closed-hours snapshot freeze

### Files:

- `frontend/src/lib/PositionStrip.svelte` — lines 415–461, three fixed `$derived.by` blocks

## 22.19. CSV export button — all ag-Grid card headers

Every data card across the app now has a one-click CSV download button.

**Component:** `GridDownloadButton.svelte` — 1.4 rem × 1.4 rem, `var(--algo-cyan-bg)` palette, down-arrow SVG. Self-hides when `onClick` is null. Sibling CSS eliminates the left margin gap produced by `GridSearchButton` + `GridDownloadButton` adjacency.

**CardControls integration:** `CardControls.svelte` gained an `onDownload = null` prop. When non-null, the button renders between Search and Collapse. Canonical toolbar order across all cards:

```
Refresh · Search · Download · Collapse · DefaultSize · Fullscreen
```

### Coverage by surface:



| Surface         | Card                      | Export target                         |
|-----------------|---------------------------|---------------------------------------|
| Pulse           | Pinned/Watchlist          | active tab grid                       |
| Pulse           | Winners / Losers          | gridWin / gridLose                    |
| Pulse           | Positions                 | gridPositions                         |
| Pulse           | Holdings                  | gridHoldings                          |
| Dashboard       | Capital (Funds + Margins) | _fundsGrid then _marginGrid           |
| Dashboard       | Equity (Pos + Hold)       | _eqPosGrid then _eqHoldGrid           |
| PerformancePage | NAV/Funds strip           | navGrid or fundsGrid (tab-aware)      |
| PerformancePage | Positions Summary         | positionsSummaryGrid                  |
| PerformancePage | Positions Breakdown       | positionsAllGrid                      |
| PerformancePage | Holdings Summary          | holdingsSummaryGrid                   |
| PerformancePage | Holdings Breakdown        | holdingsAllGrid                       |
| NavBreakdown    | NAV breakdown             | hand-rolled via exportRowsToCsv       |
| Derivatives     | Legs card                 | hand-rolled via exportRowsToCsv       |
| Derivatives     | Snapshot card             | hand-rolled via exportRowsToCsv       |
| Derivatives     | Payoff card               | excluded (SVG chart, no tabular data) |

**Hand-rolled grid utility:** `frontend/src/lib/utils/csvExport.js` exports `exportRowsToCsv(rows, columns, filename)`. Accepts a column descriptor array `{header, key, format?}`, builds RFC 4180 CSV blob, triggers browser download. Used for NavBreakdown, Derivatives Legs, and Derivatives Snapshot (none of which use ag-Grid).

**Files:**

- `frontend/src/lib/GridDownloadButton.svelte` — new component
- `frontend/src/lib/CardControls.svelte` — `onDownload` prop
- `frontend/src/lib/utils/csvExport.js` — new hand-rolled export utility
- `frontend/src/lib/MarketPulse.svelte` — Pulse card wiring
- `frontend/src/routes/(algo)/dashboard/+page.svelte` — Dashboard wiring
- `frontend/src/lib/PerformancePage.svelte` — PerformancePage wiring
- `frontend/src/lib/NavBreakdown.svelte` — NavBreakdown wiring
- `frontend/src/routes/(algo)/admin/derivatives/+page.svelte` — Derivatives wiring

22.20. docs/ folder reorganisation + exhaustive spec files

**Docs structure:** All markdown documentation (except `CLAUDE.md`, `CLAUDE_HISTORY.md`, `README.md`) moved to `docs/` :

| Path                              | Contents                                                                                                                                                     |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>docs/specs/</code>          | Feature behavioral contracts — <code>PULSE_SPEC</code> , <code>BROKER_SPEC</code> , <code>NAVSTRIP_SPEC</code> , ...                                         |
| <code>docs/guides/</code>         | Operator guides — <code>USER_GUIDE</code> , <code>ADMIN_GUIDE</code> , <code>AGENTS_GUIDE</code> , <code>LAB_MCP_GUIDE</code> , <code>SIMULATOR_GUIDE</code> |
| <code>docs/audits/</code>         | Point-in-time audit snapshots                                                                                                                                |
| <code>docs/DESIGN_GUIDE.md</code> | This file                                                                                                                                                    |
| <code>docs/MIGRATION.md</code>    | DB migration history                                                                                                                                         |
| <code>docs/deployment.md</code>   | Ops runbook                                                                                                                                                  |

**Spec files (21 total):** Every major surface now has a behavioral contract spec in `docs/specs/`. Each spec documents: API endpoint signatures and params, frontend store variables ( `$state` / `$derived` / `$effect` ), explicit SSOT chain for every displayed value, edge cases, and test coverage map.

Spec files are **not** auto-loaded — Claude reads them explicitly when working on or testing the relevant surface. They are the source of truth for expected behaviour and complement `CLAUDE.md` (which describes how to build) with what each surface does.

| Spec             | Surface                                            |
|------------------|----------------------------------------------------|
| ACTIVITY_SPEC    | Activity log modal / surface                       |
| AGENTS_SPEC      | Agent engine — condition tree, actions, grammar    |
| AUDIT_SPEC       | Audit log + history page                           |
| AUTOMATION_SPEC  | Automation page — agents, templates, tokens        |
| BROKER_SPEC      | Broker connection layer                            |
| CHART_SPEC       | Chart workspace — OHLCV, indicators, signals       |
| DASHBOARD_SPEC   | Dashboard — hero metrics, grids, agent log         |
| DERIVATIVES_SPEC | Derivatives analytics — Greeks, payoff, strategy   |
| EXECUTION_SPEC   | Execution modes 1–5 (sim/paper/live/replay/shadow) |
| GTT_SPEC         | GTT book + template attach pipeline                |
| HEDGE_SPEC       | Proxy hedges + beta regression                     |
| LAB_SPEC         | MCP server + research lab                          |
| NAV_SPEC         | NAV formula v4 + investor portal                   |
| NAVSTRIP_SPEC    | NavStrip pill cluster                              |
| ORDERS_SPEC      | Order placement — ticket + basket                  |
| PERSISTENCE_SPEC | Persistence pipeline — stores, modes, retention    |
| PULSE_SPEC       | MarketPulse page (expanded to ~840 lines, §11–24)  |
| REPLAY_SPEC      | Sim replay driver                                  |
| SETTINGS_SPEC    | DB-backed settings                                 |
| SIMULATOR_SPEC   | Market simulator                                   |
| SYMBOLS_SPEC     | Symbol resolution + virtual roots                  |

**PULSE\_SPEC expansion:** §11–24 added covering the unified pipeline, bucket sort, column definitions, row grouping, LTP tick flash, sparkline merge strategy, context menu, watchlist management, account multi-select, persistent cache layer, closed-hours snapshot behavior, and CardControls cluster.

**Files:**

- docs/specs/\*.md — 21 spec files
- docs/guides/\*.md — operator guides (moved from repo root)
- generate\_pdf.py — updated path to docs/DESIGN\_GUIDE.md
- tools/pdf-gen/README.md — updated path reference
- CLAUDE.md — docs layout table + Common Tasks doc rows added

## 22.21. Spec files expanded — Orders, Activity, Charts (modal + page coverage)

Three behavioral contract specs rewritten from v1.0 to v2.0 with exhaustive coverage of both the modal and bookmarkable-page variants of each surface.

**ORDERS\_SPEC.md v2.0** additions:

- Surface variants: OrderTicket modal (keyboard `t`, context-menu prefill, alert deep-link) vs /orders history page (filter bar, status histogram, CSV export)
- Full OrderTicket state machine: IDLE → LOADING\_PREFLIGHT → PREFLIGHT\_OK/FAIL → SUBMITTING → SUBMITTED/ERROR
- Field-level validation spec (symbol, exchange, product, order type, qty/lots, price, trigger, account)
- 15 audit checkpoints (G2 cap, DRAFT no-broker, basket partial fail, etc.)
- Playwright + pytest test coverage map with known gaps

**ACTIVITY\_SPEC.md v2.0** additions:

- 5 mount points: ActivityLogModal, /activity page, Orders card, Dashboard card, Automation inline — with shared vs isolated filter state documented
- Full `activityStore` API: `$state` variables, getters/setters, `openActivityModal(tab?)`
- 7-tab catalog with endpoint, level-parsing rule, and column list per tab
- LogPanel component props (15 configurable props), scroll behavior, lazy polling

- Deep-link override: `openActivityModal('conn')` from `BrokerHealthBadge` bypasses persisted tab; no-arg call leaves persisted tab unchanged
- Filter sharing rules: shared (modal + /activity page); isolated (Orders card, Dashboard card)

#### CHART\_SPEC.md v2.0 additions:

- `ChartModal` (keyboard `k`, optional symbol/exchange prefill) vs `/charts` page (URL param sync: `?symbol=&exchange=&range=`, `replaceState` on change, cold-start deep-link)
- Full `chartStore` SSOT: `clearData()` wipes ohlcv + lastFetched atomically on symbol change; `clearOhlcv()` wipes bars only; `isFresh()` 30s TTL guard
- All 8 indicators fully specified (SMA 20/50, EMA 20/50 Wilder, VWAP vol-guard, Bollinger  $\pm 2\sigma$  20-period, RSI 14, MACD 12/26/9)
- Known defects:
  - **P1 — Chart hang on null/unresolved symbol:** RESOLVED in commit `cafcf0f7`. Root cause was `$effect` calling `clearData()` on empty-string symbol + `_loadHistorical` early return skipping the loading-reset logic. Fix: `$effect` guards against `!symbol`; early return now explicitly resets loading state before returning. Spec: `docs/specs/CHART_SPEC.md §15`.
  - **P2 — MCX rollover stale bars:** virtual root may resolve to expiring contract on rollover day; workaround is manual range change to force refetch.

#### Files:

- `docs/specs/ORDERS_SPEC.md` — v2.0
- `docs/specs/ACTIVITY_SPEC.md` — v2.0
- `docs/specs/CHART_SPEC.md` — v2.0

## 22.22. Agent execution — actions.py updates (NCO guard + async close)

Three changes to `backend/api/algo/actions.py` ensure agent-driven orders are guarded correctly:

**NCO (NSE Commodity) added to G1/G2 exchange guard** — `run_preflight` previously bypassed the `LOT_MULTIPLE` (G1) and `FAT_FINGER` (G2) guards for F&O on "unknown" exchanges. NCO was added to the exchange set (`"MCX"`, `"NCO"`, `"NFO"`, `"BFO"`, `"CDS"`), so NCO orders now receive the same lot validation as MCX orders. Pre-fix: NCO orders bypassed both guards, risking over-placement.

**Unbound broker guard in \_action\_live\_close\_position** — `broker` variable was not initialized before the try block. On exception, `diagnose_live_failure(broker=broker)` would reference an undefined variable. Fix: `broker = None` before try; guarded `if broker is not None` before calling `diagnose`. Pre-fix: exceptions swallowed silently in Python < 3.11, or masked by unrelated `NameError` in 3.11+.

#### Files:

- `backend/api/algo/actions.py::run_preflight` — NCO added to exchange guard
- `backend/api/algo/actions.py::_action_live_close_position` — broker initialization + guard

## 22.23. Template attach — parent\_lot\_size NFO/BFO/CDS support

`backend/api/algo/template_attach.py::apply_plan_live` now resolves `parent_lot_size` for all F&O exchanges: (`"MCX"`, `"NCO"`, `"NFO"`, `"BFO"`, `"CDS"`). Previously, lot-size resolution was limited to MCX/NCO only, causing the G1 (`LOT_MULTIPLE`) guard to be dead for NFO/BFO/CDS GTT template legs.

#### How it works:

1. `apply_template_to_order` calls `await get_lot_size(symbol, exchange)` for all F&O exchanges
2. `plan.parent_lot_size` is always resolved (never 0) before `apply_plan_live` is called
3. `apply_plan_live` G1 check verifies every GTT leg + wing qty against `parent_lot_size`
4. On failure, returns `AttachResult.errors` immediately (upstream of broker call)

Pre-fix: G1 was dead for NFO/BFO/CDS because `lot_size` was unknown; orders could be placed with qty not a multiple of `lot_size`, causing rejects or silent qty truncation by the broker.

#### Files:

- `backend/api/algo/template_attach.py::apply_plan_live` — `parent_lot_size` resolution extended

## 22.24. Agent engine — topic suppression doesn't burn lifespan

`backend/api/algo/agent_engine.py::_v2_record()` call and the triggered-branch DB mutation block (increment `trigger_count`, set status to `cooldown`, clear `condition_first_true_at`) are now deferred from the per-agent loop into a post-loop survivor section (after `_compute_topic_suppression` runs). Non-triggered paths (`cooldown`→active transition, debounce latch persist) remain inline.

**Behavioral change:** a low-priority agent suppressed by a higher-tier agent on the same topic no longer has its trigger quota burned. It remains active and can fire on a future tick when the high-priority agent is not suppressing. Pre-fix: suppressed agents still consumed their quota as if they fired, leading to premature cooldown of secondary strategies.

**Example scenario:**

- Agent A (loss-hedge, tier=high): fires on -5% loss
- Agent B (loss-rebalance, tier=low, topic="loss"): fires on -3% loss
- Tick 1: A fires, suppresses B
- Tick 2: loss hits -3% but A is in cooldown → B should fire now

Pre-fix: B's quota was burned on tick 1 (suppressed), so it wouldn't fire on tick 2 even though A was cooling.

**Files:**

- `backend/api/algo/agent_engine.py::run_cycle` — survivor-section mutation deferral

## 22.25. Demo banner + Showcase page + Nav + Fullscreen + Derivatives Exp Close

### Demo banner moved to layout

`feat(demo)` commit a6d5e2f0: Demo banner ("Rambo Terminal — live production...") moved from Dashboard-only to `frontend/src/routes/(algo)/+layout.svelte` so it appears on every algo page (Pulse, Dashboard, Derivatives, Orders, Charts, Performance, Automation).

- Positioned fixed; `top: 3rem; height: 2rem; z-index: 46` — does not displace content
- `fix(layout)` commit b8a5203b: Added `:has(.demo-banner)` CSS rules to adjust `.page-header` and `.algo-content` vertical spacing (same pattern as existing `:has(.ps-strip)` rules)

### Showcase page — merged About + Tour

`feat(showcase)` commit 2479ddf1: `/showcase` page replaces separate `TourModal` + `HireMeModal` with unified recruiter/investor landing.

- Contains: hero (name, credentials, roles, contact CTAs), facts grid, 9 architecture cards, `TourModal` (60-second auto-tour still available via `/showcase?tour=1`)
- `chore` commit 417880a2: `HireMeModal.svelte` deleted (was 295 lines); `HireMe` reference removed
- `fix(nav)` commit 2f51706f: Nav link "Tour" → "About"; standalone About button removed from desktop + mobile hamburger menus

### Fullscreen button cluster reorder

`fix(fullscreen)` commit 714f3394: In fullscreen card mode (`.fs-card-on`), button cluster order is now Refresh (leftmost) → Download → Search. Changed by adjusting the `right: calc(2rem + 0.65rem + 1.7rem * N)` slot assignments in `app.css`.

### Derivatives — Exp Close per-underlying spot resolver + TOTAL row decoration

`fix(derivatives)` commits 9bb890bd + b8a5203b:

- **Exp Close spot resolution:** `annotateOptionCandidates` in `derivativesMath.js` now accepts `spot` as a number (backward-compatible) OR `(underlying: string) => number` function (v2). Internally: `const resolveSpot = typeof spot === 'function' ? spot : () => spot;`
- **Full-book expiry analysis:** `expiryCloseAnalysis` uses a `spotResolver` closure that resolves `spot` per-underlying via SSE snapshot → `batchQuote` cache → 0 fallback. The `!spot` early-return gate is removed — full-book expiry close now runs across all positions regardless of selected underlying selection. Enables mixed-underlying baskets (NIFTY + BANKNIFTY + CRUDEOIL simultaneously).
- **TOTAL row CSS convention:** Container holds `display:grid + subgrid + grid-column:1/-1` for alignment only. Amber decoration (background, border-top/bottom, font-size, font-family, text-align) lives on `> span` children — matches `.byund-row-total > span` Snapshot pattern. Container alignment is orthogonal to cell styling.

**Files changed:**

- `frontend/src/routes/(algo)/+layout.svelte` — demo banner markup + styling
- `app.css` — demo banner + fullscreen button cluster + TOTAL row CSS
- `frontend/src/routes/(algo)/admin/showcase/+page.svelte` — new page
- `frontend/src/lib/HireMeModal.svelte` — deleted
- `frontend/src/routes/(algo)/+layout.svelte` — nav Tour→About rename
- `frontend/src/lib/data/derivativesMath.js::annotateOptionCandidates` — `spot` parameter accepts function
- `frontend/src/routes/(algo)/admin/derivatives/+page.svelte` — `spotResolver` closure in `expiryCloseAnalysis`

## 22.26. Derivatives rows + LogPanel CSS Grid — border-bottom + 2-col grid layout

Two CSS refinements improve visual clarity and fix layout bugs.

### Derivatives candidate grid — row separation via border-bottom

**Problem:** `.cand-grid` used `row-gap: 0.2rem` for visual separation between rows. The gap is transparent, so the parent grid's dark navy background bled through, creating unintended dark horizontal stripes.

**Fix:** Switched to `border-bottom: 1px solid rgba(126,151,184,0.10)` on each `.cand-row` (matches `.byund-row` pattern in Snapshot grid). Result: clean row-level separation using the row's own background, not the parent's. Row visual definition is now explicit and isolated.

**Files:** `frontend/src/routes/(algo)/admin/derivatives/+page.svelte` — `.cand-grid` CSS + `.cand-row` CSS

### Unified amber TOTAL row stratum (single CSS rule, two mechanisms)

**Problem:** TOTAL rows in two grids (Legs + Snapshot) were styled separately, with subtle color inconsistencies.

**Solution:** Single unified CSS rule now covers both:

```
.cand-row.cand-row-total,
.byund-row-total > span {
  background: linear-gradient(rgba(251,191,36,0.22), rgba(251,191,36,0.22)), #1d2a44 !important;
  border-top: 2px solid rgba(251,191,36,0.70);
  border-bottom: 1px solid rgba(251,191,36,0.40);
  color: var(--c-action);
  font-weight: 700;
}
```

**Why two different mechanisms achieve the same visual:**

- **Legs grid ( `.cand-row-total` ):** Uses `display: grid; grid-template-columns: subgrid; grid-column: 1/-1` with `column-gap: 0.6rem`. Amber rule applied to the **container** so it covers gap areas. Child `> span` elements receive only typography overrides (no color/background/border).
- **Snapshot grid ( `.byund-row-total` ):** Uses `display: contents` with zero column-gap. Amber rule applied to per-span children directly (each span gets its own amber background, border-top/bottom, color, font-weight).

**Why this split design?** — The container-level rule works for subgrid (grid-gap areas need coverage from parent). Per-span rules work for `display: contents` (eliminates the parent layer, leaving bare span children). By applying the same CSS rule to both selectors, we get identical visual styling across two different grid architectures — reducing maintenance burden and ensuring TOTAL rows render consistently everywhere.

**Files:** `frontend/src/routes/(algo)/admin/derivatives/+page.svelte` — shared `.cand-row.cand-row-total + .byund-row-total > span` CSS rule in `app.css`

### LogPanel 2-column layout — CSS Grid instead of column-count

**Problem:** `LogPanel.svelte` used CSS `column-count: 2` for multi-column magazine layout on Agent/Terminal/System/Conn tabs ( $\geq 900\text{px}$ ). Magazine layouts rely on the browser computing column heights automatically — but this breaks inside `overflow-y: auto` scrollable containers. The browser can't determine column heights when the container itself can scroll, causing layout thrashing.

**Fix:** Switched to CSS Grid: `display: grid; grid-template-columns: 1fr 1fr; column-gap: 1.5rem; align-content: start`. Grid handles dynamic row heights cleanly; scroll axis is independent from layout computation. Column divider via `background-image: linear-gradient(...)` (CSS `column-rule` only works with `column-count`, not Grid).

**Breakpoint:**  $\geq 900\text{px}$  uses Grid;  $< 900\text{px}$  uses `display: block` (single-column).

**Template gate:** `{multiColumn && logTab !== 'order' ? 'lp-multicol' : ''}` — Orders tab always single-column (preserves traditional table layout).

**Files:** `frontend/src/lib/LogPanel.svelte` — `.lp-multicol` CSS + template gate

## Part VII — Operations

### 23. How to add a new template field

Templates have grown organically. The current schema is wide (5 mandatory + 7 optional fields). To add a new one:

## Backend

1. **Add the column** to `OrderTemplate` in `backend/api/models.py`.
2. **Idempotent ALTER TABLE** in `backend/api/database.py::init_db`.
3. **Schema fields** in `backend/api/schemas.py` — `OrderTemplate` (response), `OrderTemplateCreate`, `OrderTemplatePatch`. Also `TicketOrderRequest` + `BasketLeg` if you want a per-submit override.
4. **\_build\_overrides\_json** in `orders.py` (search for the function name) — add the override → JSON key.
5. **resolve\_template\_plan** in `template_attach.py` — add the `_pick()` call and the GTT spec emission.
6. **Seeded defaults** — update `SYSTEM_TEMPLATES` in `templates_seed.py` if your field should ship with a value.

## Frontend

7. **Template management UI** at `/automation/templates` — add the input.
8. **Override input** at the shell-level Template container in `SymbolPanel.svelte` — add the override field + reset on template change.
9. **Preview** — `previewTicketTemplate` should already wire it because the backend handles it; double-check the chip render handles the new shape.

## Documents

10. **Add a row** to §10 if it's a default field.
11. **Update §11** if your field has unusual merge semantics.

## 24. Testing philosophy

The codebase has fewer tests than ideal — that's a known debt. Where tests exist:

- **backend/tests/** — `pytest` + `pytest-asyncio`. Run via `pytest backend/tests/`.
- **frontend/e2e/** — `Playwright`. Run via `cd frontend && npx playwright test`.
- **No unit tests for frontend** — relies on `svelte-check` + manual flows + `e2e`.

The `Playwright` tests run against `dev.ramboq.com` (deployed dev branch). They're slow but high-confidence. Use them for any UX flow that changes; backend `pytest` for any algo/broker change.

**Rule of thumb:** if you're touching `chase.py`, `template_attach.py`, or any broker adapter, add a `pytest` test. If you're touching `SymbolPanel` / `OrderTicket` flow, add a `Playwright` spec.

⚙️ **TECH — Why Playwright over Cypress** — **WHY** Multi-tab support, native browser context isolation, better async waits. `Cypress`'s same-origin restrictions don't fit our auth flow (OAuth-like JWT). **WHAT** Specs in `frontend/e2e/*.spec.js`. Run with `--workers=1` so dev DB writes don't race. **HOW** Use `expect(...).toContainText(...)` for chip assertions; `toHaveAttribute('placeholder', ...)` for input placeholders. **WHERE** `frontend/e2e/`.

## 25. Logging discipline

Three log files matter:

- `api_log_file` — full API log (5MB rotating × 5). Read this first when debugging.
- `api_error_file` — `stdout+stderr` tee from `systemd`. Catches uncaught exceptions.
- `hook.log` — webhook listener output.

Log levels by intent:

- `DEBUG` — for trace-style detail. Verbose; filtered out in prod.
- `INFO` — operator-visible events. Order placed, agent fired, chase replaced.
- `WARNING` — recoverable failures. Broker auth retry, asymmetric GTT, partial OCO failure.
- `ERROR` — uncaught exceptions, lost state. Should also trigger Telegram.

**Don't log inside hot loops** without a rate limit. `_task_performance` ran a `logger.info` per row early on; quickly buried `api_log_file` under non-actionable noise.

## 26. Deployment notes

Both `dev` and `main` deploy via webhook. Push triggers:

```
GitHub push → webhook.ramboq.com → /etc/webhook/dispatch.sh
→ main: /opt/ramboq/webhook/deploy.sh prod main
→ other: /opt/ramboq_dev/webhook/deploy.sh dev <branch>
```

deploy.sh (per env):

1. `git pull`
2. `pip install` (production deps)
3. `npm run build` (vite)
4. `systemctl restart ramboq_api.service / ramboq_dev_api.service`
5. `notify_deploy.py` (Telegram-only since May 2026)

**Per-environment serialisation:** a host-wide `/tmp/ramboq_deploy.lock` prevents concurrent prod + dev builds from race-condition npm conflicts. `nice -n 19 ionice -c 3` on npm so background builds never starve API responsiveness.

**Manual server work after SSH:** always `chown -R www-data:www-data /opt/ramboq /opt/ramboq_dev`. Webhook deploys fail silently if file owner is wrong.

🔗 **TECH — Webhook-based deploy vs CI/CD platform** — **WHY** We're a single-server setup; GitHub Actions would add 30-60s to every deploy plus a \$/runner cost. The webhook is bash + git, zero dependencies. **WHAT** `webhook` (Adnan Hajdarbegovic's daemon) listens on port 9000, validates the HMAC, runs `dispatch.sh`. **HOW** Push to a watched branch triggers it automatically. Logs in `hook.log`. **WHERE** `/etc/webhook/hooks.json` (on server); `webhook/dispatch.sh` + `webhook/deploy.sh` (in repo).

## 26.1 Pre-commit hook for PDF sync

New file `tools/hooks/pre-commit` — bash hook that auto-regenerates `DESIGN_GUIDE.pdf` when `docs/DESIGN_GUIDE.md` is staged for commit. PDF is staged and included in the same commit. Commit is aborted if generation fails (prevents stale PDF landing with fresh .md).

**Install:**

```
cp tools/hooks/pre-commit .git/hooks/pre-commit
chmod +x .git/hooks/pre-commit
```

**Prerequisites:**

- `cd tools/pdf-gen && npm install && npx playwright install chromium`
- `python3` with `PIL` / `Pillow` (standard in venv)

**How it works:**

1. Hook runs on `git commit`
2. Checks if `docs/DESIGN_GUIDE.md` is in the staging area
3. If yes, runs `python3 tools/pdf-gen/generate_pdf.py` to rebuild PDF
4. Stages `docs/DESIGN_GUIDE.pdf` if successful
5. Commits both .md and .pdf together
6. Fails commit if PDF generation errors (forces operator to fix .md syntax before pushing)

**Operator experience:**

- `git add docs/DESIGN_GUIDE.md` (and any other changes)
- `git commit -m "docs(...): update architecture section"`
- Hook automatically regenerates PDF and stages it
- PDF is in the commit; no separate manual step needed

**Files:**

- `tools/hooks/pre-commit` — new hook script
- `tools/pdf-gen/generate_pdf.py` — unchanged PDF generation logic



## 26.5 Recent fixes and operational improvements (Jul 2026)

| Item                                                        | Issue                                                                                                                           | Fix location                                                                                                                  |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Auth:</b><br><b>email_verified</b><br><b>silent drop</b> | Non-designated actors could attempt to change email_verified field on PUT /admin/users/:username without error                  | backend/api/routes/admin.py::update_user_details — added 403 check before PATCH                                               |
| <b>Startup:</b><br><b>PerfSnapshot</b><br><b>NameError</b>  | _backfill_from_disk fired before PerfSnapshot imported                                                                          | backend/api/background.py::_backfill_from_disk — added import inside function                                                 |
| <b>F&amp;O orders: lots-</b><br><b>first API (P0)</b>       | API now accepts LOTS for lot_size > 1 instruments; converts to contracts at boundary                                            | §5.1; backend/api/routes/orders_place.py , orders_basket.py , template_attach.py                                              |
| <b>Day P&amp;L SSOT →</b><br><b>settlement delta</b>        | Compute day P&L as pnl – prev_settlement_pnl (yesterday's daily_book total); fallback to synthetic unrealized for new positions | §21.5.5; backend/api/routes/positions.py::_backfill_prev_settlement_pnl + frontend/src/lib/data/nav.js::baseDayPnlForPosition |
| <b>Sparkline: MCX/CDS</b><br><b>watchlist exchange</b>      | MCX/CDS symbols lost correct exchange during universe warmup                                                                    | §20.1; _sparkline_universe_symbols now queries (tradingsymbol, exchange) pair                                                 |
| <b>Sparkline: virtual</b><br><b>root resolution</b>         | MCX/CDS watchlist items not resolved to active contract before OHLCV fetch                                                      | §20.1; snapshot_sparkline calls symbol_resolver.resolve_symbol()                                                              |
| <b>Sparkline: Tier 4</b><br><b>fallback</b>                 | OHLCV store cold-starts had no fallback during db_only mode                                                                     | §20.1; batch_sparkline added _fill_from_daily_book_sparkline path                                                             |
| <b>BSE ticker: 50-cap</b><br><b>truncation</b>              | BSE holdings after 50th unresolved NSE entry weren't subscribed                                                                 | §14.5.9.5; _perf_subscribe_book_symbols uses chunked loop, not [:50] slice                                                    |
| <b>BSE equity token:</b><br><b>NSE→BSE fallback</b>         | Quote/sparkline lookup for equity exchanges didn't check companion immediately                                                  | §14.5.9.5; _resolve_token_for_sym pairs NSE↔BSE before derivatives walk                                                       |
| <b>BFO F&amp;O: routing</b><br><b>verified</b>              | Confirmed BSE F&O (BFO) wired correctly for subscription + gating                                                               | §14.5.9.5; BFO in sparkline exchanges, maps to NSE segment, token at index 0                                                  |
| <b>NavStrip: lifetime</b><br><b>slot SSOT fix</b>           | P slot 2 + H slots 2-3 accumulated SSE delta on lifetime values, diverging from MarketPulse TOTAL rows                          | §22.18; frontend/src/lib/PositionStrip.svelte                                                                                 |
| <b>CSV export: all ag-</b><br><b>Grid cards</b>             | No download button on Dashboard, PerformancePage, NavBreakdown, Derivatives Legs/Snapshot                                       | §22.19; GridDownloadButton.svelte + CardControls.svelte + csvExport.js                                                        |
| <b>Docs: reorganise +</b><br><b>21 spec files</b>           | All .md docs moved to docs/ ; 21 exhaustive behavioral contract specs added                                                     | §22.20; docs/specs/*.md , docs/guides/*.md                                                                                    |
| <b>Specs v2.0: Orders,</b><br><b>Activity, Chart</b>        | v1.0 specs lacked modal/page surface split, state machine, audit cases, and test map                                            | §22.21; docs/specs/ORDERS_SPEC.md , ACTIVITY_SPEC.md , CHART_SPEC.md                                                          |

## 27. Sprint history + audit fixes

Previous fixes are documented in-code via comments. Key milestones:

| Phase/Sprint         | Key fixes                                                                                                                                                                                                                                       | Lookup                        |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Phase 0–3            | Template attach pipeline (resolve → plan → GTT place)                                                                                                                                                                                           | grep Phase \d                 |
| Sprint A–E           | Reconcile paths, partial fills, Dhan/Groww OCO, rate limits                                                                                                                                                                                     | grep Sprint [A–E]             |
| Gap closure (B–L)    | 28 audit fixes across categories                                                                                                                                                                                                                | git log --grep="audit fix" -i |
| Jul 2026             | F&O lots convention, Day P&L SSOT, sparkline + BSE ticker, NavStrip SSOT, CSV export, docs/specs reorganisation, spec v2.0 (Orders/Activity/Chart)                                                                                              | See §26.5 + commit bodies     |
| Jul 2026 (CC sprint) | Cyclomatic complexity reduction across full codebase: 0 D/E/F-grade functions, C-grade reduced from 393→216. C→B helper extraction pattern applied to all layers (brokers/, api/routes/, api/algo/, shared/). TLM CCWATCH now gates all merges. | git log --grep="refactor(cc)" |

See commit bodies for specific gap IDs (e.g. B-1 = Dhan status map, C-3 = postback fallback window, H-5 = cap warnings). These are documented in code as defensive comments.



## Part VIII — Wrap-up

### 28. Reading order for a new developer

If you've got a week to onboard:

#### Day 1 — understand the shape:

- This doc end-to-end
- `CLAUDE.md` skim (it's the operator-facing manual; some route URLs may reference `/agents/*` which has been redirected to `/automation/*` — see §29)
- `docs/specs/<SURFACE>_SPEC.md` — read the spec for the surface you're working on; each one documents API endpoints, frontend stores, and SSOT chains
- `backend/api/app.py` startup wiring
- `backend/api/models.py` schema

#### Day 2 — order flow:

- `frontend/src/lib/SymbolPanel.svelte + OrderTicket.svelte` (the modal)
- `backend/api/routes/orders.py::ticket_order` (single submit path)
- `backend/api/algo/chase.py::chase_order` (the loop)

#### Day 3 — templates:

- `backend/api/algo/template_attach.py` (resolve + apply)
- `backend/api/algo/templates_seed.py` (the matrix)
- Trace one BUY CE order from click → fill → attach end-to-end

#### Day 4 — brokers:

- `backend/brokers/base.py` (the ABC)
- `backend/brokers/adapters/kite.py` (reference impl)
- `backend/brokers/adapters/dhan.py + groww.py` (vendor quirks)

#### Day 5 — background + extras:

- `backend/api/background.py` (every task)
- `backend/api/algo/actions.py` (agent action handlers)
- `frontend/src/lib/order/ChaseCard.svelte + OrderCard.svelte` (display)

If you've got a day: read §7 (chase loop) above, then read `chase.py::chase_order` source. Everything else extends from that one function.

### 29. When in doubt

Open an `Agent` with `subagent_type=audit` and ask it to trace your specific scenario. The audit agents in this codebase are well-calibrated for finding subtle issues. Don't merge a change to `chase.py` or `template_attach.py` without one.

**Known doc-drift in `CLAUDE.md`** (as of the most recent doc audit): the older operator manual still references `/agents`, `/agents/activity`, `/agents/fragments` URLs. These have been redirected to `/automation`, `/automation/activity`, etc. The redirect routes still work; the URLs in `CLAUDE.md` are just stale. The current canonical URLs are under `/automation/*`.

## 30. Operator's mental model — the one-page summary

| Action                                               | Read this section                                     |
|------------------------------------------------------|-------------------------------------------------------|
| "What happens when I click Submit on Ticket?"        | §5 — single ticket sequence                           |
| "What does the chase loop do between attempts?"      | §7 — chase lifecycle                                  |
| "How does TP/SL get attached?"                       | §9 — template attach pipeline                         |
| "Why is my SL not ratcheting on Dhan?"               | §13 — trail-stop subsystem                            |
| "How does the Default pill pick the right template?" | §10 — 4-default matrix                                |
| "When does the preview chip swap on Chain?"          | §17 — frontend modal state                            |
| "What runs in the background?"                       | §20 — task topology                                   |
| "Why does the navbar strip not match the dashboard?" | §21 — data refresh paths                              |
| "What can a demo visitor do?"                        | §22 — demo mode flow                                  |
| "How do I add a new broker?"                         | §15                                                   |
| "How do I add a new template field?"                 | §23                                                   |
| "What's the tech stack?"                             | §2 — overview; also inline ⦿ TECH callouts throughout |

## Part IX — Change recipes (cookbook)

This section turns the design knowledge above into runnable change recipes. Each recipe lists the **exact files to edit**, the **exact pattern to copy**, and the **verification step** before commit. Use these as templates — copy-paste, rename, tweak.

The cookbook is intentionally prescriptive. You do not need to read the full doc above to follow a recipe; you only need §3 (architectural principles) for the philosophy, then jump straight here.

### 31. Recipe: add a new route

**Scenario:** you want a new endpoint, e.g. `GET /api/positions/heatmap` that returns aggregated per-symbol stats.

#### Steps

1. **Pick the right route file.** `backend/api/routes/` is grouped by domain ( `orders.py` , `positions.py` , `holdings.py` , `agents.py` , etc.). Add the new route to the matching file. New domain? Create `heatmap.py` .
2. **Write the msgspec response type** in `backend/api/schemas.py` :

```
class HeatmapRow(msgspec.Struct):
    symbol: str
    pnl: float
    weight: float
class HeatmapResponse(msgspec.Struct):
    rows: list[HeatmapRow]
    refreshed_at: str
```

3. **Add the route** to the controller (mirror an existing simple route as a template — e.g. `PositionsController.list_positions` in `backend/api/routes/positions.py` ):

```
@get("/heatmap", guards=[auth_or_demo_guard])
async def heatmap(self, request: Request) -> HeatmapResponse:
    is_demo = request.state.is_demo
    rows = await self._build_heatmap()
    if is_demo:
        rows = [_mask_row(r) for r in rows]
    return HeatmapResponse(rows=rows, refreshed_at=timestamp_display())
```

4. **Register the controller** in `backend/api/app.py` if the file is new. Existing controllers don't need re-registration.
5. **Frontend wrapper** in `frontend/src/lib/api.js` :

```
export const fetchHeatmap = () => _get('/api/positions/heatmap');
```

The wrapper handles auth, retries, demo masking display, and error trimming automatically. **Never call `fetch()` directly from a component** — always go through `api.js`.

6. **Demo masking.** Read paths must mask account values for demo sessions (§22). Use `mask_column(col)` helper, never roll your own.
7. **Verify.** `pytest backend/tests/test_routes_smoke.py` (add a smoke test if the route is non-trivial) + manual curl.

## 32. Recipe: add a column to an existing table

**Scenario:** you want `algo_orders.last_chase_quote` to store the last broker depth snapshot.

### Steps

1. **Edit `backend/api/models.py`**. Add the field to the SQLAlchemy model:

```
last_chase_quote: Mapped[str | None] = mapped_column(Text, nullable=True)
```

2. **Add an idempotent ALTER** to `backend/api/database.py::init_db`:

```
await conn.execute(text(
    "ALTER TABLE algo_orders ADD COLUMN IF NOT EXISTS last_chase_quote TEXT"
))
```

The `IF NOT EXISTS` is non-negotiable — `init_db` runs on every startup, idempotency required.

3. **Update msgspec schema** in `backend/api/schemas.py` if the column should be returned over the wire:

```
class AlgoOrderInfo(msgspec.Struct, kw_only=True):
    ...
    last_chase_quote: str | None = None
```

`kw_only=True` is non-negotiable — the struct interleaves required + optional fields, and Python 3.13's stricter msgspec refuses that without it. Don't strip the modifier when copy-pasting.

4. **Populate it.** Decide which code path writes to it. For our example, `chase.py::chase_order` writes after each depth quote fetch.
5. **Frontend render** (optional). Add a column to `OrderCard.svelte` or `OrderTab.svelte` ag-Grid config; render via `cellRenderer`.
6. **Verify.** `psql -d ramboq_dev -c "\d algo_orders"` shows the new column; `pytest ; e2e` if frontend-visible.

⚠ **Never write a migration script.** The `init_db ALTER TABLE ... IF NOT EXISTS` pattern is the migration mechanism. We don't use Alembic.

## 33. Recipe: add a new background task

**Scenario:** you want `_task_unrealized_pnl_alert` that fires every 60s during market hours.

### Steps

1. **Define the coroutine** in `backend/api/background.py`. Copy the shape of `_task_oco_pair_watcher` — it's the simplest template:

```
async def _task_unrealized_pnl_alert():
    interval = 60
    while True:
        try:
            await _run_unrealized_pnl_check()
        except Exception as e:
            logger.exception(f"_task_unrealized_pnl_alert failed: {e}")
            await asyncio.sleep(interval)
```

The `try/except` around the loop body is non-negotiable — without it, an uncaught exception silently kills the task forever.

2. **Spawn it at startup.** In `backend/api/app.py::on_startup`:

```
asyncio.create_task(_task_unrealized_pnl_alert())
```

3. **Gate by market hours** if appropriate. Use `is_any_segment_open()` from `backend/shared/helpers/date_time_utils.py`:

```
if not is_any_segment_open():
    await asyncio.sleep(interval)
    continue
```

4. **Gate by capability flag.** If the task hits an external service, wrap in `is_enabled('telegram' | 'mail' | ...)` so dev branches don't spam.
5. **Verify.** Start dev ( `uvicorn backend.api.app:app` ), watch `.log/api_log_file` for the task's INFO/DEBUG logs, kill, restart, confirm it picks up cleanly.

⚠ **Do not use `time.sleep`.** Always `await asyncio.sleep(...)`. Sync sleep blocks the entire event loop.

## 34. Recipe: add a new agent action

**Scenario:** you want `square_off_underlying` so an agent can close every position on a given underlying.

### Steps

1. **Add the handler** in `backend/api/algo/actions.py`. Mirror the shape of `_action_close_position`:

```
async def _action_square_off_underlying(
    action: AgentAction,
    context: dict[str, Any],
) -> ActionResult:
    params = action.params or {}
    underlying = params.get("underlying")
    if not underlying:
        return ActionResult(ok=False, reason="missing underlying")
    ...
```

2. **Register it** in the `_ACTION_HANDLERS` map at the bottom of `actions.py`:

```
_ACTION_HANDLERS["square_off_underlying"] = _action_square_off_underlying
```

3. **Add the grammar token** in `backend/api/algo/grammar.py::_SYSTEM_TOKENS`:

```
{
    "grammar_kind": "action",
    "token_kind": "action_type",
    "token": "square_off_underlying",
    "value_type": "enum",
    "resolver": "backend.api.algo.actions._action_square_off_underlying",
    "params_schema": {
        "required": ["underlying"],
        "properties": {"underlying": {"type": "string"}},
    },
    "is_system": True,
    "is_active": True,
}
```

4. **Mode resolution.** Honor `_resolve_mode()` — never call broker directly. Use `get_broker(account)` and respect the row's mode.
5. **Verify.** Create a test agent in dev via `/admin/tokens + /automation`, fire-in-simulator, confirm event row + broker call.

## 35. Recipe: add a new template field (worked example)

**Scenario:** add `tp_breakeven_lock: bool` — when true, after TP1 fires, modify SL to entry price (free trade).

### Steps

1. **Backend column** in `backend/api/models.py::OrderTemplate`:

```
tp_breakeven_lock: Mapped[bool | None] = mapped_column(Boolean, nullable=True)
```

2. **Idempotent ALTER** in `backend/api/database.py::init_db`:

```
await conn.execute(text(
    "ALTER TABLE order_templates ADD COLUMN IF NOT EXISTS tp_breakeven_lock BOOLEAN"
))
```

3. **Schemas** in `backend/api/schemas.py`:

- `OrderTemplate` (response): `tp_breakeven_lock`: `bool | None = None`
- `OrderTemplateCreate`, `OrderTemplatePatch`: same field
- `TicketOrderRequest`: optional `tp_breakeven_lock_override`: `bool | None = None` if you want per-submit override
- `BasketLeg`: same per-leg

4. **Override JSON build** in `backend/api/routes/orders.py::_build_overrides_json` (search for the function):

```
if data.tp_breakeven_lock_override is not None:
    result["tp_breakeven_lock"] = data.tp_breakeven_lock_override
```

5. **Plan resolution** in `backend/api/algo/template_attach.py::resolve_template_plan`:

```
breakeven_lock = _pick("tp_breakeven_lock", template, overrides)
```

Then emit the appropriate behavior — for breakeven lock, you'd persist it into `attached_gtts_json` and watch for TP1 fill events.

6. **Seeded defaults** in `backend/api/algo/templates_seed.py::SYSTEM_TEMPLATES` — add to whichever defaults should ship with it ON.

7. **Frontend management UI** at `frontend/src/routes/(algo)/automation/templates/+page.svelte`:

- Add a toggle to the create/edit form
- Surface it in the listing

8. **Frontend override input** in `frontend/src/lib/SymbolPanel.svelte`:

- Add a `_sharedTpBreakevenLockOverride = $state(null)` shell-level
- Render in the Template row alongside the existing override inputs
- Reset on template change (search the `$effect` block that watches `_sharedTemplateId`)
- Pass through to `OrderTicket` and per-leg basket logic

9. **Preview chip**. Backend handles the math; frontend just renders. The preview will automatically pick up the new override because `previewTicketTemplate` passes through the full overrides dict.

10. **Update this doc**. Add a row to §10 (4-default matrix) if any default ships with it, and to §11 (merge order) if your field has unusual merge semantics.

11. **Verify**. `svelte-check`, `pytest`, `e2e` on `dev.ramboq.com`: create a template with the field, place a paper order, watch the chain of events in `.log/api_log_file`.

## 36. Recipe: add a new broker capability flag

**Scenario:** you want to track whether each broker supports iceberg orders (currently not in the matrix).

### Steps

1. **Add the field** to `backend/brokers/capabilities.py::BrokerCapabilities`:

```
@dataclass(frozen=True)
class BrokerCapabilities:
    ...                # existing fields
    iceberg_order: bool # No default — force explicit setting per broker
```

Note the discipline: real fields don't get defaults (audit B-5 lesson). Every broker constant must set every field.

2. **Set per-broker explicitly** in **all four** constants — KITE/DHAN/GROWW plus the `UNKNOWN_CAPS` fallback at the bottom of `capabilities.py`. Missing `UNKNOWN_CAPS` will crash the unknown-broker code path at runtime:

```
KITE_CAPS    = BrokerCapabilities(..., iceberg_order=True)
DHAN_CAPS    = BrokerCapabilities(..., iceberg_order=False)
GROWW_CAPS   = BrokerCapabilities(..., iceberg_order=False)
UNKNOWN_CAPS = BrokerCapabilities(..., iceberg_order=False) # conservative default
```

3. **Capability registry** in `capabilities.py::CAPS_BY_BROKER_ID` already routes by `broker_id` — no change needed.

4. **Frontend warning helper** in `frontend/src/lib/data/brokerCapWarnings.js`:

- Update the warning-aggregation logic to surface a warning when a template asks for an iceberg leg against a broker where `!caps.iceberg_order`.

5. **Consumer code** queries via `get_broker(account).capabilities.iceberg_order` or the HTTP endpoint `/api/admin/brokers/{account}/capabilities`.

6. **Verify**. Inspect `/admin/brokers` page; the new cap should surface in the row.

## 37. Recipe: add a new page

**Scenario:** new admin page at `/admin/funds-history`.

### Steps

1. **Create the route file** `frontend/src/routes/(algo)/admin/funds-history/+page.svelte`. Copy structure from `/admin/brokers/+page.svelte` — it's the simplest admin page template.
2. **Page header.** Use the canonical pattern (see "Page-header rule" in CLAUDE.md or §17 of this doc):

```
<div class="page-header">
  <span class="algo-title-group">
    <h1 class="page-title-chip">Funds History</h1>
  </span>
  <span class="algo-ts">{$nowStamp}</span>
  <span class="ml-auto"></span>
  <span class="page-header-actions">
    <RefreshButton onClick={load} loading={_loading} label="Refresh" />
    <PageHeaderActions />
  </span>
</div>
```

3. **Navbar entry** in `frontend/src/routes/(algo)/+layout.svelte` `navItems`. Pick a group (`monitor` | `analyze` | `modes` | `build` | `config`). Set `adminOnly: true` if it should hide in demo mode.
4. **Data loading.** Always:
  - Use `$effect` for mount + cleanup (not legacy `onMount`)
  - Use `marketAwareInterval` from `$lib/stores` for polling (not raw `setInterval`)
  - Read via `$lib/api.js` wrappers
5. **Verify.** Visit the route, check navbar entry, confirm Refresh works, watch console for errors. Playwright spec if non-trivial.

## 38. Recipe: add a setting

**Scenario:** add `chase.max_consecutive_errors` so operator can tune the abort threshold (currently hardcoded `_MAX_CHASE_ERRORS=5`).

### Steps

1. **Add to SEEDS** in `backend/shared/helpers/settings.py`:

```
("chase", "chase.max_consecutive_errors", "int", 5,
 "Abort a chase after this many consecutive API failures", None, "errors"),
```

2. **Read it** wherever the constant was used. Replace `_MAX_CHASE_ERRORS` with `get_int('chase.max_consecutive_errors', 5)`. The cache invalidates on every PATCH; reads are O(1).
3. **Verify.** Visit `/admin/settings` → confirm new row in the Chase bucket. Edit it, watch the change take effect on next chase iteration without restart.

⚡ **TECH** — the seeder preserves operator overrides on deploy; only the description / schema / default\_value refresh. So bumping the default in code only affects fresh installs.

## 39. Recipe: change an existing default template

**Scenario:** operator wants `default-long-option` to have SL -40% instead of no SL.

### Steps

1. **Edit SYSTEM\_TEMPLATES** in `backend/api/algo/templates_seed.py`:

```
{
  "name": "default-long-option",
  ...
  "sl_pct": 40, # was None
  "sl_type": "LIMIT",
}
```

2. **Re-seeder behavior.** On startup, `seed_templates` (in `templates_seed.py`) rebuilds system templates by `name` — operator's edits to custom templates are preserved, but system templates are overwritten. **The operator's pulls of `default-long-option` will get the new SL on next deploy.**
3. **If operator has saved-instance edits** (i.e. clicked Edit on a system template and saved), those land in a separate row keyed by `user_id`. They survive system re-seed. To force-refresh, the operator deletes their saved copy.
4. **Verify.** Restart dev, hit `/automation/templates`, confirm SL value is updated. Place a test order — fill should trigger SL GTT placement.

## 40. Recipe: wire a new notification channel

**Scenario:** add Slack as a notify channel alongside Telegram + email.

### Steps

1. **Add config keys** in `backend/config/secrets.yaml` (manually on server, gitignored):

```
slack_webhook_url: "https://hooks.slack.com/services/..."
```

2. **Add capability flag** in `backend/config/backend_config.yaml::cap_in_dev`:

```
slack: True
```

`is_enabled('slack')` returns `True` on main always, else respects this flag.

3. **Add the helper** in `backend/shared/helpers/alert_utils.py`:

```
def _send_slack(message: str) -> None:
    if not is_enabled('slack'):
        return
    webhook = secrets.get('slack_webhook_url')
    if not webhook:
        return
    requests.post(webhook, json={"text": message}, timeout=5)
```

4. **Add the grammar token** for the notify channel:

```
# backend/api/algo/grammar.py::_SYSTEM_TOKENS
{
    "grammar_kind": "notify",
    "token_kind": "channel",
    "token": "slack",
    ...
}
```

5. **Wire it in `_dispatch`** in `alert_utils.py` — for each notify event, check `channel == 'slack'` and call `_send_slack`.
6. **UI checkbox** in agent editor ( `frontend/src/routes/(algo)/automation/+page.svelte` ) — add Slack to the events grid.
7. **Verify.** Create a test agent with Slack notify, fire in simulator, confirm message lands.

## 41. Recipe: ship a fix to dev + main

**Scenario:** you've finished a small fix on `dev` branch and want to deploy to prod.

### Steps

1. **Run pre-flight checks locally.**

- `cd backend && pytest` (or scoped to affected files)
- `cd frontend && npm run check` (svelte-check)
- `cd frontend && npx playwright test --workers=1` if frontend-touching

2. **Commit on dev.** Use the conventional format:

```
<Sprint/scope>: <one-line summary>

<optional body explaining why>

Co-Authored-By: Claude Opus 4.7 (1M context) <noreply@anthropic.com>
```

3. **Push dev.** `git push origin dev` . Webhook triggers dev deploy automatically. Watch logs:

```
ssh ramboq "tail -f /opt/ramboq_dev/.log/api_log_file"
```

4. **Verify on dev.ramboq.com** . Manual smoke test of the changed feature.

5. **Merge to main.**

```
git checkout main
git merge --ff-only dev
git push origin main
git checkout dev # back to dev for next change
```

6. **Webhook deploys prod automatically.** Watch `/opt/ramboq/.log/api_log_file` .

7. **Telegram deploy ping** lands in `RamboQuant Alerts` group — confirm.

⚠️ Don't squash-merge. We keep linear history via `--ff-only` . ⚠️ Don't tag releases; we deploy on every push.

## 42. Cross-cutting checklist before every commit

Run mentally before every commit. Skipping any of these has burned us before:

- ☐ **Demo mode honored?** Read paths mask accounts; write paths return 403 or downgrade to paper (§22).
- ☐ **Idempotency on side-effects?** Anything that places orders/GTTs needs a guard (§3.2).
- ☐ **Mode resolution?** Order code reads `row.mode` instead of branching by `if live` (§3.4).
- ☐ **Logger discipline?** No `print()` ; no logger calls in hot loops (§25).
- ☐ **Hot-loop sleep?** `asyncio.sleep` , never `time.sleep` (§4.3).
- ☐ **TODO/FIXME?** None in code paths; finish or extract.
- ☐ **Old tests still pass?** `pytest` on the closest test file.
- ☐ **svelte-check clean?** No new errors introduced.
- ☐ **CLAUDE.md updated?** If a fact in CLAUDE.md changes, fix it here.
- ☐ **DESIGN\_GUIDE.md updated?** This file. Especially recipe sections + line-anchored grep targets if functions moved.
- ☐ **Sprint/audit fix labeled?** Commit body mentions the gap ID or sprint letter so `git log --grep` finds it.
- ☐ **Co-author trailer?** Co-Authored-By line at the end of the commit body.

If you can't tick every box, hold the commit. The cost of pausing is low; the cost of regression is high.

## Where to learn more

| If you want to learn...                       | Read                                        |
|-----------------------------------------------|---------------------------------------------|
| How the operator uses the platform end-to-end | <code>USER_GUIDE.md</code>                  |
| Day-to-day operations + troubleshooting       | <code>ADMIN_GUIDE.md</code>                 |
| Agent authoring + testing                     | <code>AGENTS_GUIDE.md</code>                |
| Simulator scenarios + Run-in-Simulator        | <code>SIMULATOR_GUIDE.md</code>             |
| Lab + MCP-driven research workflow            | <code>LAB_MCP_GUIDE.md</code>               |
| Operator's-eye-view of every page             | <code>CLAUDE.md</code> (large, but indexed) |
| <b>Architecture + change recipes</b>          | <b>This document</b>                        |

## Glossary

Curated index of the terms that appear repeatedly in this guide. Alphabetized so you can jump-lookup without scrolling the TOC.



| Term                  | Meaning                                                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Action                | Side-effect an agent performs when its condition fires — place order, close position, alert. See §34.                                                                         |
| AgentEngine           | Declarative rule runner. Reads <code>agents</code> table + built-in <code>BUILTIN_AGENTS</code> , evaluates conditions each cycle, dispatches actions. §22.5, §34.            |
| Alert                 | Runtime event produced when an agent condition fires. Persisted to <code>agent_events</code> ; may or may not have a Notify or Action. Distinct from <b>Notify</b> .          |
| Basket order          | Multi-account / multi-leg order dispatched atomically. <code>P05T /api/orders/basket</code> groups by account and fans out via <code>asyncio.gather</code> . §6, §22.10.      |
| Broker abstraction    | Uniform Python interface (Kite / Dhan / Groww adapters) so route handlers stay broker-agnostic. <code>backend/brokers/adapters/</code> . §14.                                 |
| Chase loop            | Adaptive limit-order engine that re-quotes toward the touch until filled or capped. Spread-aware. §7, §12.                                                                    |
| CardHeader.svelte     | Unified card title bar using CSS custom properties ( <code>--ch-*</code> ) for theming; embeds <code>CardControls</code> ; wired to 9 card header sites. §18.5.               |
| cap_in_dev            | Nested dict of capability flags in <code>backend_config.yaml</code> . All True on <code>main</code> , per-branch on dev. Gated via <code>is_enabled('&lt;cap&gt;')</code> .   |
| ChaseAggPicker.svelte | L/M/H aggression picker with <code>variant='ticket' 'panel'</code> prop. Extracted to standalone component. §18.5.                                                            |
| close_settled         | Second phase of the snapshot lifecycle — fires 15 min after <code>&lt;exch&gt;:close</code> . UPSERTs broker's weighted-avg-last-30-min close price.                          |
| conn_service          | Standalone Litestar app on <code>/tmp/ramboq_conn.sock</code> that owns broker sessions (Kite WebSocket, Dhan/Groww tokens). Restart independent of the main API.             |
| Demo mode             | Signed-out + prod branch — read-only + PII-masked. Write paths return 403 or degrade to paper. §22.                                                                           |
| Firm NAV              | <code>cash_sod + option_premium + Σ position.unrealised + Σ holdings.cur_val</code> . Canonical formula (v4) in <code>backend/api/algo/nav.py:compute_firm_nav</code> .       |
| GTT                   | Good-Till-Triggered order — broker-side conditional order. Kite native; Dhan OCO leg; Groww emulated. §9.                                                                     |
| @for_all_accounts     | Decorator that fans out a broker call across every account and concatenates the results. <code>pd.concat(..., ignore_index=True)</code> at call site.                         |
| Idempotency guard     | Column like <code>attached_gtts_json</code> IS NULL that ensures a side-effect fires at most once even under postback retries. §3.2.                                          |
| KiteTicker            | Persistent Kite WebSocket. One per <code>conn_service</code> process. Ticks land in <code>/dev/shm/ramboq_ticks</code> mmap; main API reads via <code>MmapTickReader</code> . |
| LP unit               | Limited-Partner accounting unit. <code>units_held × nav_per_unit = slice</code> . §22.6.                                                                                      |
| MarketPulse           | Two-side ag-Grid page ( <code>/pulse</code> ) — pinned watchlists + movers left, positions + holdings right. Canonical operator surface.                                      |
| MCP                   | Model Context Protocol — Claude-Code integration exposing 25 platform tools (17 read-only + 2 persist + 6 write-gated).                                                       |
| Notify                | Delivery channel wrapping an Alert — telegram / email / websocket / log. Vocabulary chain: Agent → Alert → Notify → Action.                                                   |
| OHLCV                 | Open-High-Low-Close-Volume daily bars. Cached in <code>ohlcv_store</code> (3-tier: LRU → PostgreSQL → broker). 5-year retention.                                              |
| Paper mode            | Live-quote execution against <code>PaperTradeEngine</code> — no broker orders. Mode 2 of the confidence ladder (sim → paper → shadow → live).                                 |
| PBKDF2                | Password hash algorithm (SHA-256, 210k iters). JWT signing separate — HS256.                                                                                                  |
| Preflight             | Fat-finger + lot-multiple guards before order placement (G1, G2). Parallelized via <code>asyncio.gather</code> . §22.10.                                                      |
| Proxy hedge           | Cross-reference between holdings and option roots. $\beta$ -regressed. <code>hedge_proxies</code> table.                                                                      |
| Refresh cycle mode    | Persistence override — <code>off / soft / hard</code> . Bypasses cache tiers when defect-recovering. Runtime-only, resets on restart.                                         |
| Shadow mode           | Log-only mode — validates payload against real broker but does not execute. Mode 5, prod-only.                                                                                |
| snapshot_extras       | Payload block in <code>daily_book.payload_json</code> carrying open/high/low/close_settled/day_change_val/... for closed-hours reads.                                         |
| Snapshot lifecycle    | Per-exchange event sequence — <code>open</code> → <code>close</code> (first-cut snapshot) → <code>close_settled</code> (broker's settled close).                              |
| SSOT                  | Single Source Of Truth. See <code>baseDayPnlForPosition</code> (Day P&L), <code>compute_firm_nav</code> (NAV), <code>resolve_current_price</code> (LTP resolver).             |
| Template attach       | Post-fill automation — attaches GTT exits and take-profit legs to a filled parent. Idempotent via <code>parent_order_id</code> . §9.                                          |
| TemplateBar.svelte    | TP/SL/wing override row extracted from <code>SymbolPanel</code> ; uses <code>\$bindable()</code> for all four overrides. §18.5.                                               |
| Ticker mmap           | <code>/dev/shm/ramboq_ticks</code> — fixed 4096-slot shared-memory buffer, version-word atomic, lock-free reads. Main API tails it via <code>MmapTickReader</code> .          |
| Trail stop            | Stop-loss that ratchets toward the touch on favorable moves. Broker-side (Kite <code>trail_gtt</code> ) or emulated. §13.                                                     |
| Virtual root          | MCX/CDS synthetic symbol (e.g. <code>CRUDEOIL</code> = front-month, <code>CRUDEOIL_NEXT</code> = back-month). Resolver: <code>symbol_resolver.py</code> .                     |

## Alphabetical section index

Quick-jump index by first significant word — useful when you remember a name but not the section number.

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